

Probability

QuantProf revision · each problem: the idea in plain words with a live diagram, then a numbered proof.

- | | |
|--|---|
| 1 · Basic terms | 21 · The normal distribution |
| 2 · Probability of an event | 22 · Total probability, continuous case |
| 3 · Inclusion–exclusion | 23 · Expected value |
| 4 · Counting in probability | 24 · Law of total expectation |
| 5 · Conditional probability | 25 · LOTUS |
| 6 · Multiplication rule & independence | 26 · Tail sum (discrete) |
| 7 · Law of total probability | 27 · Tail integral |
| 8 · Bayes | 28 · Linearity of expectation |
| 9 · Paradoxes | 29 · Indicator variables |
| 10 · Reduced sample space | 30 · Products of independent variables |
| 11 · Geometric probability | 31 · Means of common distributions |
| 12 · Recursion in probability | 32 · Recursion in expectation |
| 13 · Symmetry I: ad hoc | 33 · Symmetry in expectation I |
| 14 · Symmetry II: fewer states | 34 · Symmetry in expectation II: fewer states |
| 15 · Symmetry III: continuous → discrete | 35 · Random points on a segment |
| 16 · Random variables | 36 · Conditional expectation |
| 17 · PMF and CDF (discrete) | 37 · Inclusion–exclusion for waiting times |
| 18 · Common discrete distributions | 38 · Summation via expectation |
| 19 · PDF and CDF (continuous) | 39 · Variance |
| 20 · Common continuous distributions | |

1 · Basic terms

- 1. Sample space Ω :** the set of all outcomes. **Event:** a subset of Ω .

2. $A \cap B$: both happen. $A \cup B$: at least one happens.
 A^c : A does not happen.
3. **Mutually exclusive**: $A \cap B = \emptyset$. **Exhaustive**: $A_1 \cup \dots \cup A_n = \Omega$. A **partition** is both.
4. **Independent**: $P(A \cap B) = P(A)P(B)$ (see §6).

2 · Probability of an event

1. If all outcomes are **equally likely**: $P(A) = |A|/|\Omega|$. This formula only holds under that condition.
2. $P(A^c) = 1 - P(A)$.
3. Trap: {red, blue} is not an equally likely sample space for 9 red + 1 blue balls. Label the balls 1–10 instead: $P(\text{blue}) = \frac{1}{10}$.

Warm-ups

Question: (a) One card is drawn from a well-shuffled 52-card deck. What is the probability that it is a king? (b) A fair coin is flipped 3 times. What is the probability of getting exactly 2 heads?

Key idea

List outcomes so that each is equally likely.

Picture

List outcomes that are equally likely, then count. A card: 52 equally likely cards, 4 kings. Three flips: 8 equally likely sequences, 3 with exactly two H.

Equally likely outcomes, favourable ones highlighted.

A♠	2♠	3♠	4♠	5♠	6♠	7♠	8♠	9♠	10♠	J♠	Q♠	K♠
A♥	2♥	3♥	4♥	5♥	6♥	7♥	8♥	9♥	10♥	J♥	Q♥	K♥
A♦	2♦	3♦	4♦	5♦	6♦	7♦	8♦	9♦	10♦	J♦	Q♦	K♦
A♣	2♣	3♣	4♣	5♣	6♣	7♣	8♣	9♣	10♣	J♣	Q♣	K♣

(a) 4 kings of 52 cards: $\frac{1}{13}$.

(b) The 8 flip sequences: HHH **HHT HTH** HTT **THH** THT TTH TTT – 3 of 8 have exactly 2 heads: $\frac{3}{8}$.

Solution

Setup. (a) One card from a shuffled 52-card deck: all 52 cards are equally likely. (b) Three fair flips: all $2^3 = 8$ sequences HHH, HHT, ..., TTT are equally likely.

- (a) There are 4 kings, so $P(\text{king}) = \frac{4}{52} = \frac{1}{13} \approx 0.07692$.

Why: favourable outcomes over all outcomes, valid because the 52 outcomes are equally likely.

- (b) Exactly 2 heads means choosing which 2 of the 3 flips are heads: $\binom{3}{2} = 3$ sequences (HHT, HTH, THH). So $P = \frac{3}{8}$.

Why: 3 favourable out of 8 equally likely sequences.

Conclusion. $\frac{1}{13}$ and $\frac{3}{8}$. ■

3 • Inclusion–exclusion

The same as counting PIE, with $|\cdot|$ replaced by $P(\cdot)$ (divide by $|\Omega|$, or use additivity):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C).$$

Sports and drinks

(a) In a group of people, the probability that a person plays sport S is 0.7, plays sport B is 0.5, and plays both is 0.4.

(b) In another group, the probabilities that a person drinks:

tea, coffee, juice: 0.6, 0.5, 0.4;

tea and coffee, tea and juice, coffee and juice:
0.30, 0.20, 0.15;

all three: 0.10.

Question: (a) what is the probability that a person plays at least one of the two sports? (b) What is the probability that a person drinks at least one of the three drinks?

Key idea

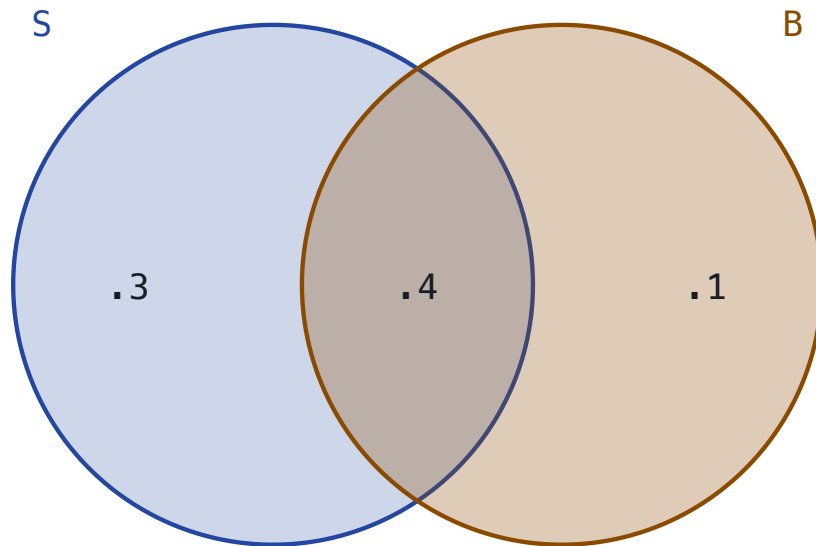
Plug into PIE.

Picture

Adding the circles counts the overlaps too often; inclusion-exclusion fixes that. Fill the Venn

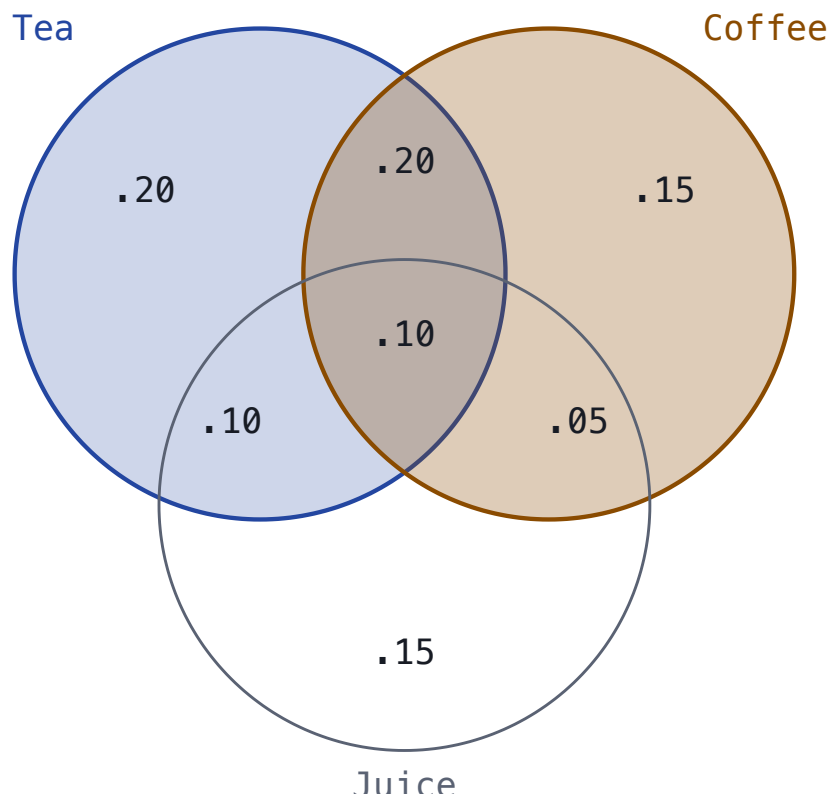
diagram from the centre out and the regions add to the answer.

(a) Probabilities of each region.



(a) Regions $.3 + .4 + .1 = 0.8 = .7 + .5 - .4$.

(b) Each region, worked out from the centre (.10) outwards.



(b) All seven regions add to $0.95 = 1.5 - .65 + .10$.

Solution

1. (a) $P(S \cup B) = P(S) + P(B) - P(S \cap B) = .7 + .5 - .4 = 0.8$.

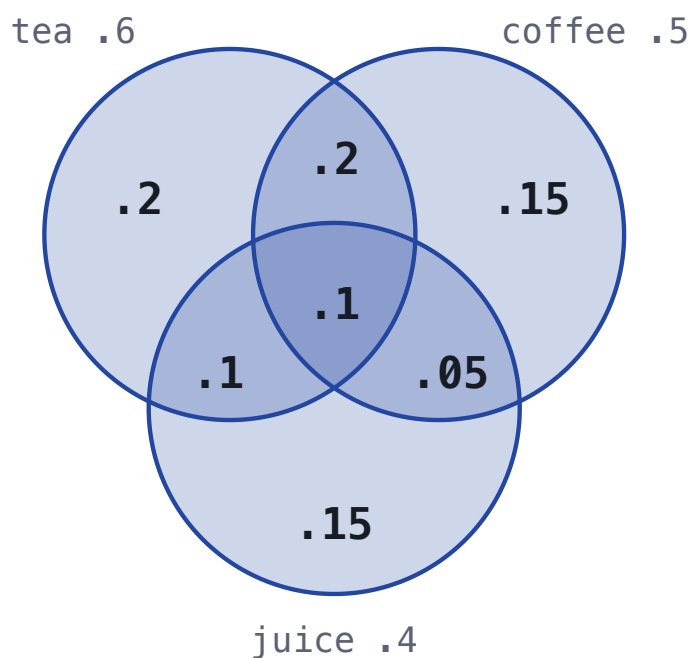
Why: inclusion-exclusion for two events: $P(S) + P(B)$ counts $S \cap B$ twice, so subtract it once.

2. (b) With T, C, J for tea, coffee, juice: $P(T \cup C \cup J) = (.6 + .5 + .4) - (.30 + .20 + .15) + .10$.

Why: inclusion-exclusion for three events: add the singles, subtract the pairs, add back the triple.

3. $= 1.5 - .65 + .10 = 0.95$.

Why: arithmetic. The Venn diagram below shows the 7 regions adding to 0.95.



Conclusion. 0.8 and 0.95. ■

4 · Counting in probability

Ask: can I count both all outcomes and the favourable ones, in an equally likely model? If so, $P = \text{favourable/all}$.

A and B not adjacent

5 people, including A and B, stand in a row in a random order (all orders equally likely).

Question: what is the probability that A and B are *not* standing next to each other?

Key idea

Complement of the block count.

Picture

Count the opposite, then subtract. Glue A and B into one block: 4 units in $4!$ orders, times 2 inside the block = 48 of the 120 rows. So apart = $1 - 48/120$.

Run once

Run 50,000

Solution

Setup. 5 people in a random row: all $5! = 120$ orders are equally likely.

1. Rows with A and B adjacent: glue A and B into a block, giving 4 units in $4! = 24$ orders, times 2 orders inside the block: 48.

Why: each such row comes from exactly one order of the units and one order inside the block.

2. $P(\text{adjacent}) = \frac{48}{120} = \frac{2}{5}$.

Why: favourable over equally likely.

3. $P(\text{not adjacent}) = 1 - \frac{2}{5} = \frac{3}{5}$.

Why: complement.

Conclusion. A and B are apart with probability $\frac{3}{5}$. ■

First HH exactly at toss 10

A fair coin is tossed until two heads in a row (HH) appear for the first time.

Question: what is the probability that this first HH is completed exactly on toss 10? Write it as a fraction p/q in lowest terms and report $p + q$.

Picture

Remember one thing: was the last toss H? Go toss by toss. An H after an H ends the game (a win only if it is toss 10). The target toss is fixed, so the answer depends on which toss you are at: fill a table from toss 10 backwards.

Rows i = toss about to happen; columns h = previous toss was H.
Click a cell to see its equation.

$i \setminus h$	0	1
1	$\frac{17}{512}$	$\frac{21}{1024}$
2	$\frac{21}{512}$	$\frac{13}{512}$
3	$\frac{13}{256}$	$\frac{1}{32}$
4	$\frac{1}{16}$	$\frac{5}{128}$

5	5/64	3/64
6	3/32	1/16
7	1/8	1/16
8	1/8	1/8
9	1/4	0
10	0	1/2
11	0	0

Previous toss T (or start). $\text{rec}(1,0) = \frac{1}{2} \cdot \text{rec}(2,1) + \frac{1}{2} \cdot \text{rec}(2,0) = \mathbf{17/512}$

Solution

Form 1 (i → end) · **Turn** = one toss. State h = was the previous toss H? HH → ends. There is a fixed deadline (toss 10), so MEMORYLESS keeps i .

1. **LEVEL.** $\text{rec}(i, h) = P(\text{the first HH is completed exactly on toss } 10), \text{ given tosses } 1 \text{ to } i-1 \text{ had no HH, and } h = 1 \text{ if toss } i-1 \text{ was H (} h = 0 \text{ if it was T, or if } i = 1).$ The answer is $\text{rec}(1, 0)$.

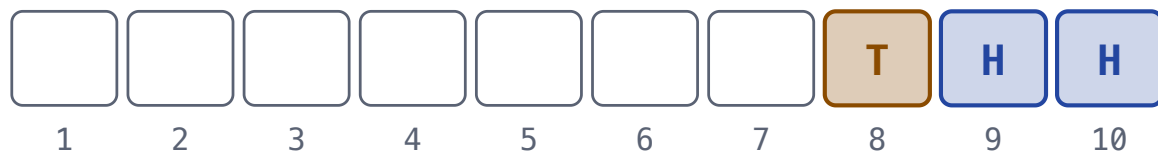
Why: to know whether toss i completes an HH you must remember only whether toss $i-1$ was H.

2. **CHOICE.** Toss i is H or T, each $\frac{1}{2}$.

Why: a fair coin.

3. **CHECK.** H with $h = 1$: HH is completed on toss i , ends, value 1 if $i = 10$ and 0 otherwise. H with $h = 0$: continue to $(i+1, 1)$. T: continue to $(i+1, 0)$. If $i = 11$, toss 10 has passed with no HH: value 0.

Why: the definition of 'first HH exactly on toss 10'.



no HH: $f_7 = 34$ ways

forced

$$P = 34 / 2^{10}$$

4. **MOVE.** $\text{rec}(i, 1) = \frac{1}{2} [i = 10] + \frac{1}{2} \text{rec}(i + 1, 0)$. $\text{rec}(i, 0) = \frac{1}{2} \text{rec}(i + 1, 1) + \frac{1}{2} \text{rec}(i + 1, 0)$.

Why: law of total probability over toss i .

5. **MEMORYLESS (check it).** No: the coin forgets, but the target 'toss 10' is fixed, so from toss $i+1$ there is one toss less to go. $\text{rec}(i+1, h) \neq \text{rec}(i, h)$; keep i . Since i only grows, the recursion stops at the base case $i = 11$: plain recursion, no equation to solve.

Why: a fixed deadline makes the turn number part of the state.

6. **SOLVE.**

```
double dp[12][2]; // all -1 before the first call
double rec(ll i, ll h) {
    if (i == 11) return 0; // base case: toss 10 passed
    if (dp[i][h] != -1) return dp[i][h]; // cache check
    double H = h ? (i == 10) : rec(i + 1, 1); // toss i = H
    double ans = 0.5 * H + 0.5 * rec(i + 1, 0); // toss i = T
    return dp[i][h] = ans; // save and return
} // rec(1, 0) cost: 20 states x 2 transitions
```

Filling the table from $i = 10$ down to $i = 1$ (each row uses only the row above it):

i	$\text{rec}(i, 0)$	$\text{rec}(i, 1)$
10	0	$\frac{1}{2}$
9	$\frac{1}{4}$	0

8	$\frac{1}{8}$	$\frac{1}{8}$
7	$\frac{1}{8}$	1/16
6	3/32	1/16
5	5/64	3/64
4	1/16	5/128
3	13/256	1/32
2	21/512	13/512
1	17/512	21/1024

So $P = \frac{17}{512}$, and $p + q = 529$.

Why: e.g. $\text{rec}(10, 1) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = \frac{1}{2}$; $\text{rec}(10, 0) = \frac{1}{2} \cdot \text{rec}(11, 1) + \frac{1}{2} \cdot \text{rec}(11, 0) = 0$; $\text{rec}(9, 0) = \frac{1}{2} \cdot \text{rec}(10, 1) + \frac{1}{2} \cdot \text{rec}(10, 0) = \frac{1}{4}$. 17 is prime and 512 is a power of 2, so 17/512 is in lowest terms.

Answer. $P = \frac{17}{512}$, $p + q = 529$. ■

3×2 matrix, rows and columns $\equiv 0 \pmod{3}$

Each of the 6 entries of a 3×2 matrix (3 rows, 2 columns) is chosen independently and uniformly from $\{0, 1, 2\}$.

Question: what is the probability that every row sum and every column sum is divisible by 3?

Key idea

The two top-left entries determine everything.

Picture

Count the free choices. Once the first column's top two entries are chosen, every other entry is forced

by the "multiple of 3" rules, and the last column checks itself. 9 free choices out of 729.

Click the two free cells (blue) to change them. Everything else is forced so that every row and column sum is a multiple of 3.

1	2	row 3
2	1	row 3
0	0	row 0
col 3	col 3	

$3 \times 3 = 9$ free choices out of $3^6 = 729$ matrices: [1/81](#).

Solution

Setup. A 3×2 matrix (a_{ij}) with each entry uniform on $\{0, 1, 2\}$, independently. All $3^6 = 729$ matrices are equally likely. We want every row sum and every column sum to be $\equiv 0 \pmod{3}$.

- Each row i needs $a_{i1} + a_{i2} \equiv 0$, i.e. $a_{i2} \equiv -a_{i1}$. So a_{i2} is determined by a_{i1} .

Why: for each value of a_{i1} exactly one value in $\{0,1,2\}$ works.

2. Column 1 needs $a_{11} + a_{21} + a_{31} \equiv 0$, so a_{31} is determined by a_{11} and a_{21} .

Why: same reason.

3. Column 2 is then automatically fine: $a_{12} + a_{22} + a_{32} \equiv -(a_{11} + a_{21} + a_{31}) \equiv 0$.

Why: step 1 gives each $a_{i2} \equiv -a_{i1}$, and step 2 gives the column 1 sum $\equiv 0$.

4. So the valid matrices match the free choices of (a_{11}, a_{21}) : $3 \cdot 3 = 9$ of them.

Why: everything else is forced (steps 1–3), and every choice of the two free entries works.

a_{11}	a_{12}	free: $3 \times 3 = 9$ forced: rows and column 1 sum to 0 column 2: automatic
a_{21}	a_{22}	
a_{31}	a_{32}	

5. $P = \frac{9}{729} = \frac{1}{81} \approx 0.01235$.

Why: favourable over equally likely.

Conclusion. The probability is $\frac{1}{81}$. ■

Partial sums of 1..13 never $\equiv 0 \pmod{3}$

The numbers 1, 2, ..., 13 are put in a random order.

Reading from left to right, write down the running

total after each number (13 running totals in all).

Question: what is the probability that none of the 13 running totals is divisible by 3? Write it as p/q in lowest terms and report $p + q$.

Picture

Only remainders mod 3 matter. Multiples of 3 never change the running total (they just can't go first). The other nine numbers are forced: after a total of 1 you must add a 1-type, after 2 a 2-type – the pattern 1, 1, 2, 1, 2, 1, 2, 1, 2. Count those orders and divide by $13!$.

A random order of 1..13 with its running totals mod 3 (orange = a total divisible by 3, which kills the order). Good orders must follow the pattern 1, 1, 2, 1, 2, ... among the non-multiples of 3.

6	5	7	8	1	9	3	2	10	4	11	12	13
0	2	0	2	0	0	0	2	0	1	0	0	1

Random order

Find a good order

Try 300 000 orders

Some running total is divisible by 3.

Solution

Form 1 ($i \rightarrow \text{end}$) · **Level** = one position of the order. State = how many numbers of each residue are left, and the running total mod 3. Count the good orders, then divide by $13!$.

1. **LEVEL.** Sort 1..13 by residue mod 3:

five are $\equiv 1$ (1, 4, 7, 10, 13), four are $\equiv 2$ (2, 5, 8, 11), four are $\equiv 0$ (3, 6, 9, 12).

$\text{rec}(a, b, z, s)$ = the number of ways to order the numbers still left (a of residue 1, b of residue 2, z of residue 0) so that no running total is $\equiv 0$, given the running total so far is $\equiv s$.

The answer is $\text{rec}(5, 4, 4, 0) / 13!$.

Why: to check the next running total we only need the current total mod 3 and which residues are still available.

2. **CHOICE.** The next number: one of the a numbers $\equiv 1$, one of the b numbers $\equiv 2$, or one of the z numbers $\equiv 0$.

Why: any number still left can go next.

3. **CHECK.** The new running total $s + \text{residue}$ must not be $\equiv 0 \pmod{3}$. All numbers used: base case, 1 way.

Why: the event.

4. **MOVE.** $\text{rec}(a, b, z, s) = a \cdot \text{rec}(a-1, b, z, s+1) + b \cdot \text{rec}(a, b-1, z, s+2) + z \cdot \text{rec}(a, b, z-1, s)$, keeping only the terms that pass CHECK (residues taken mod 3).

Why: sum rule over the next number; the factor a (or b, z) counts which actual number of that residue goes next.

5. **SOLVE (code).**

```
ll dp[6][5][5][3]; // all -1
ll rec(ll a, ll b, ll z, ll s) {
    if (a == 0 && b == 0 && z == 0) return 1; // base case: all placed
    if (dp[a][b][z][s] != -1) return dp[a][b][z][s]; // cache check
    ll ans = 0;
    if (a && (s + 1) % 3) ans += a * rec(a - 1, b, z, (s + 1) % 3);
    if (b && (s + 2) % 3) ans += b * rec(a, b - 1, z, (s + 2) % 3);
    if (z && s % 3) ans += z * rec(a, b, z - 1, s);
}
```

```

    return dp[a][b][z][s] = ans; // save and return
} // rec(5, 4, 4, 0) = 34214400 cost: 6*5*5*3 states x 3 transitions

```

$$\text{rec}(5, 4, 4, 0) = 34\,214\,400, \text{ and } \frac{34\,214\,400}{13!} = \frac{34\,214\,400}{6\,227\,020\,800} = \frac{1}{182}.$$

Why: $6\,227\,020\,800 / 34\,214\,400 = 182.$

6. BY HAND (why the count is $5! \cdot 4! \cdot 9 \cdot 10 \cdot 11 \cdot 12$).

Zeros: a 0-residue never changes the total, so it can go anywhere except first.

Nonzeros: from total $s \in \{1, 2\}$ the next nonzero residue must equal s (the other would make 0). So the 9 of them are forced to be 1, 1, 2, 1, 2, 1, 2, 1, 2 (starting with 2 would need five 2s; there are only four).

Count: the 1s fill their slots in $5!$ ways, the 2s in $4!$ ways, and 3, 6, 9, 12 are inserted one at a time never in front: 9, 10, 11, 12 places.

Why: this is the CHECK read by hand: it leaves only one residue pattern.

residue	1	1	2	1	2	1	2	1	2
running sum mod 3	1	2	1	2	1	2	1	2	1

each next residue must equal the running sum
0s (never first) slot in anywhere: $9 \cdot 10 \cdot 11 \cdot 12$

Answer. $P = \frac{1}{182}, p + q = 183. \blacksquare$

5 · Conditional probability

1. $P(A | B) = \frac{P(A \cap B)}{P(B)}$, for $P(B) > 0$. When outcomes are equally likely, this is $|A \cap B|/|B|$: the sample space shrinks to B .
2. The template for probability = [Form 1](#) of the [DP template](#): LEVEL → CHOICE → CHECK → MOVE → MEMORYLESS → SOLVE. For any process that repeats a turn until something ends it:
 1. **LEVEL**: $\text{rec}(i, s)$ = probability / expected value from turn i onward, given state s before turn i . s = whatever you must remember (a score, a suffix, a position); leave it out when nothing carries over.
 2. **CHOICE**: every outcome of turn i , with its probability.
 3. **CHECK**: for each outcome: does the process end (value known), or continue to turn $i+1$ in state s' ?
 4. **MOVE**: write the honest equation first: ending outcome → $\text{prob} \times \text{value}$; continuing outcome → $\text{prob} \times \text{rec}(i+1, s')$ (plus $\text{prob} \times \text{cost}$, for an expectation).
 5. **MEMORYLESS** (only now): is the future from $(i+1, s)$ exactly the future from (i, s) ?
Yes → drop i and write $\text{rec}(s)$: equations to solve; a rec on both sides (self-loop) is solved by algebra.
No, fixed deadline ("after exactly 6 moves") → keep i ; i only grows, so it is plain recursion

down to the deadline.

No, for another reason → work out how they differ ([HH ends on an even toss](#)).

6. **SOLVE:** line by line. No loops left → a C++ rec.

3. **"A before B" lemma.** Independent identical turns. Each turn: A with prob a (end, value 1), B with prob b (end, value 0), else continue.

MOVE: $\text{rec}(i) = a + (1 - a - b)\text{rec}(i + 1)$.

MEMORYLESS: $\text{rec}(i + 1) = \text{rec}(i) = x$.

SOLVE: $(a + b)x = a$, so

$$P(A \text{ first}) = \frac{a}{a + b}.$$

Trigger: "given...", or "one thing before another".

Red → diamond; even → > 3

(a) One card is drawn from a shuffled 52-card deck.

(b) A fair die is rolled.

Question: (a) find $P(\text{diamond})$ and $P(\text{diamond} \mid \text{the card is red})$, and add them. (b) Given that the die shows an even number, what is the probability that it is greater than 3?

Key idea

Shrink the sample space to the condition.

Picture

"Given" shrinks the sample space. Given red, only the 26 red cards are left, and 13 of them are

diamonds. Given even, only 2, 4, 6 are left, and 4, 6 are above 3.

Conditioning shrinks the sample space. Press the buttons: faded cards are ruled out.



All cards

Given: red

No condition: 13 diamonds of 52: $\frac{1}{4}$.

Solution

Setup. (a) One card from a shuffled deck. (b) One roll of a fair die.

1. (a) $P(\diamond) = \frac{13}{52} = \frac{1}{4}$.

Why: 13 diamonds among 52 equally likely cards.

2. $P(\diamond \mid \text{red}) = \frac{|\diamond \cap \text{red}|}{|\text{red}|} = \frac{13}{26} = \frac{1}{2}$. Their sum is 0.75.

Why: given red, the sample space shrinks to the 26 red cards, equally likely, and all 13 diamonds are red.

3. (b) $P(> 3 \mid \text{even}) = \frac{|\{4, 6\}|}{|\{2, 4, 6\}|} = \frac{2}{3}$.

Why: given even, the sample space is $\{2, 4, 6\}$, equally likely; the ones above 3 are 4 and 6.

Conclusion. 0.75 and $\frac{2}{3}$. ■

First ace at 27, then 2♥?

A standard 52-card deck is shuffled and dealt face up one card at a time. Suppose you are told that the first ace to appear is the 27th card.

Question: given this, what is the probability that the 28th card is the 2 of hearts?

Key idea

Given the condition, the first 26 cards are a uniform choice of non-aces, and the 25 cards after position 27 are in uniform random order.

Picture

Picture the condition. The first 26 cards are non-aces, card 27 is an ace, and the remaining 25 cards are in random order. $2\heartsuit$ is among those 25 with chance $22/48$, and then sits at spot 28 with chance $1/25$.

Deals where the first ace is card 27: 26 non-aces, then an ace, then 25 shuffled cards. $2\heartsuit$ is in the back part with chance $22/48$, and then at spot 28 with chance $1/25$.



Deal one

Deal 200 000

$2\heartsuit$ (blue) is at position 40. Aces orange.

Solution

Setup. A shuffled 52-card deck; all $52!$ orders are equally likely. E = 'the first ace is at position 27'. We want $P(\text{position 28 is } 2\heartsuit \mid E)$.

1. Given E : positions 1–26 hold 26 of the 48 non-aces, position 27 holds an ace, and positions 28–52 hold the remaining 25 cards (3 aces and 22 non-aces).

Why: that is what 'first ace at 27' means.

2. Given E , each non-ace is equally likely to be among the first 26, and the 25 cards after position 27 are in uniformly random order.

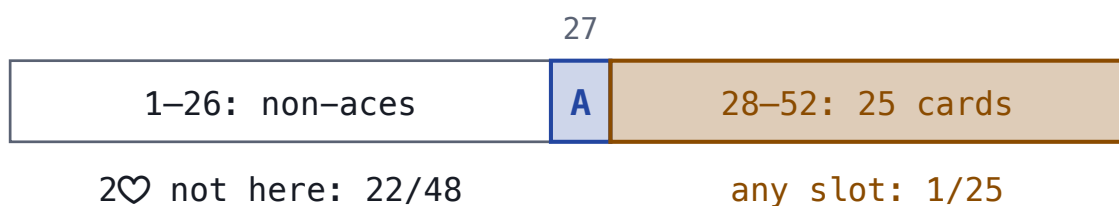
Why: swapping two non-aces everywhere, or rearranging positions 28–52, maps orders in E to orders in E one-to-one, and all orders are equally likely.

3. $P(2\heartsuit \text{ is not in positions 1–26} \mid E) = \frac{22}{48}$.

Why: 26 of the 48 non-aces are in the first 26 positions, each non-ace equally likely to be one of them (step 2); so $2\heartsuit$ is there with probability $\frac{26}{48}$ and not there with $\frac{22}{48}$.

4. If $2\heartsuit$ is among the last 25 cards, it is at position 28 with probability $\frac{1}{25}$.

Why: uniform order of those 25 cards (step 2).



$$P = 22/48 \cdot 1/25 = 11/600$$

5. $P = \frac{22}{48} \cdot \frac{1}{25} = \frac{22}{1200} = \frac{11}{600} \approx 0.018$.

Why: multiplication rule: steps 3 and 4.

Conclusion. The probability is $\frac{11}{600}$. ■

Robot clicks H

The 26 letters A to Z are written in a row. A robot repeatedly chooses one of the letters still present, uniformly at random, and clicks it: the clicked letter and every letter to its right are deleted. This goes on until no letters are left.

Question: what is the probability that the letter H gets clicked at some point?

Picture

Only the letters A to H matter. Clicks to the right of H just shorten the row. The first click that lands on A...H decides everything: H is clicked exactly if that click is H itself, and each of the 8 letters is equally likely.

The robot clicks random letters; each click deletes that letter and everything to its right. H is clicked only if it is the first letter picked among A...H.



Run the robot

Run 100 000

Clicks (numbered): Z → A. H was not clicked.

Solution

Form 1 (i → end) · Turn = one click. State n = letters still present (A up to the n-th letter). Click H → ends (1). Click left of H → ends (0, H is deleted). Click right of H → next turn with fewer letters.

1. **LEVEL.** $\text{rec}(i, n) = P(\text{H gets clicked at some point})$, given that before click i the letters present are the first n letters (A, B, ...) with $n \geq 8$, so H (the 8th letter) is still there. The answer is $\text{rec}(1, 26)$.

Why: a click deletes a letter and everything to its right, so what is left is always A up to some letter.

2. **CHOICE.** Click letter k , for each k in $\{1, \dots, n\}$, with probability $\frac{1}{n}$.

Why: each click is uniform over the letters present.

3. **CHECK.** $k = 8$ (H): ends, value 1. $k \leq 7$: H is to the right of k , so it is deleted: ends, value 0. $k \geq 9$: letters 1 to $k-1$ remain, continue to $(i+1, k-1)$.

Why: the rules.



4. **MOVE.** $\text{rec}(i, n) = \frac{1}{n} \cdot 1 + \sum_{k=9}^n \frac{1}{n} \text{rec}(i+1, k-1) = \frac{1}{n} \left(1 + \sum_{m=8}^{n-1} \text{rec}(i+1, m) \right)$.

Why: law of total probability over the click; put $m = k - 1$.

5. **MEMORYLESS.** $\text{rec}(i+1, n) = \text{rec}(i, n) = \text{rec}(n)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** $\text{rec}(8) = \frac{1}{8}$ (the sum is empty). If $\text{rec}(8)$, ..., $\text{rec}(n-1)$ are all $\frac{1}{8}$, then $\text{rec}(n) = \frac{1}{n} \left(1 + \frac{n-8}{8}\right) = \frac{1}{n} \cdot \frac{n}{8} = \frac{1}{8}$. So $\text{rec}(26) = \frac{1}{8}$.

Why: there are $n - 8$ terms in the sum; n only goes down, so build up from $n = 8$. (Same shape as the Falling frog.)

Answer. H is clicked with probability $\frac{1}{8}$. ■

△ Course's "1 way to click H / 8 ways to remove it" is only valid because every click is uniform over the letters present, which it never states.

7 before 11

Two dice are rolled over and over. If the total is 7 you win; if the total is 11 you lose; any other total means you roll again.

Question: what is the probability that you win?

Picture

Ignore the rolls that change nothing. Only 7 (6 ways) and 11 (2 ways) end the game, so the game ends on a 7 with chance $6 \div (6 + 2) = \frac{3}{4}$.

Run once

Run 50,000

Solution

Form 1 (i → end) · Turn = one roll of two dice. 7 → ends (win). 11 → ends (lose). Anything else → next turn.

1. **LEVEL.** $\text{rec}(i) = P(\text{you win, looking from turn } i \text{ onward}), \text{ given that turns } 1 \text{ to } i-1 \text{ were neither } 7 \text{ nor } 11. \text{ The answer is } \text{rec}(1).$

Why: the game is a sequence of turns; $\text{rec}(i)$ is the probability looking forward from turn i .

2. **CHOICE.** At turn i the total is 7 with probability $\frac{6}{36}$, 11 with probability $\frac{2}{36}$, neither with probability $1 - \frac{6}{36} - \frac{2}{36} = \frac{28}{36}$.

Why: 7 comes from (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 of the 36 equally likely pairs. 11 comes from (5,6), (6,5): 2 pairs. The three cases are disjoint and cover every roll.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

7: 6 cells
11: 2 cells
 all others: reroll, ignore them
 $P = 6 / (6 + 2)$

3. **CHECK.** 7: ends, you win, value 1. 11: ends, you lose, value 0. Neither: continue to turn $i+1$.

Why: the rules.

4. **MOVE.** $\text{rec}(i) = \frac{6}{36} \cdot 1 + \frac{2}{36} \cdot 0 + \frac{28}{36} \text{rec}(i+1).$

Why: law of total probability over the outcome of turn i . A continuing outcome contributes (its probability) $\times \text{rec}(i+1)$.

5. **MEMORYLESS.** At turn $i+1$ the next roll is again 7 with $\frac{6}{36}$, 11 with $\frac{2}{36}$, neither with $\frac{28}{36}$, under the

same rules. So $\text{rec}(i+1) = \text{rec}(i)$.

Why: each turn is independent of earlier turns, has the same probabilities and the same rules, and nothing from earlier turns is carried forward.

6. **SOLVE.** Let $x = \text{rec}(i)$. $x = \frac{6}{36} + \frac{28}{36}x$. So $x - \frac{28}{36}x = \frac{6}{36}$, which is $\frac{8}{36}x = \frac{6}{36}$, so $x = \frac{6}{8} = \frac{3}{4}$.

Why: substitute MEMORYLESS into MOVE; move the x terms left; divide by $\frac{8}{36}$.

Answer. You win with probability $\frac{3}{4}$. (The 'A before B' lemma: $\frac{a}{a+b}$ with $a = \frac{6}{36}$, $b = \frac{2}{36}$.) ■

6 · Multiplication rule & independence

1. $P(A \cap B) = P(A)P(B | A)$. Chained:

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2 | A_1) \dots P(A_n | A_1 \cap \dots \cap A_{n-1})$$

2. **Independent** A, B : $P(A \cap B) = P(A)P(B)$, equivalently $P(A | B) = P(A)$ when $P(B) > 0$.

3. **Mutually independent** $A_1, \dots, A_n$: the product rule holds for every sub-collection, not just all n together.

4. If A, B are independent, so are A^c, B^c : $P(A^c \cap B^c) = 1 - P(A) - P(B) + P(A)P(B) = (1 - P(A))(1 - P(B))$.

△ Course defines independence of n events by the single equation $P(\bigcap A_i) = \prod P(A_i)$. That is not enough. Counterexample: $A = B$, $P(A) =$

$\frac{1}{2}$, $C = \emptyset$; the full product holds ($0 = 0$), but A and B are not independent.

Draws without replacement

Balls or cards are drawn one after another *without replacement* (a drawn item is not put back).

Question:

- (a) A bag has 5 green and 3 red balls, and 2 are drawn. What is the probability that the first is red and the second is green?
- (b) Two cards are drawn from a 52-card deck. What is the probability that the first is an ace and the second is a queen?
- (c) Three cards are drawn. What is the probability of an ace, then a queen, then a king?

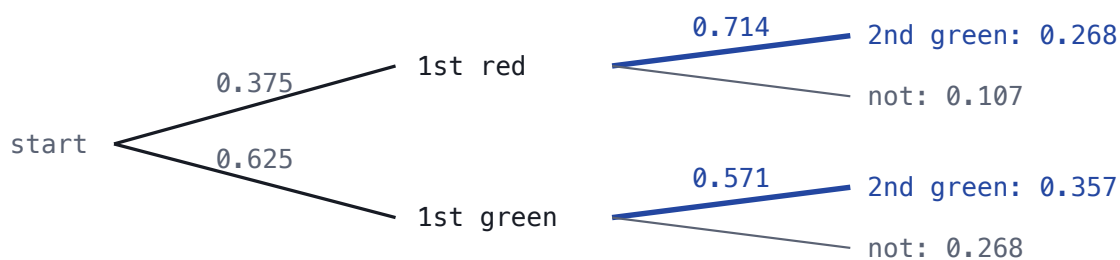
Key idea

Chain rule: each factor is computed on the remaining deck.

Picture

Multiply along the path, updating the deck each time. First red: $3/8$. Then 7 balls are left and 5 are green: $5/7$.

- (a) The two draws as a tree. The top branch (red, then green) has probability $3/8 \times 5/7$.



$$P(\text{2nd green}) = 0.375 \cdot 0.714 + 0.625 \cdot 0.571 = \mathbf{0.625}.$$

$$P(\text{1st red} \mid \text{2nd green}) = 0.268 \div 0.625 = \mathbf{0.429}$$

$$P(\text{1st green} \mid \text{2nd green}) = 0.357 \div 0.625 = \mathbf{0.571}$$

Solution

- (a) 5 green, 3 red. $P(\text{1st red}) = \frac{3}{8}$. Given that, 7 balls remain and 5 are green: $P(\text{2nd green} \mid \text{1st red}) = \frac{5}{7}$. Multiply: $\frac{15}{56} \approx 0.26786$.

Why: chain rule $P(A \cap B) = P(A)P(B \mid A)$; each factor is computed on what is left.

- (b) $P(\text{ace}) = \frac{4}{52}$, then $P(\text{queen} \mid \text{ace first}) = \frac{4}{51}$.
Product: $\frac{16}{2652} = \frac{4}{663} \approx 0.00603$.

Why: after one card, 51 remain and all 4 queens are still there.

- (c) Then $P(\text{king} \mid \text{ace, queen}) = \frac{4}{50}$. Product: $\frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} = \frac{64}{132600} \approx 0.00048$.

Why: 50 cards remain, all 4 kings among them.

Conclusion. $\frac{15}{56}$, $\frac{4}{663}$ and about 0.00048. ■

Ant on an icosahedron

An icosahedron is a solid with 12 corners (vertices); each vertex is joined by edges to 5 others. An ant starts at vertex A. At each move it

walks along one of the 5 edges at its current vertex, chosen at random. B is the vertex directly opposite A.

Question: what is the probability that the ant is at B after exactly 3 moves?

Picture

Only the distance from A matters, not which vertex.

Group the 12 vertices into 4 levels (A, its 5 neighbours, B's 5 neighbours, B) and follow the level for exactly 3 moves.

To reach B in 3 moves the ant must go down a level every time: $1 \times 2/5 \times 1/5$.

Levels by distance from A. Press Step 3 times and read the bar at B: $2/25 = 0.08$.



$P(X_n = j)$ for $n = 0$

1

0

0

0

Step

Run until stuck

Reset

click a state to start there

Step 0: at L0.

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one move. State L = level (distance from A). Fixed deadline (3 moves), so MEMORYLESS keeps i .

1. **LEVEL.** Group the vertices by distance from A:

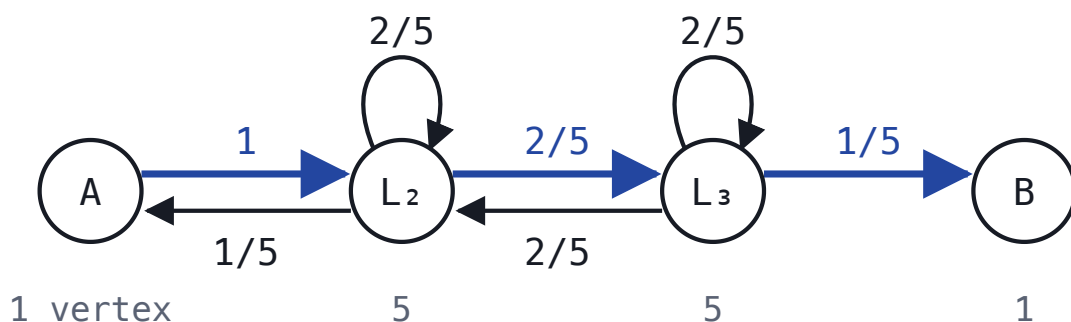
level 0 = A; level 1 = the 5 neighbours of A;
level 2 = the 5 neighbours of B; level 3 = B.
 $\text{rec}(i, L) = P(\text{ant is at B after move } 3), \text{ given it is at level } L \text{ after } i \text{ moves. The answer is } \text{rec}(0, 0).$

Why: $12 = 1 + 5 + 5 + 1$ vertices; only the level matters for where the ant can go next.

2. **CHOICE.** From level 0: all 5 edges go to level 1.

From level 1: 1 edge to level 0, 2 inside level 1, 2 to level 2. From level 2: 2 to level 1, 2 inside level 2, 1 to level 3. Each edge has probability $\frac{1}{5}$.

Why: the neighbours of A form a ring of 5, and so do the neighbours of B; the edge counts add up to 5 at every vertex.



3. **CHECK.** $i = 3$: the walk is over, value 1 if $L = 3$ and 0 otherwise. $i < 3$: continue to $(i+1, L')$.

Why: the question asks about the position after exactly 3 moves.

4. **MOVE.** $\text{rec}(i, L) = \sum_{L'} P(L \rightarrow L') \text{rec}(i + 1, L').$

Why: law of total probability over the move.

5. **MEMORYLESS (check it).** No: the deadline 'after 3 moves' is fixed, so keep i . i only grows, so this is plain recursion down to $i = 3$.

Why: a fixed deadline makes the turn number part of the state.

6. **SOLVE.** $\text{rec}(3, L) = [L = 3]$. $\text{rec}(2, L) = P(L \rightarrow 3): \frac{1}{5}$ for $L = 2$, 0 otherwise. $\text{rec}(1, 1) = P(1 \rightarrow 2) \cdot \frac{1}{5} = \frac{2}{5} \cdot \frac{1}{5} = \frac{2}{25}$. $\text{rec}(0, 0) = P(0 \rightarrow 1) \cdot \text{rec}(1, 1) = 1 \cdot \frac{2}{25} = \frac{2}{25}$.

Why: MOVE, read backwards from the deadline; from level 0 the ant surely reaches level 1.

Answer. The ant is at B after 3 moves with probability $\frac{2}{25}$. ■

Shooting stars

(a) The probability of seeing a shooting star in any given hour is 0.6, and different hours are independent. (b) The probability of seeing at least one shooting star in an hour is 0.8. Think of the hour as four 15-minute blocks that are independent and alike.

Question: (a) what is the probability of seeing at least one shooting star in 2 hours? (b) What is the probability of seeing a star in the first 15 minutes?

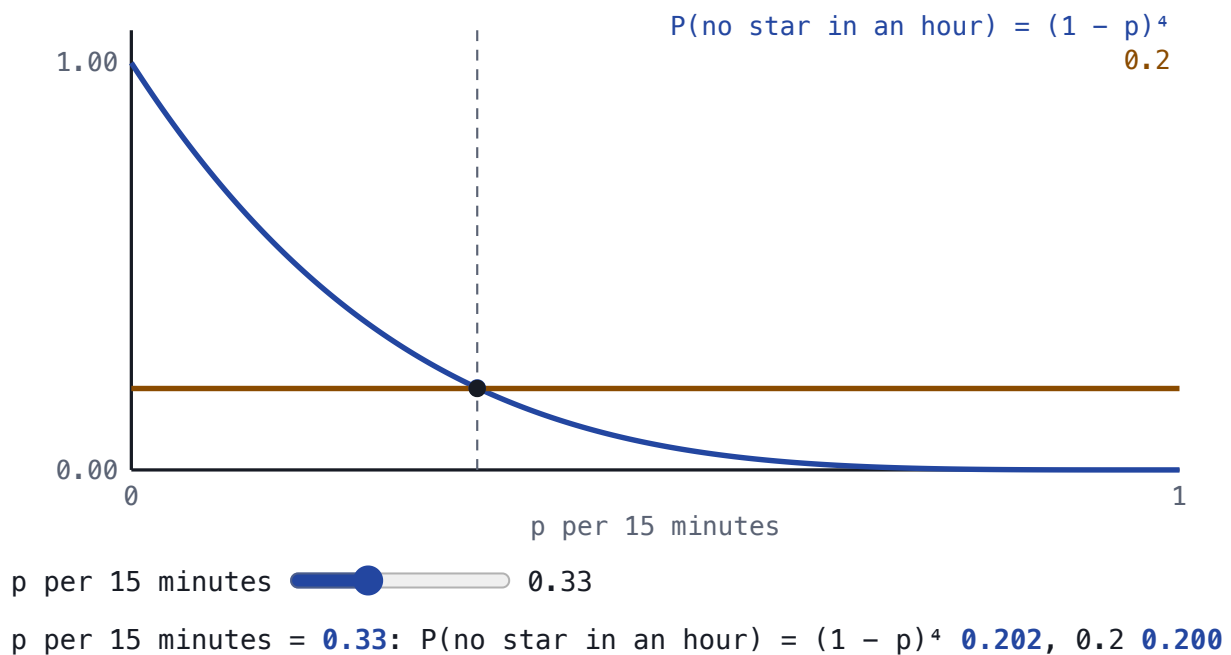
Key idea

Work with "none": the complements multiply.

Picture

Work with "no star at all": independent pieces multiply. Two hours with no star: 0.4×0.4 . An hour with no star is four quarter-hours with no star: $(1 - p)^4 = 0.2$.

Slide p until the curve meets 0.2.

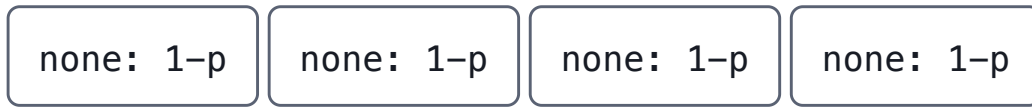


Solution

- (a) $P(\text{no star in one hour}) = 1 - 0.6 = 0.4$, so
 $P(\text{no star in 2 hours}) = 0.4^2 = 0.16$.
Why: complement, then the two hours are independent, so the 'none' probabilities multiply.
- $P(\text{at least one in 2 hours}) = 1 - 0.16 = 0.84$.
Why: complement.
- (b) Let $p = P(\text{star in one 15-min block})$. The hour is 4 independent identical blocks, so
 $P(\text{no star in the hour}) = (1 - p)^4$.

Why: no star in the hour means none in each of the 4 blocks.

one hour = 4 independent 15-min blocks



$$P(\text{no star in the hour}) = (1-p)^4 = 0.2$$

4. $(1-p)^4 = 1 - 0.8 = 0.2$, so $1-p = 0.2^{1/4} \approx 0.66874$ and $p \approx 0.33126$.

Why: take the fourth root: $\sqrt[4]{0.2} \approx 0.44721$ and $\sqrt{0.44721} \approx 0.66874$.

Conclusion. 0.84 and about 0.331. ■

7 · Law of total probability

If $A_1, \dots, A_n$ partition Ω , then $B = \bigcup (B \cap A_i)$, a disjoint union. So

$$P(B) = \sum_i P(B \mid A_i) P(A_i).$$

In practice: list the disjoint ways B can happen and add them up.

Defects and coins

(a) Machine A makes 60% of a factory's items and machine B makes the other 40%. 2% of A's items are defective and 5% of B's items are defective. (b)

There are three coins whose probabilities of heads

are 0.5, 0.75 and 1. One coin is picked at random and flipped.

Question: (a) what is the probability that a randomly chosen item is defective? (b) What is the probability of heads?

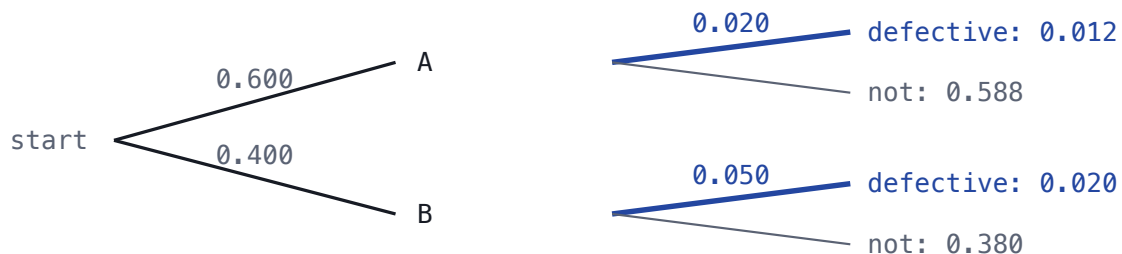
Key idea

Split on which machine or coin.

Picture

Split by where the item came from, then add the branches. Defective from A: 0.6×0.02 ; defective from B: 0.4×0.05 .

(a) Tree: machine, then defective or not.



$$P(\text{defective}) = 0.600 \cdot 0.020 + 0.400 \cdot 0.050 = 0.032.$$

$$P(A \mid \text{defective}) = 0.012 \div 0.032 = 0.375$$

$$P(B \mid \text{defective}) = 0.020 \div 0.032 = 0.625$$

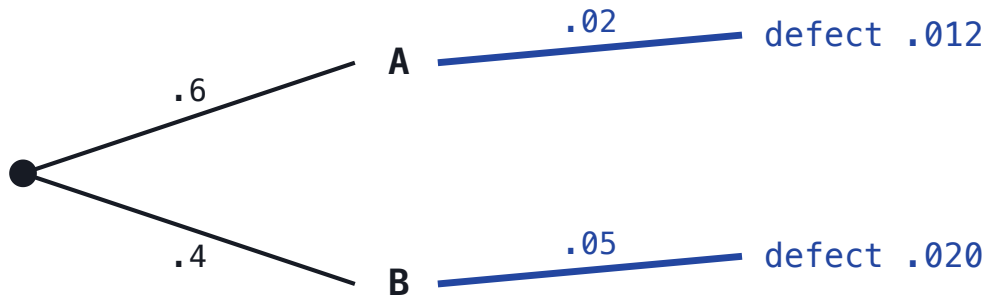
Solution

- (a) Every item comes from machine A (0.6) or B (0.4), and never both. So $P(D) = P(D \mid A)P(A) + P(D \mid B)P(B)$.

Why: law of total probability over the partition $\{A, B\}$.

$$2. = 0.02 \cdot 0.6 + 0.05 \cdot 0.4 = 0.012 + 0.020 = 0.032.$$

Why: substitute.



$$P(\text{defect}) = .012 + .020 = .032$$

3. (b) Each coin is picked with probability $\frac{1}{3}$:

$$P(H) = \frac{1}{3}(0.5) + \frac{1}{3}(0.75) + \frac{1}{3}(1) = \frac{2.25}{3} = 0.75.$$

Why: total probability over which coin was picked.

Conclusion. 0.032 and 0.75. ■

8 · Bayes

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum_j P(B | A_j) P(A_j)}.$$

This is just $P(A_i \cap B)/P(B)$, with the numerator from the chain rule and the denominator from total probability.

Three bags

Bag 1 has 3 red and 4 black balls, bag 2 has 2 red and 11 black, and bag 3 has 8 red and 1 black. A bag

is chosen at random (each with probability $\frac{1}{3}$), and one ball is drawn from it. The ball is red.

Question: what is the probability that it came from bag 3?

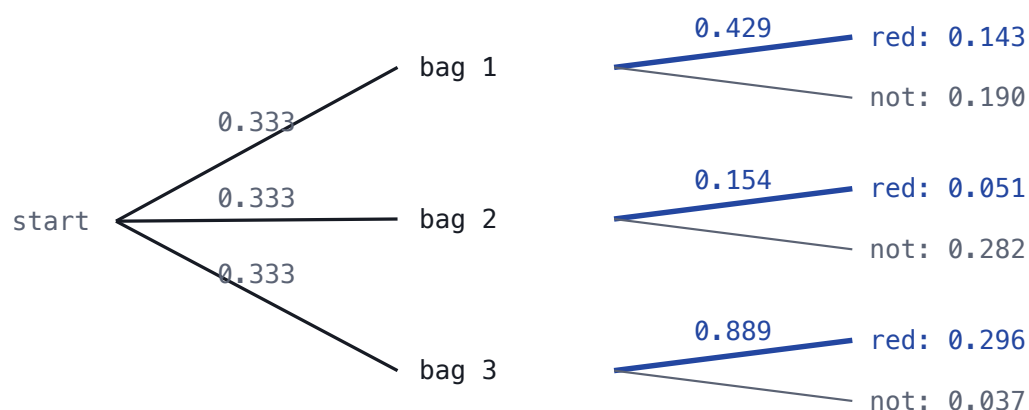
Key idea

The equal priors cancel: the answer is the ratio of the likelihoods.

Picture

Among all the ways to draw red, what share goes through bag 3? The bags are equally likely, so compare the chances of red: $8/9$ against $3/7 + 2/13 + 8/9$.

Tree: bag, then colour. Given red, keep only the red leaves.



$$P(\text{red}) = 0.333 \cdot 0.429 + 0.333 \cdot 0.154 + 0.333 \cdot 0.889 = \mathbf{0.490}.$$

$$P(\text{bag 1} \mid \text{red}) = 0.143 \div 0.490 = \mathbf{0.291}$$

$$P(\text{bag 2} \mid \text{red}) = 0.051 \div 0.490 = \mathbf{0.105}$$

$$P(\text{bag 3} \mid \text{red}) = 0.296 \div 0.490 = \mathbf{0.604}$$

Solution

Setup. A bag is picked uniformly ($\frac{1}{3}$ each), then a ball from it. $P(\text{red} \mid \text{bag 1}) = \frac{3}{7}$, $P(\text{red} \mid \text{bag 2}) = \frac{2}{13}$, $P(\text{red} \mid \text{bag 3}) = \frac{8}{9}$.

$$1. \text{ Bayes: } P(\text{bag 3} \mid \text{red}) = \frac{P(\text{red} \mid 3) P(3)}{\sum_k P(\text{red} \mid k) P(k)}.$$

Why: numerator from the chain rule, denominator from total probability.

$$2. \text{ The priors are all } \frac{1}{3} \text{ and cancel: } = \frac{8/9}{3/7 + 2/13 + 8/9}.$$

Why: factor $\frac{1}{3}$ out of the numerator and every denominator term.

$$3. \text{ Over the common denominator 819: } \frac{3}{7} = \frac{351}{819}, \frac{2}{13} = \frac{126}{819}, \frac{8}{9} = \frac{728}{819}. \text{ So } P = \frac{728}{351+126+728} = \frac{728}{1205} \approx 0.604.$$

Why: $819 = 7 \cdot 9 \cdot 13$.

$P(\text{red} \mid \text{bag}), \text{ equal priors}$

bag 1 3/7

bag 2 2/13

bag 3 8/9

$$P(\text{bag 3} \mid \text{red}) = \text{bag 3's bar} / \text{all bars} \approx 0.604$$

Conclusion. Given red, bag 3 has probability about 0.604. ■

9 fair + 1 double-headed coin

A box holds 10 coins: 9 are fair and 1 has heads on both sides. A coin is picked at random and flipped, and it shows heads.

Question: what is the probability that it is the two-headed coin?

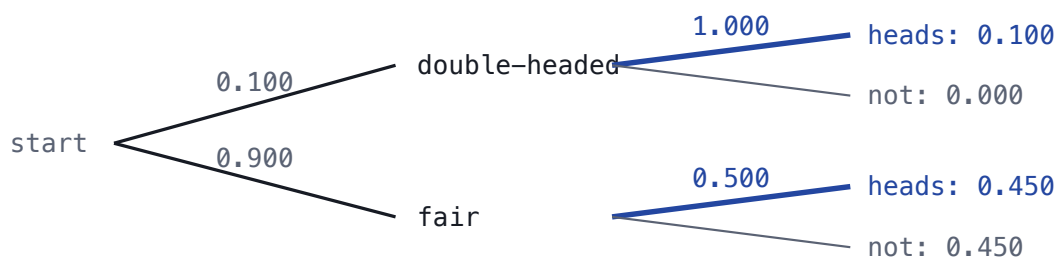
Key idea

Prior times likelihood, then normalise.

Picture

Among all the ways to see heads, what share uses the trick coin? Trick coin and heads: $1/10$. Fair coin and heads: $9/10 \times 1/2$. So $1/10 \div (1/10 + 9/20)$.

Move the sliders to see how the answer depends on the numbers.



P(double-headed) 0.1

P(heads | double-headed) 1

P(heads | fair) 0.5

$P(\text{heads}) = 0.100 \cdot 1.000 + 0.900 \cdot 0.500 = 0.550$.

$P(\text{double-headed} | \text{heads}) = 0.100 \div 0.550 = 0.182$

$P(\text{fair} | \text{heads}) = 0.450 \div 0.550 = 0.818$

Solution

Setup. 10 coins: 9 fair and 1 double-headed (DH). One is picked uniformly and flipped; it shows H.

$$1. \quad P(\text{DH and H}) = \frac{1}{10} \cdot 1 = \frac{1}{10}.$$

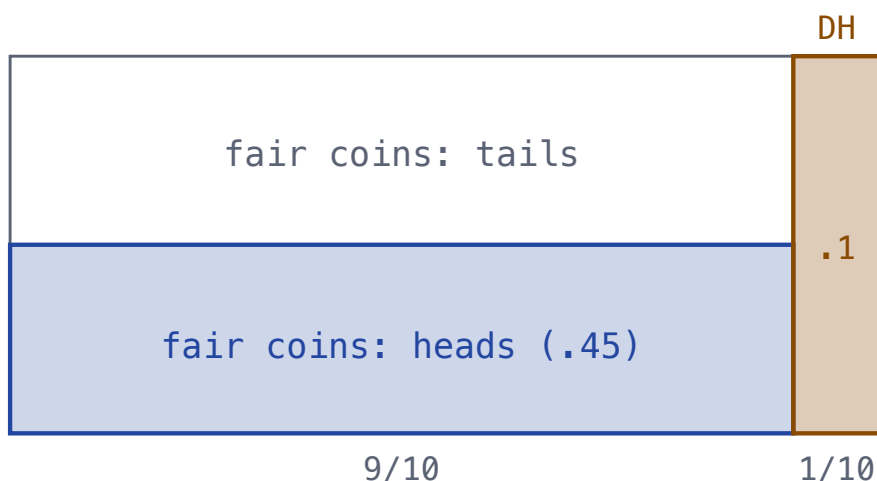
Why: pick DH with probability $\frac{1}{10}$; it always shows H.

$$2. \quad P(\text{fair and H}) = \frac{9}{10} \cdot \frac{1}{2} = \frac{9}{20}.$$

Why: pick a fair coin with probability $\frac{9}{10}$, then H with $\frac{1}{2}$.

$$3. P(H) = \frac{1}{10} + \frac{9}{20} = \frac{2}{20} + \frac{9}{20} = \frac{11}{20}.$$

Why: the two cases are disjoint and cover every way to get H.



$$\text{given H: DH share} = .1 / (.45 + .1) = 2/11$$

$$4. P(\text{DH} | H) = \frac{1/10}{11/20} = \frac{2}{11} \approx 0.182.$$

Why: definition of conditional probability.

Conclusion. The coin is double-headed with probability $\frac{2}{11}$. ■

Disease test

1% of people have a certain disease. A test for it is positive for 90% of people who have the disease, but it is also (wrongly) positive for 5% of people who don't. A randomly chosen person tests positive.

Question: what is the probability that this person actually has the disease?

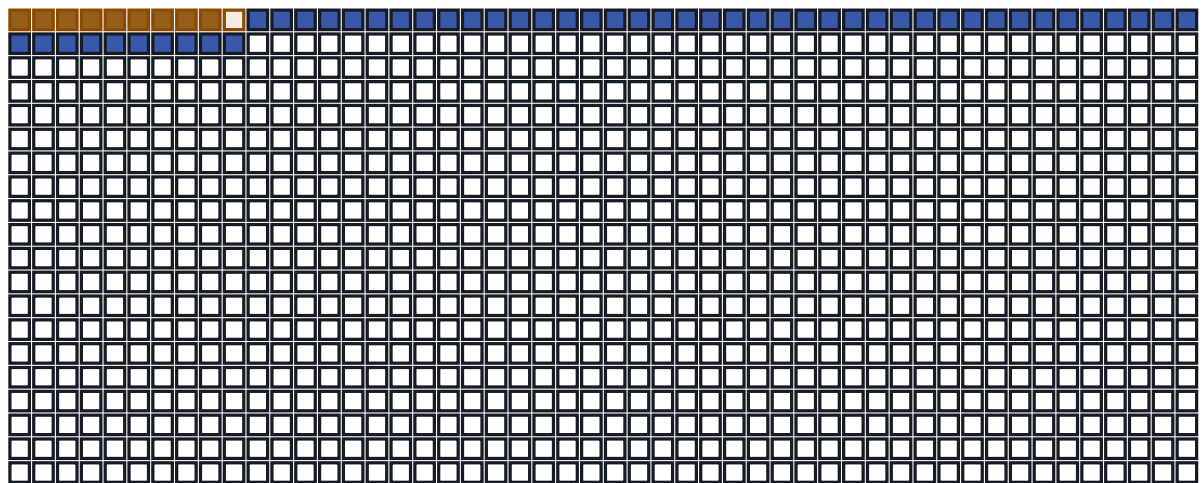
Key idea

The rare disease makes false positives outnumber true positives.

Picture

Picture 1000 people. 10 are sick and 9 of them test positive. 990 are healthy and about 50 of them test positive anyway. So of ~59 positives only 9 are sick: about 15%.

1000 people. Orange = has the disease; filled = tests positive.
Among all the filled squares, what share is orange?



$P(\text{disease})$ 0.01 $P(+ \mid \text{disease})$ 0.9

$P(+ \mid \text{healthy})$ 0.05

Positives: 9 sick + 50 healthy = 59. $P(\text{sick} \mid \text{positive}) \approx 9 \div 59 = \mathbf{0.153}$ (exact 0.154).

Solution

Setup. D = has the disease, T = tests positive.

$P(D) = 0.01$, $P(T \mid D) = 0.9$, $P(T \mid \text{not } D) = 0.05$.

1. $P(D \cap T) = 0.01 \cdot 0.9 = 0.009$.

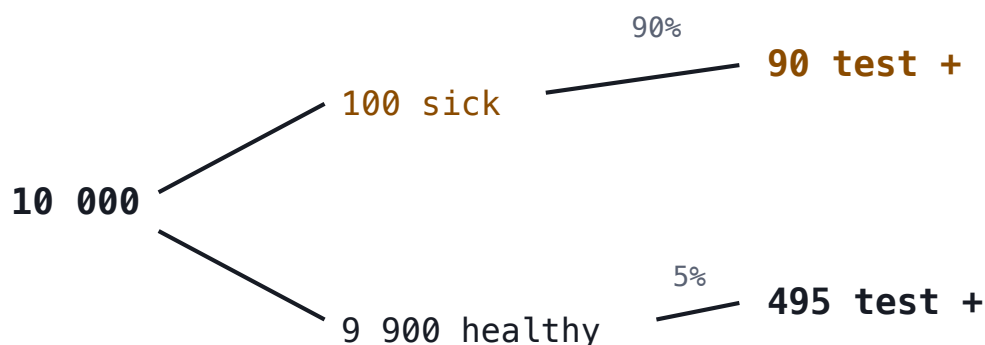
Why: chain rule.

2. $P(\text{not } D \cap T) = 0.99 \cdot 0.05 = 0.0495$.

Why: chain rule; $P(\text{not } D) = 0.99$.

3. $P(T) = 0.009 + 0.0495 = 0.0585$.

Why: a positive test comes from a sick or a healthy person, never both.



sick among positives: $90 / (90 + 495) \approx 0.154$

4. $P(D | T) = \frac{0.009}{0.0585} \approx 0.154$.

Why: definition of conditional probability. In counts: of 10 000 people, 90 sick people test positive but so do 495 healthy people.

Conclusion. A positive test means disease with probability only about 0.154. ■

9 · Paradoxes

1. Start from the **full, equally likely** sample space. Delete what the information rules out. Only then count.
2. Paths or stories are **not** equally likely by default: multiply the probabilities along each one.

3. If someone *chooses* what to reveal, model how they choose. That rule is part of the information.

Three coins (HH, HT, TT)

A bag holds three coins: one with heads on both sides (HH), one with heads and tails (HT), and one with tails on both sides (TT). You pick a coin at random and look at one random side of it. That side is heads.

Question: what is the probability that you picked the HH coin (so the other side is also heads)?

Key idea

Count faces, not coins: there are 3 equally likely heads-faces, and 2 of them belong to HH.

Picture

Count faces, not coins. There are 3 heads-faces you could be looking at, all equally likely. Two of them are on the HH coin.

(The tempting $\frac{1}{2}$ forgets that the HH coin shows heads twice as often as the HT coin.)

Pick a coin and a face at random. Keep only the runs where the face is H. How often was it the HH coin?

Run once

Run 100 000

Solution

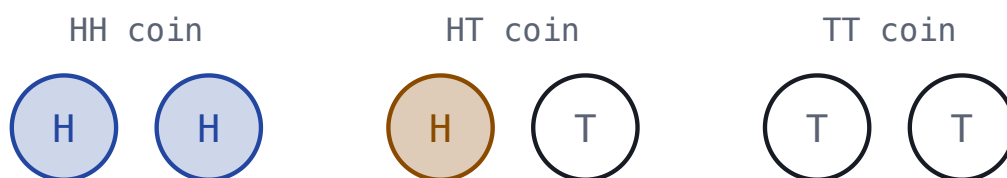
Setup. Three coins: HH (heads on both faces), HT, TT. A coin is picked uniformly, then one of its faces uniformly, and that face shows H.

1. Name the faces: HH coin A_1, A_2 (both H); HT coin B_1 (H), B_2 (T); TT coin C_1, C_2 (T). Each of the 6 (coin, face) outcomes has probability $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

Why: the coin is uniform, then the face is uniform.

2. 'The face shows H' is the event $\{A_1, A_2, B_1\}$: 3 equally likely outcomes.

Why: these are the heads faces.



3 heads faces; 2 are on the HH coin $\Rightarrow 2/3$

3. Two of them, A_1 and A_2 , are on the HH coin, so $P = \frac{2}{3}$.

Why: conditioning on an event made of equally likely outcomes: count within it.

4. The tempting answer $\frac{1}{2}$ is wrong because it treats the HH and HT coins as equally likely *after* seeing H. But the HH coin shows H every time and the HT coin only half the time.

Why: the observation carries information about the coin.

Conclusion. Given a head, the coin is double-headed with probability $\frac{2}{3}$. ■

Children and coins

Each child is a boy or a girl with probability $\frac{1}{2}$, independently of the others.

Question:

(a) A family has 2 children, and you learn that at least one is a girl. What is the probability that both are girls?

(b) Instead you learn that the *older* child is a girl. Now what is the probability that both are girls?

(c) 3 fair coins are tossed and at least one shows heads. What is the probability that all three show heads?

Key idea

Remove the ruled-out outcomes from the equally likely list.

Picture

List the equal-chance families, cross out what the information rules out, then count. "At least one girl" rules out only BB; "the older is a girl" rules out BB and BG.

The four equally likely families (older, younger). Each condition crosses some out.



At least one girl

Older is a girl

Left: BG, GB, GG $\rightarrow P(GG) = \frac{1}{3}$.**Solution**

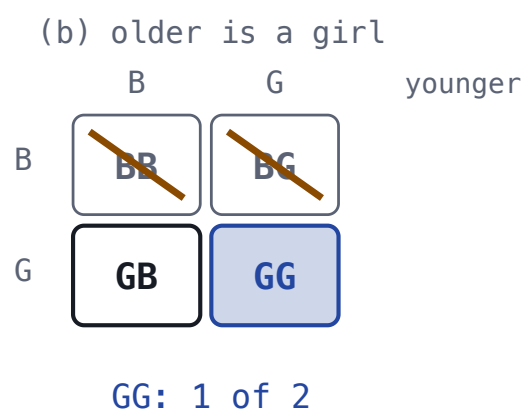
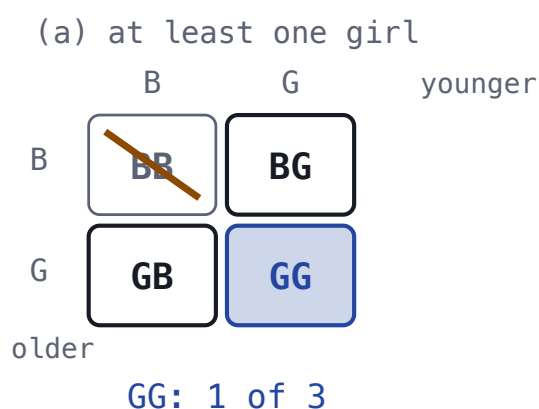
Setup. Two children, each independently a boy or girl with probability $\frac{1}{2}$. Writing (older, younger), the outcomes BB, BG, GB, GG are equally likely.

1. (a) 'At least one girl' leaves $\{BG, GB, GG\}$, so $P(GG) = \frac{1}{3}$.

Why: remove BB and count within the 3 remaining equally likely outcomes.

2. (b) 'The older is a girl' leaves $\{GB, GG\}$, so $P(GG) = \frac{1}{2}$.

Why: the older child is listed first.



3. (c) Three coins have 8 equally likely outcomes. 'At least one H' removes only TTT, leaving 7, of which 1 is HHH: $\frac{1}{7} \approx 0.143$.

Why: same method.

Conclusion. $\frac{1}{3}$, $\frac{1}{2}$ and $\frac{1}{7}$. ■

△ Course leaves this implicit: (a) assumes we learn exactly the event " ≥ 1 girl". If instead a random child is revealed to be a girl, the answer is $\frac{1}{2}$.

Falling frog

A frog sits on step 10 of a staircase with steps numbered 1 to 10. From step i (with $i \geq 2$) it jumps down to a step chosen uniformly at random from $1, 2, \dots, i-1$. It stops when it reaches step 1.

Question: what is the probability that the frog lands on step 5 at some point?

Picture

Only the first landing below 6 matters. Jumps that land on 6 or higher just start again from a lower step. The first time the frog lands on one of the steps 1–5, each of those 5 steps is equally likely. So $1/5$.

Run once

Run 100,000

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one jump. State j = current step. Landing on 5 $\rightarrow$ ends (1). Landing below 5 $\rightarrow$ ends (0). Landing on 6 or higher $\rightarrow$ next turn.

- LEVEL.** $\text{rec}(i, j) = P(\text{the frog lands on 5 at some point from now on}), \text{ given that after } i \text{ jumps it is on step } j \geq 6$. The answer is $\text{rec}(0, 10)$.

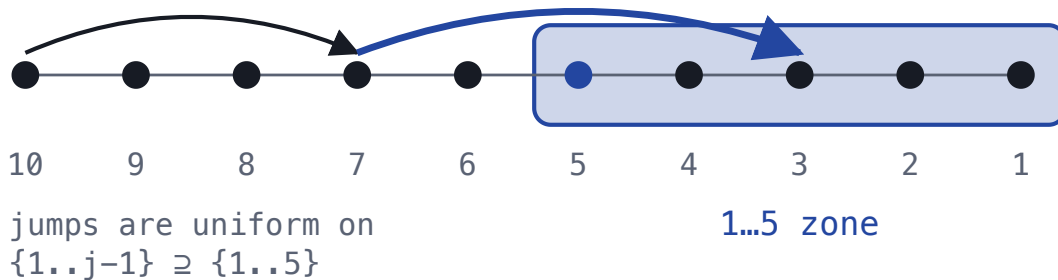
Why: the frog starts on 10 after 0 jumps.

2. **CHOICE.** From step j it lands on k , for each k in $\{1, \dots, j-1\}$, with probability $\frac{1}{j-1}$.

Why: the rules.

3. **CHECK.** $k = 5$: ends, value 1. $k \leq 4$: value 0. $k \geq 6$: continue to $(i+1, k)$.

Why: positions only go down, so from $k \leq 4$ the frog can never reach 5.



first landing in the zone: uniform $\Rightarrow P(5) = 1/5$

4. **MOVE.** $\text{rec}(i, j) = \frac{1}{j-1} \cdot 1 + \sum_{k=6}^{j-1} \frac{1}{j-1} \text{rec}(i+1, k)$.

Why: law of total probability over the jump.

5. **MEMORYLESS.** The jump rule depends only on the current step j , not on how many jumps came before, so $\text{rec}(i+1, j) = \text{rec}(i, j) = \text{rec}(j)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** $\text{rec}(6) = \frac{1}{5}$ (the sum is empty). $\text{rec}(7) = \frac{1}{6} \left(1 + \frac{1}{5}\right) = \frac{1}{6} \cdot \frac{6}{5} = \frac{1}{5}$. $\text{rec}(8) = \frac{1}{7} \left(1 + \frac{1}{5} + \frac{1}{5}\right) = \frac{1}{7} \cdot \frac{7}{5} = \frac{1}{5}$. In general, if $\text{rec}(6), \dots, \text{rec}(j-1)$ are all $\frac{1}{5}$, then $\text{rec}(j) = \frac{1}{j-1} \left(1 + \frac{j-6}{5}\right) = \frac{1}{j-1} \cdot \frac{j-1}{5} = \frac{1}{5}$. So $\text{rec}(10) = \frac{1}{5}$.

```
double dp[11]; // all -1
double rec(ll j) { // j >= 6
    if (dp[j] != -1) return dp[j]; // cache check
```

```
double ans = 1; // k = 5: lands on 5
for (ll k = 6; k < j; k++) ans += rec(k); // k >= 6: keeps falling
return dp[j] = ans / (j - 1); // save and return
} // rec(10) (j = 6: the loop is empty, that is the base case)
```

Why: steps only go down, so there are no loops: build up from $j = 6$. There are $j - 6$ terms in the sum.

Answer. The frog lands on 5 with probability $\frac{1}{5}$. ■

△ Trap: “subsets of $\{2..9\} \rightarrow \frac{1}{2}$ ” treats unequal paths as equally likely. For example, $10 \rightarrow 1$ has probability $\frac{1}{9}$, but $10 \rightarrow 9 \rightarrow 1$ has $\frac{1}{72}$.

Three prisoners

Prisoners A, B and C know that two of them, chosen at random, will be released.

A asks the warden to name one of the *other two* who will be released. So the warden names B or C; if both B and C are being released, he picks one of them at random.

He says “B”.

Question: given this, what is the probability that A will be released?

Key idea

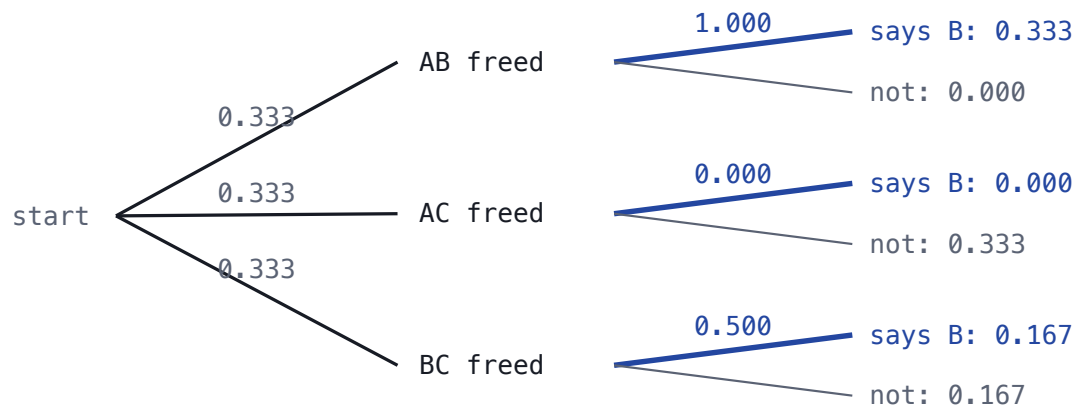
AB forces “B”, but BC says “B” only half the time.

Picture

The warden's rule matters. If AB go free he must say “B”. If BC go free he says “B” only half the time.

So "B" is twice as likely under AB as under BC, and A (free in AB) keeps probability 2/3.

Tree: who is freed, then what the warden says. A is freed in the AB branch (and AC, which can't say B).



$$P(\text{says B}) = 0.333 \cdot 1.000 + 0.333 \cdot 0.000 + 0.333 \cdot 0.500 = 0.500.$$

$$P(\text{AB freed} \mid \text{says B}) = 0.333 \div 0.500 = 0.667$$

$$P(\text{AC freed} \mid \text{says B}) = 0.000 \div 0.500 = 0.000$$

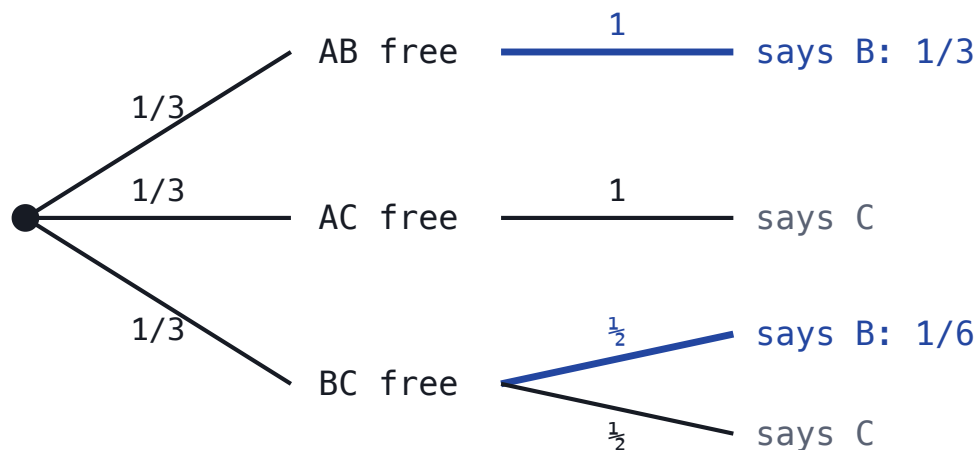
$$P(\text{BC freed} \mid \text{says B}) = 0.167 \div 0.500 = 0.333$$

Solution

Setup. Two of A, B, C will be released: AB, AC or BC, each with probability $\frac{1}{3}$. The warden names a released prisoner other than A: with AB he says B, with AC he says C, with BC he says B or C with probability $\frac{1}{2}$ each. He says "B".

1. $P(AB \text{ and "B"}) = \frac{1}{3} \cdot 1 = \frac{1}{3}$; $P(AC \text{ and "B"}) = 0$;
 $P(BC \text{ and "B"}) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

Why: chain rule with the warden's rule.



$$P(A \text{ free} \mid \text{says B}) = (1/3) / (1/3 + 1/6) = 2/3$$

$$2. \quad P(\text{"B"}) = \frac{1}{3} + 0 + \frac{1}{6} = \frac{1}{2}.$$

Why: the three release pairs are disjoint and cover everything.

3. A is released in the cases AB and AC, so

$$P(A \text{ released} \mid \text{"B"}) = \frac{1/3 + 0}{1/2} = \frac{2}{3}.$$

Why: definition of conditional probability.

Conclusion. Hearing "B" leaves A's chance at $\frac{2}{3}$: it doesn't drop to $\frac{1}{2}$, because BC produces "B" only half the time. ■

10 · Reduced sample space

In repeated independent trials, drop the outcomes that change nothing. The subsequence of relevant trials is again independent and identically distributed, with the relevant probabilities renormalised ($p_i / \sum_{\text{relevant}} p$). This is the "A before B" lemma from §5 in general form.

Tennis from 30–30

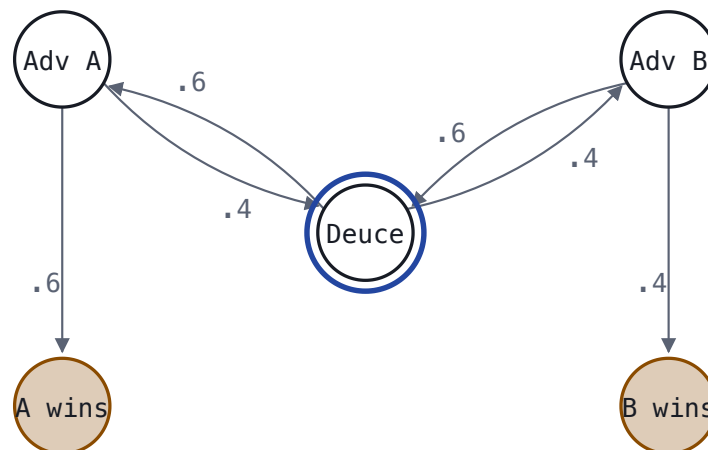
In a tennis game at 30–30 (or at deuce), a player must get two points ahead of the other to win the game. Alice wins each point with probability 0.6, independently of all other points.

Question: starting from 30–30, what is the probability that Alice wins the game?

Picture

Look at points in pairs from a level score. Alice wins both (0.36): she wins. Bob wins both (0.16): he wins. Split (0.48): level again, nothing changed. So only AA vs BB matters: $0.36 \div (0.36 + 0.16)$.

30–30 behaves like deuce. "Run 10 000": Alice wins about $9/13 = 0.692$.



Step

Run until stuck

Reset

Run 10 000

click a state to start there

Step 0: at **Deuce**.

Solution

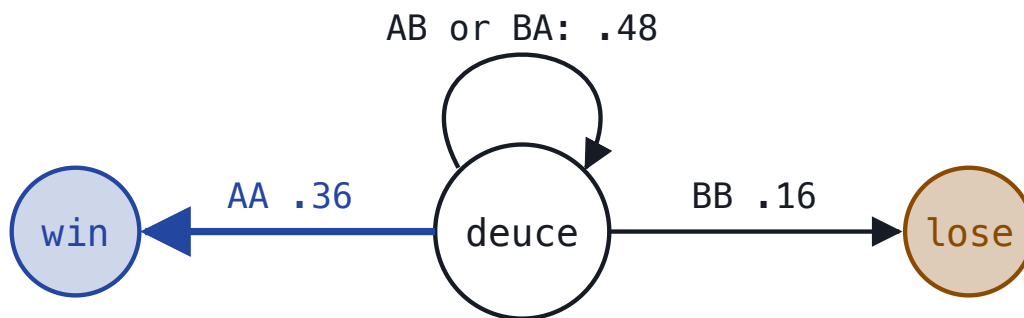
Form 1 ($i \rightarrow \text{end}$) · **Turn** = one pair of points played from a level score. AA $\rightarrow$ ends (win). BB $\rightarrow$ ends (lose). AB or BA $\rightarrow$ level again $\rightarrow$ next turn.

1. **LEVEL.** $\text{rec}(i) = P(\text{Alice wins the game, looking from turn } i \text{ onward})$, where turn i is the i -th pair of points played from a level score. 30–30 is level, so the answer is $\text{rec}(1)$.

Why: from a level score one point puts someone only 1 ahead, which is not enough, so the game can only end at the end of a pair. That makes a pair of points the natural turn.

2. **CHOICE.** At turn i : AA with probability $0.6 \cdot 0.6 = 0.36$; BB with $0.4 \cdot 0.4 = 0.16$; AB or BA with $2 \cdot 0.6 \cdot 0.4 = 0.48$.

Why: points are independent; the four orders AA, AB, BA, BB cover everything, and $0.36 + 0.16 + 0.48 = 1$.



ignore the loop: $P(\text{win}) = .36 / (.36 + .16) = 9/13$

3. **CHECK.** AA: Alice is 2 ahead, ends, value 1. BB: Bob is 2 ahead, ends, value 0. AB or BA: level again, continue to turn $i+1$.

Why: the rules.

4. **MOVE.** $\text{rec}(i) = 0.36 \cdot 1 + 0.16 \cdot 0 + 0.48 \text{rec}(i + 1)$.

Why: law of total probability over turn i .

5. **MEMORYLESS.** At turn $i+1$ the score is level again and Alice still wins each point with 0.6. So $\text{rec}(i+1) = \text{rec}(i)$.

Why: points are independent with the same probability, and only the difference in points decides the future; that difference is 0 at both turns.

6. **SOLVE.** Let $x = \text{rec}(i)$. $x = 0.36 + 0.48x$, so $0.52x = 0.36$, so $x = \frac{36}{52} = \frac{9}{13} \approx 0.69231$.

Why: substitute MEMORYLESS into MOVE; divide top and bottom of $\frac{36}{52}$ by 4.

Answer. From 30–30 Alice wins with probability $\frac{9}{13}$. ■

Two 2s and two 3s before the first 6

A fair die is rolled over and over until the first 6 appears.

Question: what is the probability that, before that first 6, exactly two 2s and exactly two 3s were rolled? (Any number of 1s, 4s and 5s may appear too.)

Picture

Throw away the rolls that change nothing (1, 4, 5). What is left is a fair 3-sided die: 2, 3 or 6. You need exactly two 2s and two 3s in the first four rolls of it, then a 6.

Run once

Run 200,000

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one roll. State (a, b) = 2s and 3s so far. 6 $\rightarrow$ ends. A third 2 or 3 $\rightarrow$ ends (fail). 1, 4, 5 $\rightarrow$ same state (a self-loop).

1. **LEVEL.** $\text{rec}(i, a, b) = P(\text{exactly two 2s and two 3s come before the first 6}), \text{ given no 6 yet and a 2s and b 3s among rolls 1 to } i-1 (a, b \leq 2)$. The answer is $\text{rec}(1, 0, 0)$.

Why: only the counts of 2s and 3s so far affect the event.

2. **CHOICE.** At roll i : 1, 4 or 5 ($\frac{3}{6} = \frac{1}{2}$); 2 ($\frac{1}{6}$); 3 ($\frac{1}{6}$); 6 ($\frac{1}{6}$).

Why: a fair die.



1, 4, 5 change nothing: drop them. 2, 3, 6: $\frac{1}{3}$ each
need 2,2,3,3 in any order, then 6: $6 \cdot (\frac{1}{3})^5$

3. **CHECK.** 6: ends, value 1 if $a = b = 2$, else 0. 2: continue to $(i+1, a+1, b)$, but if $a+1 = 3$ there are too many 2s: value 0. 3: likewise with b . 1, 4, 5: continue to $(i+1, a, b)$.

Why: the event asks for exactly two of each.

4. **MOVE.** $\text{rec}(i, a, b) = \frac{1}{2} \text{rec}(i+1, a, b) + \frac{1}{6} \text{rec}(i+1, a+1, b) + \frac{1}{6} \text{rec}(i+1, a, b+1) + \frac{1}{6} [a = b = 2]$, with $\text{rec} = 0$ whenever a count reaches 3.

Why: law of total probability over roll i .

5. **MEMORYLESS.** $\text{rec}(i+1, a, b) = \text{rec}(i, a, b) = \text{rec}(a, b)$. Then the 1/4/5 case is a self-loop: $\text{rec}(a, b) =$

$\frac{1}{2}\text{rec}(a, b) + \frac{1}{6}(\text{rec}(a+1, b) + \text{rec}(a, b+1) + [a = b = 2])$.
 Subtract $\frac{1}{2}\text{rec}(a, b)$ and multiply by 2: $\text{rec}(a, b) = \frac{1}{3}(\text{rec}(a+1, b) + \text{rec}(a, b+1) + [a = b = 2])$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) . A rec on both sides is solved by algebra, not by recursing into it.

6. **SOLVE.** $\text{rec}(2, 2) = \frac{1}{3}(0 + 0 + 1) = \frac{1}{3}$. $\text{rec}(2, 1) = \text{rec}(1, 2) = \frac{1}{3}(0 + \frac{1}{3}) = \frac{1}{9}$. $\text{rec}(2, 0) = \text{rec}(0, 2) = \frac{1}{3} \cdot \frac{1}{9} = \frac{1}{27}$. $\text{rec}(1, 1) = \frac{1}{3}(\frac{1}{9} + \frac{1}{9}) = \frac{2}{27}$. $\text{rec}(1, 0) = \text{rec}(0, 1) = \frac{1}{3}(\frac{1}{27} + \frac{2}{27}) = \frac{1}{27}$. $\text{rec}(0, 0) = \frac{1}{3}(\frac{1}{27} + \frac{1}{27}) = \frac{2}{81} \approx 0.02469$.

```
double dp[3][3]; // all -1
double rec(ll a, ll b) {
    if (a == 3 || b == 3) return 0; // pruning: too many 2s
    if (dp[a][b] != -1) return dp[a][b]; // cache check
    double ans = (rec(a + 1, b) + rec(a, b + 1) + (a == 2 && b == 2)) / 3;
    return dp[a][b] = ans; // save and return
} // rec(0, 0) cost: 9 states x 3 transitions
```

Why: counts only go up, so fill from $(2, 2)$ down to $(0, 0)$; by the $2 \leftrightarrow 3$ symmetry $\text{rec}(a, b) = \text{rec}(b, a)$.

Answer. The probability is $\frac{2}{81}$. (The self-loop factor $\frac{1}{3}$ is exactly the 'ignore 1, 4, 5' reduced die in the figure.) ■

11 · Geometric probability

If a point is uniform in a region Ω , then $P(A) = \text{vol}(A)/\text{vol}(\Omega)$. Independent uniform variables give a point uniform in a box. Draw the region, then use areas, volumes or integrals.

$$1 \leq y/x \leq 2$$

x and y are chosen independently and uniformly at random from the interval $[0,1]$.

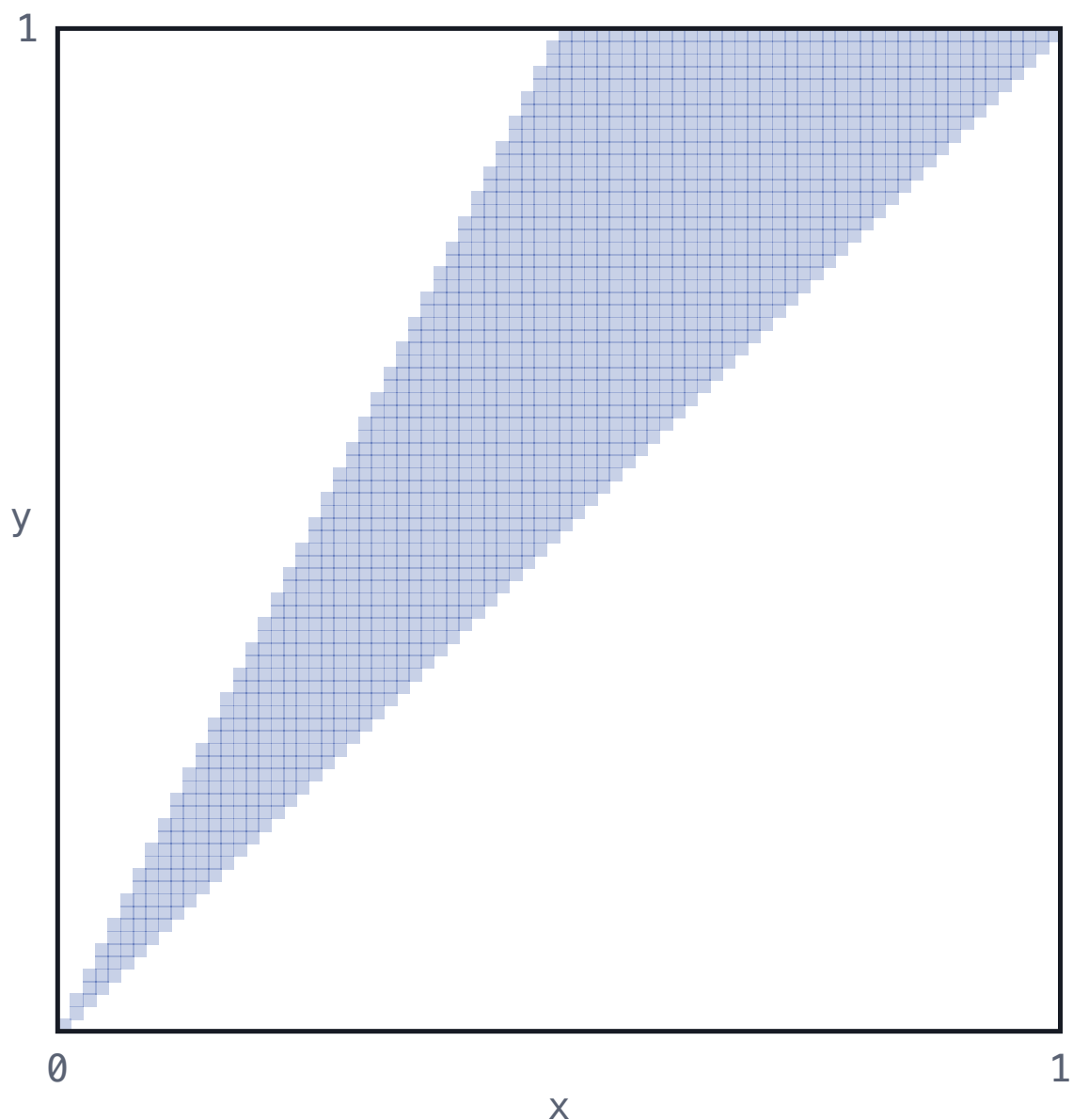
Question: what is the probability that y/x lies between 1 and 2, i.e. $1 \leq y/x \leq 2$?

Key idea

The region between the lines $y = x$ and $y = 2x$ inside the unit square is a triangle.

Picture

Probability = area. The condition is the wedge between the lines $y = x$ and $y = 2x$. Inside the unit square that wedge is a triangle of area $\frac{1}{4}$.



The shaded part is the event. Its share of the box is the probability.

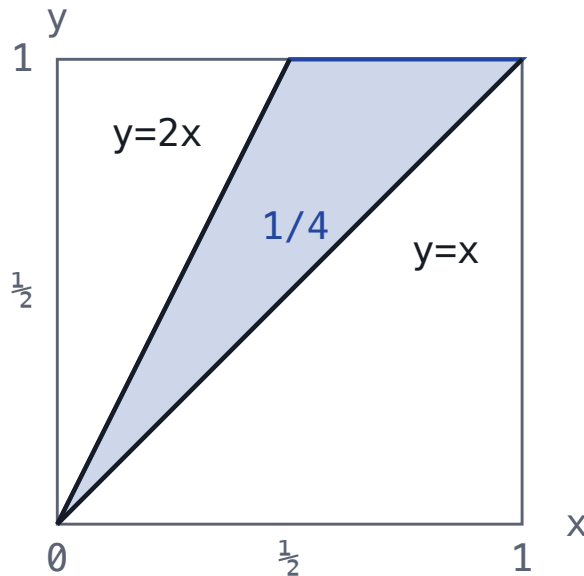
Solution

Setup. x, y are independent and uniform on $[0, 1]$, so (x, y) is uniform on the unit square and probability equals area.

1. $1 \leq y/x \leq 2 \iff x \leq y \leq 2x$ (for $x > 0$; $x = 0$ has probability 0).

Why: multiply the inequalities by $x > 0$.

2. Inside the square, this is the region between the lines $y = x$ and $y = 2x$. The line $y = 2x$ leaves the square through the top at $x = \frac{1}{2}$, so the region is the triangle $(0,0), (\frac{1}{2},1), (1,1)$.



Why: $2x = 1$ at $x = \frac{1}{2}$; $y = x$ reaches the top at $(1,1)$.

3. Its area is $\frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{1}{4}$: base from $(\frac{1}{2},1)$ to $(1,1)$ along the top edge, height 1 down to $(0,0)$.

Why: triangle area formula.

4. $P = \frac{1}{4}$.

Why: probability = area.

Conclusion. The probability is $\frac{1}{4}$. ■

△ Course writes the area as $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$ and then uses $\frac{1}{4}$. The triangle's base is $\frac{1}{2}$ and its height is 1, so the area is $\frac{1}{4}$ (step 3).

Two friends, 30-minute stays

Two friends each arrive at a café at a random time between 12:00 and 13:00 (uniform and independent).

Each stays for exactly 30 minutes and then leaves.

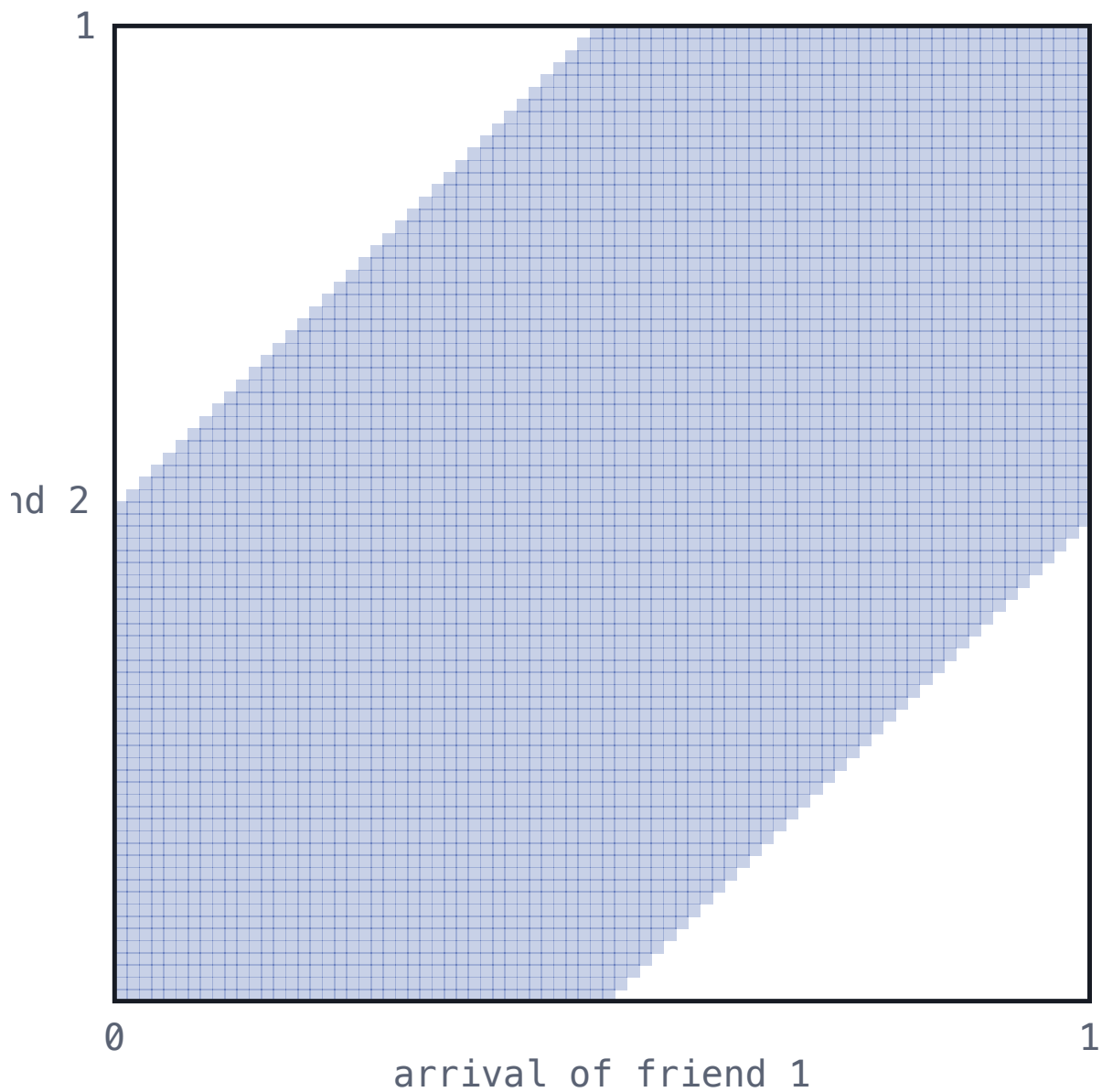
Question: what is the probability that the two friends are at the café at the same time?

Key idea

They meet $\iff |x - y| \leq \frac{1}{2}$. The complement is two corner triangles.

Picture

They meet unless one arrives more than half an hour after the other. Those "miss" cases are two corner triangles of area $\frac{1}{8}$ each; everything else is a meeting.



The shaded part is the event. Its share of the box is the probability.

Solution

Setup. Arrival times x, y are independent and uniform on $[0, 1]$ hour; each friend stays for half an hour. The point (x, y) is uniform on the unit square.

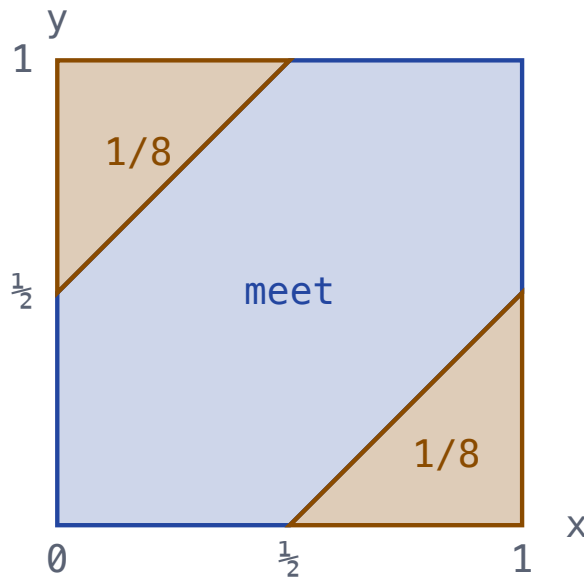
1. They meet exactly when $|x - y| \leq \frac{1}{2}$.

Why: say $x \leq y$. The first friend is there during $[x, x + \frac{1}{2}]$; they meet iff the second arrives by then, $y \leq x + \frac{1}{2}$. The case $y \leq x$ is the same with the roles swapped.

2. The complement $|x - y| > \frac{1}{2}$ is two corner triangles, $y > x + \frac{1}{2}$ and $x > y + \frac{1}{2}$, each a right triangle with legs $\frac{1}{2}$: area $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ each.

Why: for example, $y > x + \frac{1}{2}$ inside the square is the triangle $(0, \frac{1}{2}), (0, 1), (\frac{1}{2}, 1)$.

3. $P(\text{meet}) = 1 - 2 \cdot \frac{1}{8} = \frac{3}{4}$.



Why: complement of the two triangles.

Conclusion. They meet with probability $\frac{3}{4}$. ■

$$xy \leq \frac{1}{2}$$

x and y are chosen independently and uniformly from $[0, 1]$.

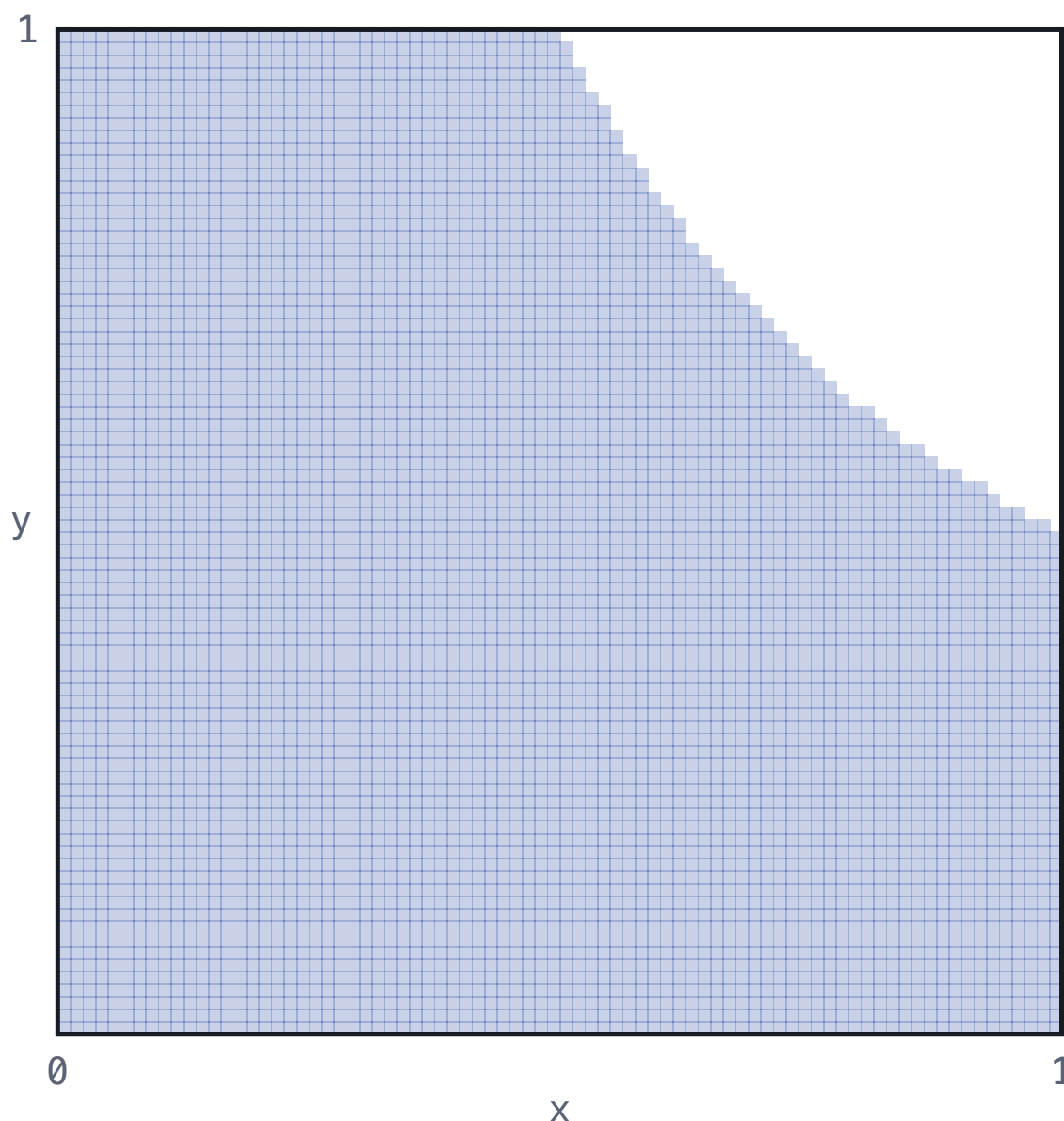
Question: what is the probability that $xy \leq \frac{1}{2}$?

Key idea

For $x \leq \frac{1}{2}$ every y works. For $x > \frac{1}{2}$ you need $y \leq \frac{1}{2x}$.

Picture

For x up to $\frac{1}{2}$ every y works; beyond that y must stay under the curve $y = 1/(2x)$. Area = $\frac{1}{2}$ + (area under the curve from $\frac{1}{2}$ to 1).



The shaded part is the event. Its share of the box is the probability.

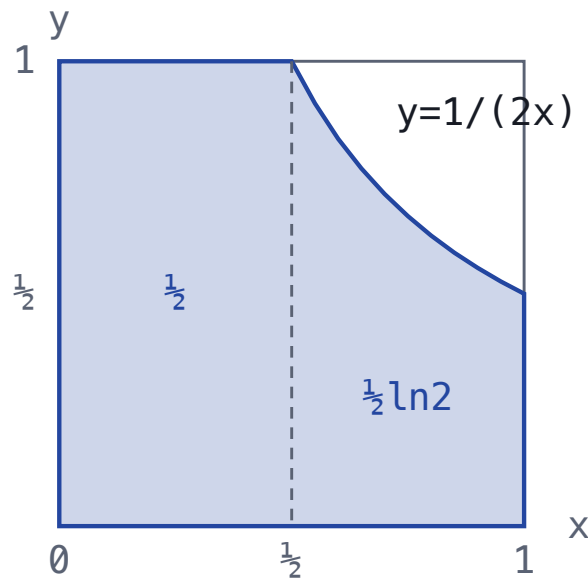
Solution

Setup. x, y independent uniform on $[0, 1]$; probability = area of $\{xy \leq \frac{1}{2}\}$ in the unit square.

1. For a fixed x , the allowed y are $y \leq \frac{1}{2x}$.

Why: divide $xy \leq \frac{1}{2}$ by $x > 0$.

2. If $x \leq \frac{1}{2}$, then $\frac{1}{2x} \geq 1$, so every $y \in [0, 1]$ works: a vertical strip of length 1. If $x > \frac{1}{2}$, the allowed y fill $[0, \frac{1}{2x}]$, of length $\frac{1}{2x} < 1$.



Why: compare $\frac{1}{2x}$ with 1.

$$3. \text{ Area} = \int_0^{1/2} 1 \, dx + \int_{1/2}^1 \frac{dx}{2x} = \frac{1}{2} + \frac{1}{2} [\ln x]_{1/2}^1 = \frac{1}{2} + \frac{1}{2} (0 - \ln \frac{1}{2}) = \frac{1}{2} + \frac{1}{2} \ln 2.$$

Why: add the strip lengths; $-\ln \frac{1}{2} = \ln 2$.

$$4. P = \frac{1}{2} + \frac{1}{2} \ln 2 \approx 0.84657.$$

Why: probability = area.

Conclusion. $P(xy \leq \frac{1}{2}) = \frac{1}{2} + \frac{1}{2} \ln 2.$ ■

Sum of n uniforms < 1

$x_1, \dots, x_n$ are chosen independently and uniformly from $[0, 1]$.

Question: what is the probability that $x_1 + \dots + x_n < 1$ when (a) $n = 2$, (b) $n = 3$, (c) n is general?

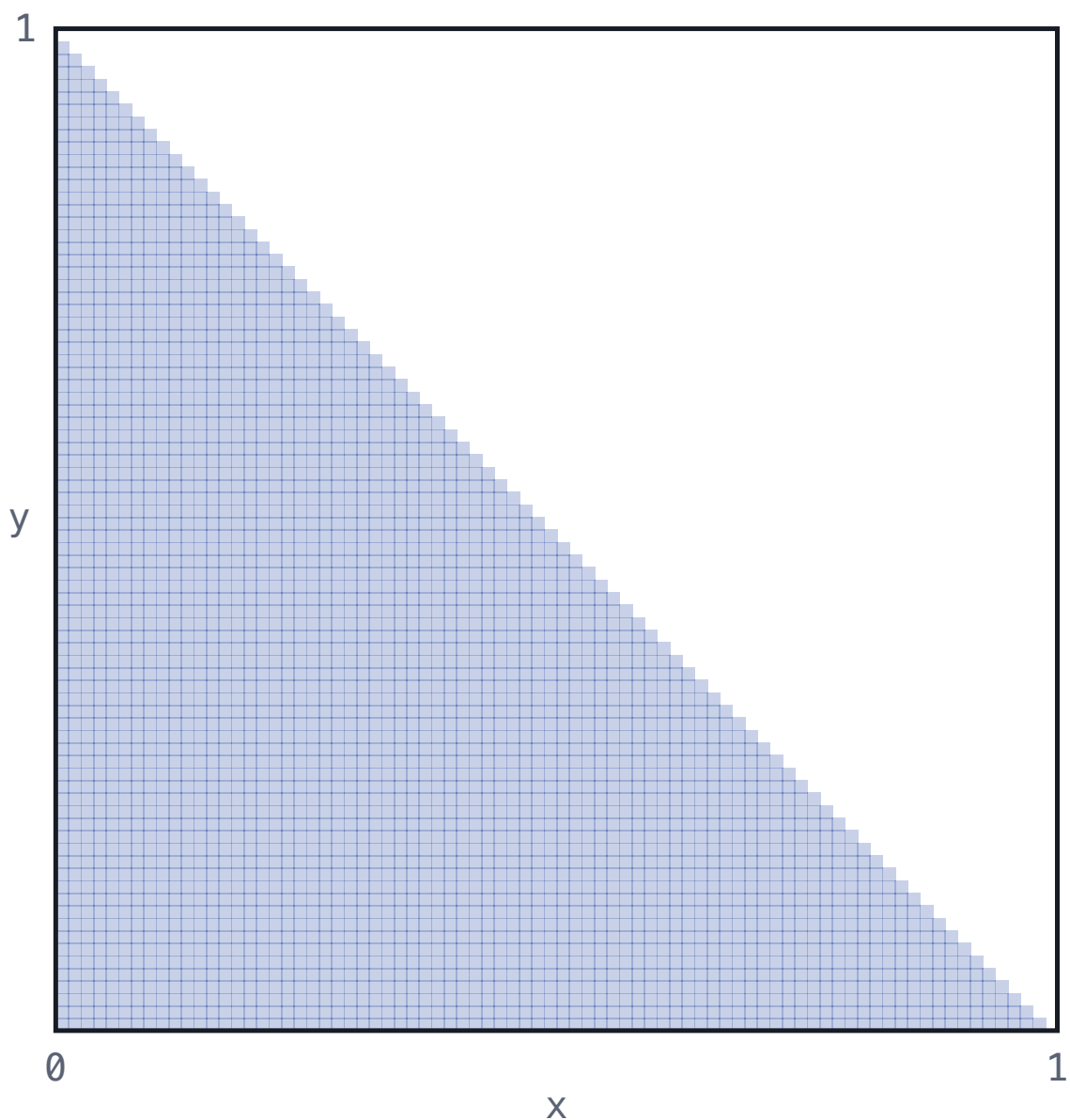
Key idea

The region is a simplex of volume $1/n!$. Slice at $x_n = t$ and use induction.

Picture

The region "sum below 1" is a corner of the cube: a triangle of area $\frac{1}{2}$ in 2-D, a pyramid of volume $\frac{1}{6}$ in 3-D, and $1/n!$ in n dimensions.

$n = 2$: the triangle below $x + y = 1$.



Throw 100 points

Throw 10 000

Clear

The shaded part is the event. Its share of the box is the probability.

Throw n random numbers in $[0, 1]$ many times and count how often their sum is below 1.

n  3

$n = 3$: sum below 1 in **0.1672** of 200 000 tries ($1/3! = 0.1667$).

Solution

Setup. $x_1, \dots, x_n$ are independent uniforms on $[0, 1]$, so $P(\sum x_i < 1)$ is the volume of $\{x \in [0, 1]^n : \sum x_i < 1\}$. For $0 \leq s \leq 1$ define $V_n(s)$ = the volume of $\{x \in [0, 1]^n : \sum x_i < s\}$.

Claim. $V_n(s) = \frac{s^n}{n!}$, so $P = \frac{1}{n!}$.

1. $n = 1$: $V_1(s) = s$, the length of $[0, s]$.

Why: one variable.

2. For $s \leq 1$ the upper bounds $x_i \leq 1$ never bind: if all $x_i \geq 0$ and $\sum x_i < s \leq 1$, each $x_i < 1$.

Why: each term is at most the sum.

3. Slice at $x_n = t$, for $0 \leq t < s$: the other coordinates must satisfy $\sum_{i < n} x_i < s - t$, a cross-section of volume $V_{n-1}(s - t)$. So $V_n(s) = \int_0^s V_{n-1}(s - t) dt$.

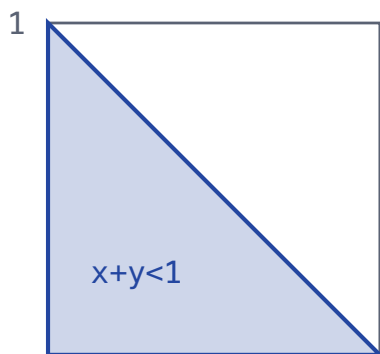
Why: volume = integral of cross-section areas (Cavalieri).

4. By induction, $V_n(s) = \int_0^s \frac{(s - t)^{n-1}}{(n - 1)!} dt = \left[-\frac{(s - t)^n}{n!} \right]_0^s = \frac{s^n}{n!}$.

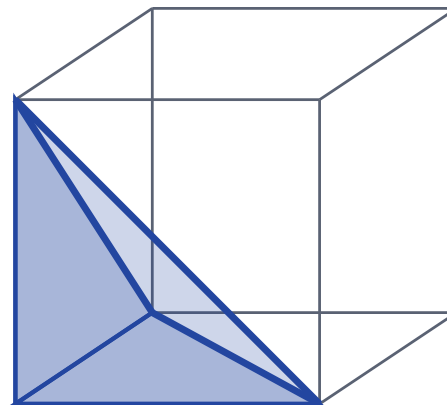
Why: assume the formula for $n - 1$, substitute, and integrate.

5. $P = V_n(1) = \frac{1}{n!}$: (a) $\frac{1}{2}$, (b) $\frac{1}{6} \approx 0.167$, (c) $\frac{1}{n!}$.

Why: $s = 1$.



$n = 2$: area $\frac{1}{2}$



$n = 3$: volume $\frac{1}{6}$

Conclusion. The sum of n uniforms is below 1 with probability $\frac{1}{n!}$. ■

Crawling ant (max 3 steps)

An ant moves in steps.

For each step it draws a random time t and a random distance x , each uniform on $[0,1]$, with all draws independent, and walks distance x in time t .

It stops at the end of the step during which its total time first goes past 1, but it never takes more than 3 steps.

Question: what is the probability that the total distance it walks is greater than 1?

Key idea

Step 1 never stops the ant ($t_1 < 1$). So it stops after 2 steps ($t_1 + t_2 > 1$) or after 3 steps.

Distances and times are independent.

Picture

Split on how many steps the ant takes. The first step never uses up the time. It takes 2 steps if $t_1 + t_2 > 1$ (chance $\frac{1}{2}$), else 3. Distances are independent of times, so multiply: $\frac{1}{2} \times P(2 \text{ distances pass } 1) + \frac{1}{2} \times P(3 \text{ distances pass } 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{5}{6}$.

Run once

Run 100,000

Solution

Setup. Step k takes time t_k and covers distance x_k ; all the t_k, x_k are independent uniforms on $[0, 1]$. The ant stops at the end of the step during which the total time passes 1, or after 3 steps, whichever comes first. Its final position is the total distance walked.

1. The ant never stops after step 1.

Why: $t_1 < 1$ (with probability 1), so the time hasn't passed 1, and it has taken fewer than 3 steps.

2. It stops after step 2 exactly when $t_1 + t_2 > 1$; otherwise it stops after step 3.

Why: the time rule, and the 3-step cap.

3. $P(\text{stops after 2 and ends past } 1) = P(t_1 + t_2 > 1) \cdot P(x_1 + x_2 > 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Why: times are independent of distances;

$P(\text{sum of 2 uniforms} < 1) = \frac{1}{2}$ (previous problem), so $P(> 1) = \frac{1}{2}$.

4. $P(\text{stops after 3 and ends past } 1) = P(t_1 + t_2 \leq 1) \cdot P(x_1 + x_2 + x_3 > 1) = \frac{1}{2} \cdot \left(1 - \frac{1}{6}\right) = \frac{5}{12}$.

Why: same independence; $P(\text{sum of 3 uniforms} < 1) = \frac{1}{6}$.

5. The two cases are disjoint, so $P = \frac{1}{4} + \frac{5}{12} = \frac{3}{12} + \frac{5}{12} = \frac{2}{3} \approx 0.667$.

Why: the ant stops after exactly one of 2 or 3 steps (steps 1–2).

Conclusion. The ant ends past 1 with probability $\frac{2}{3}$ (simulated). ■

$a_1 < a_2 < a_3$ with $a_k \sim U[0, k]$

a_1 is uniform on $[0, 1]$, a_2 is uniform on $[0, 2]$, and a_3 is uniform on $[0, 3]$, all independent.

Question: what is the probability that $a_1 < a_2 < a_3$?

Key idea

Volume of the ordered region inside a $1 \times 2 \times 3$ box.

Picture

The three numbers are a random point in a $1 \times 2 \times 3$ box. The answer is the volume of the part where they are in order, divided by 6. Integrate: for each $a_1 < a_2$, a_3 has room $3 - a_2$.

Run once

Run 100,000

Solution

Setup. a_k is uniform on $[0, k]$, independently, so (a_1, a_2, a_3) is uniform on the box $[0, 1] \times [0, 2] \times [0, 3]$ of volume 6. $P = (\text{volume of the ordered part}) / 6$.

1. For fixed $a_1 < a_2$, the allowed a_3 lie in $(a_2, 3]$, of length $3 - a_2$. So the volume is $\int_0^1 \int_{a_1}^2 (3 - a_2) da_2 da_1$.

Why: integrate the allowed length over a_2 from a_1 up to 2, then over a_1 .

2. Inner integral: $\left[3a_2 - \frac{a_2^2}{2}\right]_{a_1}^2 = (6 - 2) - \left(3a_1 - \frac{a_1^2}{2}\right) = 4 - 3a_1 + \frac{a_1^2}{2}$.

Why: antiderivative of $3 - a_2$.

3. Outer integral: $\int_0^1 \left(4 - 3a_1 + \frac{a_1^2}{2}\right) da_1 = 4 - \frac{3}{2} + \frac{1}{6} = \frac{24-9+1}{6} = \frac{8}{3}$.

Why: integrate term by term.

4. $P = \frac{8/3}{6} = \frac{4}{9} \approx 0.444$.

Why: divide by the box volume.

Conclusion. The three are in increasing order with probability $\frac{4}{9}$. ■

12 · Recursion in probability

Every problem here uses the [§5 template](#): LEVEL rec(i, s) → CHOICE → CHECK → MOVE → MEMORYLESS → SOLVE.

Choose the **state** s as the smallest description that decides the future, and merge states that are symmetric.

First to roll a 6

Two players take turns rolling a fair die. The first player to roll a 6 wins.

Question: what is the probability that the player who rolls first wins?

Picture

After a miss, the other player is in exactly your old position. If the roller wins with x , then $x = \frac{1}{6} + \frac{5}{6}(1 - x)$, so $x = 6/11$.

Run once

Run 100,000

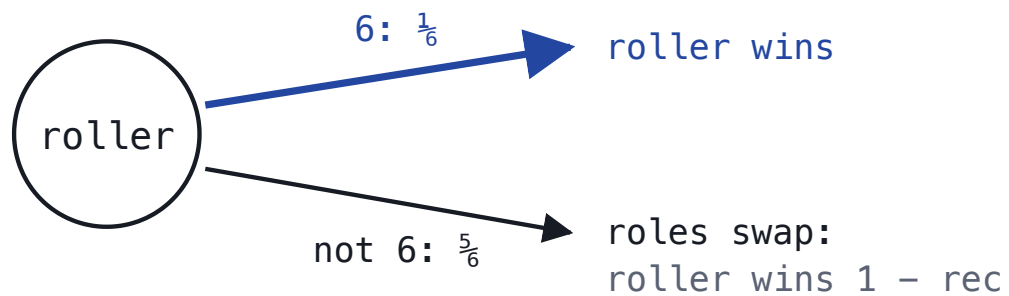
Solution

Form 1 (i → end) + Form 5 (game) · Turn = one roll.
6 → ends (the roller wins). Anything else → next turn, and the *other* player rolls.

1. **LEVEL.** $\text{rec}(i) = P(\text{the player who rolls at turn } i \text{ wins the game}), \text{ given no 6 in turns 1 to } i-1.$ The answer is $\text{rec}(1)$.
Why: the first player is the roller at turn 1. rec is about 'the current roller', not a fixed player.
2. **CHOICE.** At turn i the roll is a 6 with probability $\frac{1}{6}$, not a 6 with probability $\frac{5}{6}$.
Why: a fair die.
3. **CHECK.** 6: ends, the roller wins, value 1. Not 6: continue to turn $i+1$, where the other player rolls.
Why: the rules; players alternate.

4. **MOVE.** After a non-6, the roller at turn $i+1$ wins with $\text{rec}(i+1)$, so the roller at turn i wins with $1 - \text{rec}(i+1)$. So $\text{rec}(i) = \frac{1}{6} \cdot 1 + \frac{5}{6}(1 - \text{rec}(i+1))$.

Why: this is the Form 5 (game) move: my chance = 1 - (the chance of the player who moves next). Exactly one of the two players wins; law of total probability over turn i .



$$\text{rec} = \frac{1}{6} + \frac{5}{6} (1 - \text{rec}) \Rightarrow \text{rec} = 6/11$$

5. **MEMORYLESS.** At turn $i+1$, the player rolling faces exactly what the roller at turn i faced. So $\text{rec}(i+1) = \text{rec}(i)$.

Why: each turn is independent of earlier turns, has the same probabilities and the same rules, and nothing from earlier turns is carried forward. This works because rec is about 'the current roller'; for a fixed player it would alternate.

6. **SOLVE.** Let $x = \text{rec}(i)$. $x = \frac{1}{6} + \frac{5}{6} - \frac{5}{6}x = 1 - \frac{5}{6}x$. So $\frac{11}{6}x = 1$, so $x = \frac{6}{11} \approx 0.54545$.

Why: substitute MEMORYLESS into MOVE; expand; move the x terms left.

Answer. The first player wins with probability $\frac{6}{11}$. ■

Strictly first to heads

Several players each toss a fair coin once per round, all at the same time, round after round. A player's 'time' is the round in which they get their first head.

Question: (a) You play with 9 friends. What is the probability that your first head comes strictly earlier than every friend's? (b) What is the probability that Alice's first head comes strictly before Bob's? (c) What is the probability that Alice's first head comes strictly before Bob's, and Bob's strictly before Charlie's?

Picture

Ignore all-tails rounds; they change nothing. Then only the first round with any head matters. (b) Of the 3 outcomes (H,T), (T,H), (H,H), only (H,T) is good: $\frac{1}{3}$. (c) Alice must be alone first (1 of 7 non-all-tails outcomes), then Bob before Charlie ($\frac{1}{3}$): $\frac{1}{7} \times \frac{1}{3}$.

Run once

Run 100,000

Solution

Form 1 (i → end) · **Turn** = one round (everyone tosses). All tails → next turn. Anything else → ends (or, in (c), hands over to problem (b)).

1. **LEVEL.** For each part, $\text{rec}(i) = P(\text{the event happens, looking from round } i \text{ onward})$, given every

toss in rounds 1 to $i-1$ was tails. The answer is $\text{rec}(1)$.

Why: before round 1 nobody has a head.

2. **(a) CHOICE + CHECK.** Round i has $2^{10} = 1024$ equally likely outcomes. Only you get H ($\frac{1}{1024}$): ends, value 1. All tails ($\frac{1}{1024}$): continue. Anything else ($\frac{1022}{1024}$): some friend got H this round, so they are not later than you: ends, value 0.

Why: 10 independent fair coins; 'strictly first' fails as soon as a friend has a head no later than you.

3. **(a) MOVE.** $\text{rec}(i) = \frac{1}{1024} \cdot 1 + \frac{1}{1024} \text{rec}(i+1) + \frac{1022}{1024} \cdot 0$.

Why: law of total probability over round i .

4. **(a) MEMORYLESS.** At round $i+1$ still nobody has a head and the coins are the same, so $\text{rec}(i+1) = \text{rec}(i)$.

Why: each turn is independent of earlier turns, has the same probabilities and the same rules, and nothing from earlier turns is carried forward.

5. **(a) SOLVE.** $x = \frac{1}{1024} + \frac{1}{1024}x$, so $\frac{1023}{1024}x = \frac{1}{1024}$, so $x = \frac{1}{1023}$.

Why: move the x term left; multiply by $\frac{1024}{1023}$.

6. **(b) CHOICE + CHECK.** Round i , (Alice, Bob): (H, T) $\frac{1}{4}$: ends, value 1. (T, T) $\frac{1}{4}$: continue. (T, H) or (H, H) $\frac{1}{2}$: Bob first or a tie, value 0.

Why: two independent fair coins give 4 equally likely outcomes.

7. **(b) MOVE + MEMORYLESS.** $\text{rec}(i) = \frac{1}{4} + \frac{1}{4} \text{rec}(i+1)$, and $\text{rec}(i+1) = \text{rec}(i)$.

Why: total probability over round i ; same memorylessness as (a).

8. **(b) SOLVE.** $x = \frac{1}{4} + \frac{1}{4}x$, so $\frac{3}{4}x = \frac{1}{4}$, so $x = \frac{1}{3}$.

Why: move the x term left; multiply by $\frac{4}{3}$.

9. **(c) CHOICE + CHECK.** Round i , (Alice, Bob, Charlie): (H, T, T) $\frac{1}{8}$: Alice is strictly first; what remains is 'Bob strictly before Charlie' from a fresh round, which is problem (b): value $\frac{1}{3}$. (T, T, T) $\frac{1}{8}$: continue. Anything else: Bob or Charlie got H no later than Alice, value 0.

Why: three independent fair coins give 8 equally likely outcomes; (b) is memoryless, so its value from any fresh round is $\frac{1}{3}$.

10. **(c) MOVE + MEMORYLESS.** $\text{rec}(i) = \frac{1}{8} \cdot \frac{1}{3} + \frac{1}{8} \text{rec}(i+1)$, and $\text{rec}(i+1) = \text{rec}(i)$.

Why: total probability over round i ; same memorylessness.

11. **(c) SOLVE.** $x = \frac{1}{24} + \frac{1}{8}x$, so $\frac{7}{8}x = \frac{1}{24}$, so $x = \frac{1}{24} \cdot \frac{8}{7} = \frac{8}{168} = \frac{1}{21} \approx 0.04762$.

Why: move the x term left; multiply by $\frac{8}{7}$.

Answer. $\frac{1}{1023}$, $\frac{1}{3}$ and $\frac{1}{21}$. ■

HH ends on an even toss

A fair coin is tossed until two heads in a row appear.

Question: what is the probability that this happens on an even-numbered toss (toss 2, 4, 6, ...)?

Picture

After a T you start afresh, but the toss count has moved on by 1 (or 2 after HT). Moving by 1 flips "even" and "odd"; moving by 2 doesn't. That gives $x = \frac{1}{2}(1 - x) + \frac{1}{4}x + \frac{1}{4}$, so $x = 3/5$.

Run once

Run 100,000

Solution

Form 1 (i → end) · Turn = a 'fresh start' before toss i (no toss yet, or the last toss was T). T or HT → fresh again. HH → ends. The MEMORYLESS check *fails* here, and that is the lesson.

1. **LEVEL.** $\text{rec}(i) = P(\text{the first HH is completed on an even toss number})$, given we are fresh just before toss i . The answer is $\text{rec}(1)$.

Why: before toss 1 nothing has been tossed, so we are fresh.

2. **CHOICE.** From fresh before toss i : toss i is T ($\frac{1}{2}$); or toss i is H and toss $i+1$ is T ($\frac{1}{4}$); or tosses $i, i+1$ are HH ($\frac{1}{4}$).

Why: toss i is T, or it is H and then toss $i+1$ is T or H. Disjoint, and $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$.



$$\text{rec} = \frac{1}{2}(1 - \text{rec}) + \frac{1}{4} \text{rec} + \frac{1}{4} \Rightarrow 3/5$$

3. **CHECK.** HH: ends on toss $i+1$, which is even exactly when i is odd. T: fresh before toss $i+1$. HT: fresh before toss $i+2$.

Why: a T can never be part of a later HH, so after a T nothing carries over.

4. **MOVE.** $\text{rec}(i) = \frac{1}{2} \text{rec}(i+1) + \frac{1}{4} \text{rec}(i+2) + \frac{1}{4} \cdot [i \text{ odd}]$, where $[i \text{ odd}]$ is 1 if i is odd and 0 otherwise.

Why: law of total probability over the three cases.

5. **MEMORYLESS (check it, don't assume it).** Let M = the number of tosses a brand-new game takes. Fresh before toss j , the rest is a brand-new game, so it ends at toss $j-1+M$: $\text{rec}(j) = P(j-1+M \text{ even})$. So:

$$\text{rec}(i+1) = P(i+M \text{ even}) = 1 - P(i-1+M \text{ even}) = 1 - \text{rec}(i)$$

$$\text{rec}(i+2) = P(i+1+M \text{ even}) = P(i-1+M \text{ even}) = \text{rec}(i)$$

Why: the coin forgets, but the question is about the toss *number*: its parity flips when the start moves by 1 and stays when it moves by 2. So here $\text{rec}(i+1) \neq \text{rec}(i)$.

6. **SOLVE.** Take $i = 1$ (odd), $x = \text{rec}(1)$. $x = \frac{1}{2}(1-x) + \frac{1}{4}x + \frac{1}{4} = \frac{1}{2} - \frac{1}{2}x + \frac{1}{4}x + \frac{1}{4} = \frac{3}{4} - \frac{1}{4}x$. So $\frac{5}{4}x = \frac{3}{4}$, so $x = \frac{3}{5}$.

Why: substitute MEMORYLESS into MOVE; collect x terms; multiply by $\frac{4}{5}$. (Simulated: 0.600.)

Answer. The first HH ends on an even toss with probability $\frac{3}{5}$. ■

Parcel around 5 seats

5 people sit around a round table; you are one of them, and you are holding a parcel.

On each turn, whoever holds the parcel does one of three things, each with probability $\frac{1}{3}$: passes it left, passes it right, or keeps it.

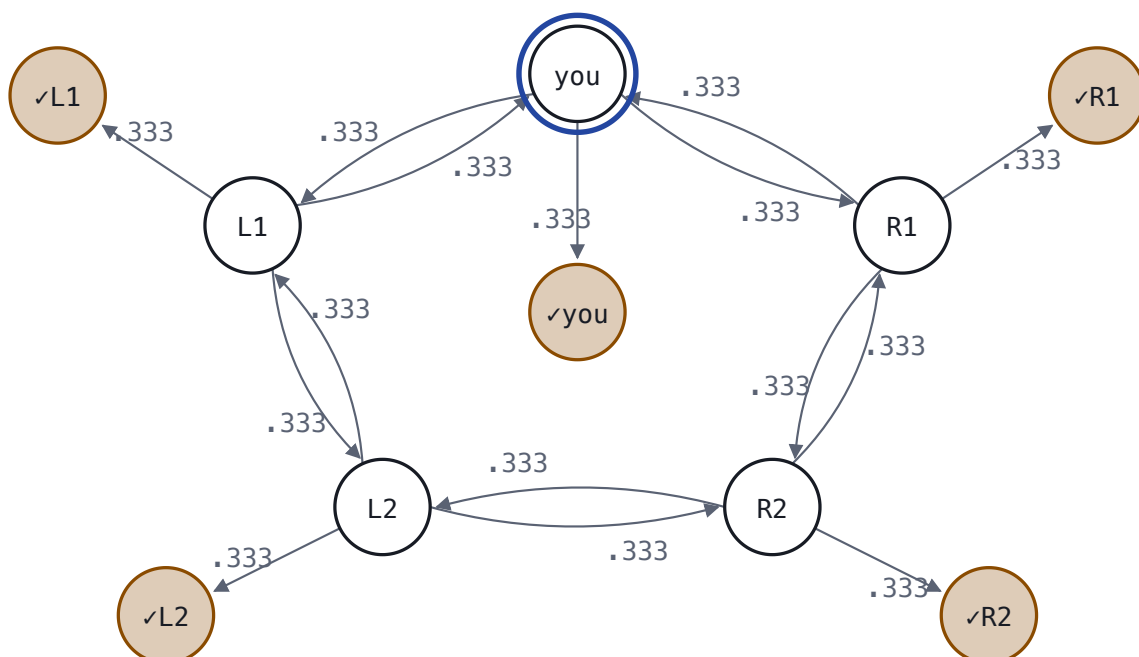
As soon as someone keeps it, the game ends and that person gets the parcel.

Question: what is the probability that you end up with the parcel?

Picture

Only the distance from you matters: 0, 1 or 2 seats away. That leaves 3 unknowns, one equation each (keep, pass towards, pass away). Check: your share plus the others' must add to 1.

Five seats (blue) and the matching "keeps it" endings (orange). "Run 10 000": you keep it about $5/11 = 0.455$.



Step

Run until stuck

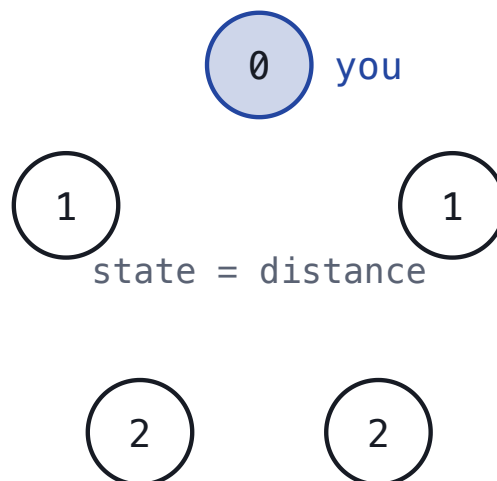
Reset

Run 10 000

click a state to start there

Step 0: at **you**.**Solution****Form 1 ($i \rightarrow \text{end}$)** · **Turn** = one turn of the holder.State d = holder's distance from you (0, 1 or 2).Keep $\rightarrow$ ends. Pass $\rightarrow$ next turn at the new distance.

- LEVEL.** $\text{rec}(i, d) = P(\text{you end up keeping the parcel})$, given that before turn i the holder is at distance d from you. Around the circle the seats are at distances 0 (you), 1, 2, 2, 1. The answer is $\text{rec}(1, 0)$.

Why: by symmetry only the distance matters.

- CHOICE.** Keep, pass left, pass right: each $\frac{1}{3}$.
Why: the rules.
- CHECK.** Keep: ends, value 1 if $d = 0$, else 0. From $d = 0$ either pass goes to distance 1. From $d = 1$: toward you $\rightarrow 0$, away $\rightarrow 2$. From $d = 2$: toward you $\rightarrow$

1, the other way $\rightarrow$ the other distance-2 seat, still 2.

Why: read off the circle; the two distance-2 seats are next to each other.

4. **MOVE.** $\text{rec}(i, 0) = \frac{1}{3} + \frac{2}{3} \text{rec}(i+1, 1)$. $\text{rec}(i, 1) = \frac{1}{3} \cdot 0 + \frac{1}{3} \text{rec}(i+1, 0) + \frac{1}{3} \text{rec}(i+1, 2)$. $\text{rec}(i, 2) = \frac{1}{3} \cdot 0 + \frac{1}{3} \text{rec}(i+1, 1) + \frac{1}{3} \text{rec}(i+1, 2)$.

Why: law of total probability over the holder's action.

5. **MEMORYLESS.** $\text{rec}(i+1, d) = \text{rec}(i, d) = \text{rec}(d)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** From $\text{rec}(2) = \frac{1}{3} \text{rec}(1) + \frac{1}{3} \text{rec}(2)$: $\frac{2}{3} \text{rec}(2) = \frac{1}{3} \text{rec}(1)$, so $\text{rec}(2) = \frac{1}{2} \text{rec}(1)$. Into $\text{rec}(1) = \frac{1}{3} \text{rec}(0) + \frac{1}{3} \text{rec}(2) = \frac{1}{3} \text{rec}(0) + \frac{1}{6} \text{rec}(1)$: $\frac{5}{6} \text{rec}(1) = \frac{1}{3} \text{rec}(0)$, so $\text{rec}(1) = \frac{2}{5} \text{rec}(0)$. Into $\text{rec}(0) = \frac{1}{3} + \frac{2}{3} \text{rec}(1) = \frac{1}{3} + \frac{4}{15} \text{rec}(0)$: $\frac{11}{15} \text{rec}(0) = \frac{1}{3}$, so $\text{rec}(0) = \frac{5}{11} \approx 0.45455$.

Why: solve from the far side inwards.

7. **CHECK THE ANSWER.** $\text{rec}(1) = \frac{2}{11}$, $\text{rec}(2) = \frac{1}{11}$, and $\frac{5}{11} + 2 \cdot \frac{2}{11} + 2 \cdot \frac{1}{11} = 1 \checkmark$.

Why: by symmetry these are also the chances that each seat ends up keeping it when you start, and somebody always keeps it.

Answer. You keep the parcel with probability $\frac{5}{11}$. ■

△ Course states $p_0 = p_1 = p_2 = \frac{5}{11}$. Only $\text{rec}(0)$ is $\frac{5}{11}$.

HHT before HTT

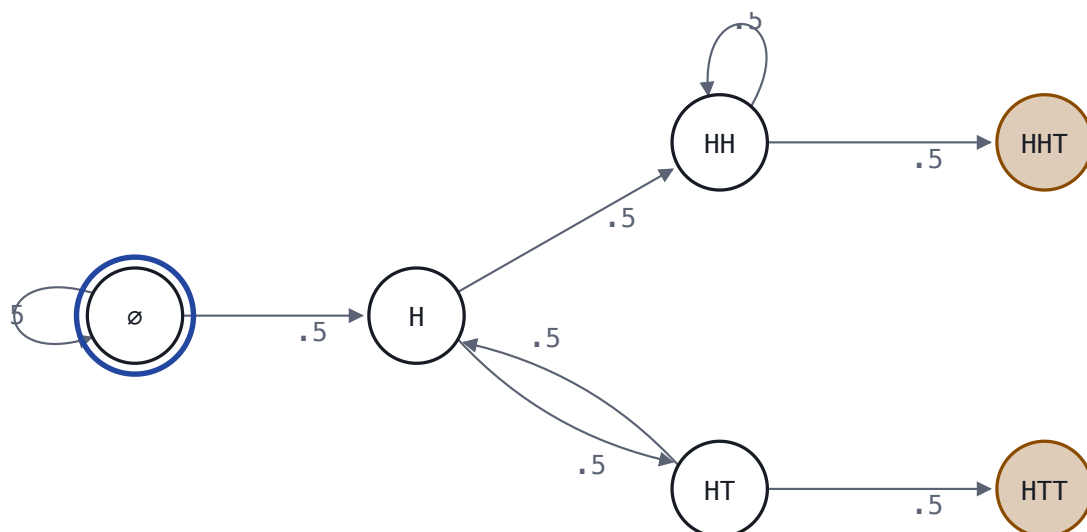
A fair coin is tossed over and over until either the pattern HHT or the pattern HTT appears (three tosses in a row).

Question: what is the probability that HHT appears before HTT?

Picture

Once you have HH, HHT is guaranteed: more H's keep you at HH, and the first T finishes HHT. HTT needs H, T, T in a row without another H getting in the way. Track the useful ending: $\emptyset$, H, HH, HT.

States are the useful endings. "Run 10 000": HHT first about 2/3.



Step

Run until stuck

Reset

Run 10 000

click a state to start there

Step 0: at $\emptyset$.

Solution

Form 1 ($i \rightarrow \text{end}$) · Turn = one toss. State s = the useful ending so far: $\emptyset$, H, HH or HT. HHT $\rightarrow$ ends

(win). HTT → ends (lose).

1. **LEVEL.** $\text{rec}(i, s) = P(\text{HHT comes before HTT})$, given that before toss i the useful ending is s : the longest ending of the tosses so far that is still the start of a pattern ($\emptyset$, H, HH or HT). The answer is $\text{rec}(1, \emptyset)$.

Why: only the useful ending can become part of a pattern.

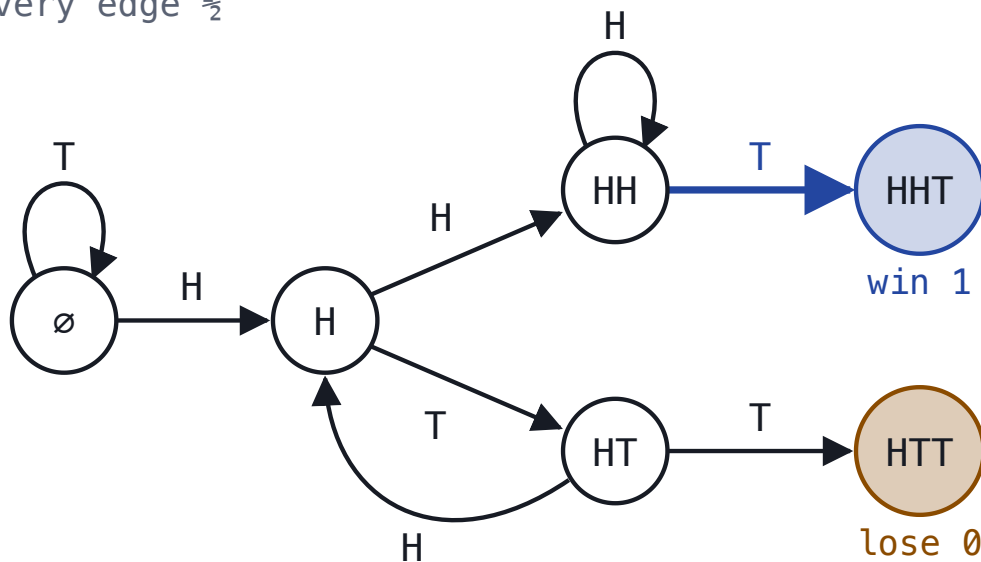
2. **CHOICE.** Toss i is H or T, each $\frac{1}{2}$.

Why: a fair coin.

3. **CHECK.** $\emptyset$: H → H, T → $\emptyset$. H: H → HH, T → HT. HH: H → HH, T → HHT, ends, value 1. HT: H → H, T → HTT, ends, value 0.

Why: after HTH the useful ending is just H; after HHH it is still HH.

every edge $\frac{1}{2}$



4. **MOVE.** $\text{rec}(i, \emptyset) = \frac{1}{2}\text{rec}(i+1, H) + \frac{1}{2}\text{rec}(i+1, \emptyset)$.
 $\text{rec}(i, H) = \frac{1}{2}\text{rec}(i+1, HH) + \frac{1}{2}\text{rec}(i+1, HT)$. $\text{rec}(i, HH) = \frac{1}{2}\text{rec}(i+1, HH) + \frac{1}{2} \cdot 1$. $\text{rec}(i, HT) = \frac{1}{2}\text{rec}(i+1, H) + \frac{1}{2} \cdot 0$.

Why: law of total probability over toss i .

5. **MEMORYLESS.** $\text{rec}(i+1, s) = \text{rec}(i, s) = \text{rec}(s)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** $\text{rec}(\text{HH}) = \frac{1}{2}\text{rec}(\text{HH}) + \frac{1}{2}$, so $\frac{1}{2}\text{rec}(\text{HH}) = \frac{1}{2}$, $\text{rec}(\text{HH}) = 1$. $\text{rec}(\text{HT}) = \frac{1}{2}\text{rec}(\text{H})$. $\text{rec}(\text{H}) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2}\text{rec}(\text{H}) = \frac{1}{2} + \frac{1}{4}\text{rec}(\text{H})$, so $\frac{3}{4}\text{rec}(\text{H}) = \frac{1}{2}$, $\text{rec}(\text{H}) = \frac{2}{3}$. $\text{rec}(\emptyset) = \frac{1}{2}\text{rec}(\text{H}) + \frac{1}{2}\text{rec}(\emptyset)$, so $\frac{1}{2}\text{rec}(\emptyset) = \frac{1}{2}\text{rec}(\text{H})$, $\text{rec}(\emptyset) = \frac{2}{3}$.

Why: from HH you can never lose (T wins, H stays at HH); then substitute upwards.

Answer. HHT comes before HTT with probability $\frac{2}{3}$. ■

Score-dependent best-of-7

Alice and Bob play rounds until one of them has won 4 rounds. When the score is tied, Alice wins the next round with probability $\frac{2}{3}$. When the score is not tied, the player who is *behind* wins the next round with probability $\frac{2}{3}$.

Question: what is the probability that Alice wins the match?

Picture

The score only goes up, so fill a table from the end. Each cell = (chance Alice wins this round) $\times$ (cell below) + (chance Bob wins) $\times$ (cell to the right).

Rows = Alice's wins, columns = Bob's wins. Click a cell.

Alice a \ Bob b	0	1	2	3	4
0	2/3	428/729	104/243	16/81	0
1	515/729	2/3	44/81	8/27	0
2	191/243	59/81	2/3	4/9	0
3	73/81	23/27	7/9	2/3	0
4	1	1	1	1	—

Alice tied or behind: she wins the round with $2/3$. $\text{rec}(0,0) = 2/3 \cdot \text{rec}(1,0) + 1/3 \cdot \text{rec}(0,1) = 2/3 \cdot 515/729 + 1/3 \cdot 428/729 = 2/3$

Solution

Form 1 (i → end) · **Turn** = one round. State (a, b) = the score. Someone reaches 4 → ends. Scores only go up, so no loops: plain recursion.

1. **LEVEL.** $\text{rec}(i, a, b) = P(\text{Alice wins the match})$, given that before round i Alice has won a rounds and Bob b . The answer is $\text{rec}(1, 0, 0)$.

Why: the score decides the round's probabilities and who has won.

2. **CHOICE.** Alice wins round i with $w = \frac{2}{3}$ if $a \leq b$ (tied, or Alice behind) and $w = \frac{1}{3}$ if $a > b$ (Bob behind, so Bob has $\frac{2}{3}$); Bob wins it with $1 - w$.

Why: the rules.

3. **CHECK.** $a = 4$: ends, value 1. $b = 4$: ends, value 0. Otherwise continue to $(i+1, a+1, b)$ or $(i+1, a, b+1)$.

Why: the match is over at 4 wins.

4. **MOVE.** $\text{rec}(i, a, b) = w \text{rec}(i+1, a+1, b) + (1 - w) \text{rec}(i+1, a, b+1)$.

Why: law of total probability over round i .

5. **MEMORYLESS.** $\text{rec}(i+1, a, b) = \text{rec}(i, a, b) = \text{rec}(a, b)$. (Here $i = a + b + 1$ anyway.)

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. SOLVE.

```
double dp[5][5]; // all -1
double rec(ll a, ll b) {
    if (a == 4) return 1; // base case: Alice has 4
    if (b == 4) return 0; // base case: Bob has 4
    if (dp[a][b] != -1) return dp[a][b]; // cache check
    double w = (a <= b) ? 2.0 / 3 : 1.0 / 3;
    double ans = w * rec(a + 1, b) + (1 - w) * rec(a, b + 1);
    return dp[a][b] = ans; // save and return
} // rec(0, 0) cost: 16 states x 2 transitions
```

Filled exactly, from $a + b = 6$ back to $(0, 0)$:

a \ b	0	1	2	3
0	2/3	428/729	104/243	16/81
1	515/729	2/3	44/81	8/27
2	191/243	59/81	2/3	4/9
3	73/81	23/27	7/9	2/3

So $\text{rec}(0, 0) = \frac{2}{3}$.

Why: e.g. $\text{rec}(3, 3) = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{2}{3}$; $\text{rec}(3, 2) = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot \text{rec}(3, 3) = \frac{1}{3} + \frac{4}{9} = \frac{7}{9}$ (Alice leads, so she wins the round with $\frac{1}{3}$).

Answer. Alice wins the match with probability $\frac{2}{3}$. ■

△ Course's statement contradicts itself: it says both “trailing player wins w.p. $\frac{2}{3}$ ” and “leader wins w.p. $\frac{2}{3}$ ”. Its recurrence uses the trailing rule, as here.

13 · Symmetry I: ad hoc

1. If the process treats labels interchangeably, every label (pair, ordering, ...) has the same probability. If exactly one of k such cases happens, each has probability $1/k$.
2. For i.i.d. continuous variables, ties have probability 0, and all $n!$ orderings are equally likely.

Marbles 3 and 98

Marbles numbered 1 to 100 are in a bag. Each move: draw 2 marbles at random, throw one of them away (using any rule that does not depend on the numbers written on them), and put the other back. This continues until one marble is left, which takes 99 moves.

Question: (a) what is the probability that marbles 3 and 98 are the two marbles drawn on move 50? (b) What is the probability that 3 and 98 are drawn together on some move?

Key idea

On any fixed move, every one of the $\binom{100}{2}$ label pairs is equally likely to be the pair drawn.

Picture

The rule never looks at labels, so all labels are treated alike. On any given move, each of the 4950

possible pairs is equally likely to be the one drawn. The 99 moves are 99 disjoint chances: 99/4950.

Run once

Run 20,000

Solution

Setup. 100 marbles labelled 1–100.

Each move draws 2 of the remaining marbles uniformly at random and discards one of them by a rule that does not look at the labels; the other goes back. Each move removes one marble, so there are 99 moves.

- (a) $P(\text{marbles 3 and 98 are the pair drawn on move 50})$.
- (b) $P(\text{they are drawn together on some move})$.

1. On any fixed move k , every pair of labels $\{i, j\}$ has the same probability of being the pair drawn.

Why: relabelling the marbles by any permutation doesn't change the law of the process: draws are uniform, and the discard rule ignores labels. So the pair $\{3, 98\}$ is no more or less likely than any other pair.

2. On move k exactly one pair is drawn, and there are $\binom{100}{2} = 4950$ possible pairs, so each has probability $\frac{1}{4950}$. (a) $\frac{1}{4950} \approx 0.00020$.

Why: the probabilities of the 4950 equally likely pairs add up to 1.

3. The events 'drawn together on move k ', $k = 1, \dots, 99$, are disjoint.

Why: once 3 and 98 are drawn together, one of them is discarded, so they can never be drawn together again.

4. (b) $P = 99 \cdot \frac{1}{4950} = \frac{1}{50} = 0.02.$

Why: add the probabilities of the disjoint events (steps 2 and 3).

Conclusion. $\frac{1}{4950}$ and $\frac{1}{50}.$ ■

△ Course doesn't state that the discard rule must ignore the labels (otherwise the symmetry in step 1 fails).

Growing die

A die has 99 faces showing $1, 2, \dots, 99$. You roll it, and then you add one more face to the die showing the number you just rolled. You repeat: roll (now with 100 faces), add a face with the result, and so on.

Question: what is the probability that the 99th roll shows 99?

Key idea

Swapping two labels everywhere leaves the process unchanged.

Picture

Relabel. Swap the names of two numbers everywhere: the game looks exactly the same. So no number is special, and on roll 99 each of the 99 values is equally likely.

Run the growing die many times and look at roll 99: every value 1–99 comes up about equally often.

Run 20 000 times

Solution

Setup. A die starts with 99 faces showing $1, \dots, 99$. Each roll is uniform over the current faces, and after each roll a new face with the value just rolled is added.

1. Swapping two values v and w everywhere (on the starting faces and in every roll) turns each possible history into another history with the same probability.

Why: the starting die has one face of each value, and the growth rule treats all values alike.

2. So on roll 99 every value is equally likely. There are 99 values, so $P(\text{roll 99 shows } 99) = \frac{1}{99} \approx 0.01010$.

Why: by step 1, the probabilities of the 99 values are equal, and they add up to 1.

Conclusion. Every value is equally likely on every roll, so the answer is $\frac{1}{99}$. ■

X_6 smaller than X_5 and X_7

X_5, X_6, X_7 are independent random numbers from the same continuous distribution.

Question: what is the probability that X_6 is smaller than both X_5 and X_7 ?

Key idea

Each of the three is equally likely to be the minimum.

Picture

Each of the three is equally likely to be the **smallest** (they are interchangeable, and ties don't happen). So $\frac{1}{3}$.

Solution

Setup. X_5, X_6, X_7 are i.i.d. continuous random variables.

1. Ties happen with probability 0.

Why: continuous variables.

2. Each of the three is equally likely to be the smallest.

Why: they are i.i.d., so relabelling them doesn't change the joint law.

3. Exactly one is the smallest, so

$$P(X_6 \text{ is the smallest}) = \frac{1}{3}.$$

Why: three equal probabilities adding to 1.

Conclusion. $\frac{1}{3}$. ■

More HT than TH in 10 tosses

A fair coin is tossed 10 times. Count the places where an H is immediately followed by a T (an 'HT') and the places where a T is immediately followed by an H (a 'TH').

Question: what is the probability that there are more HTs than THs?

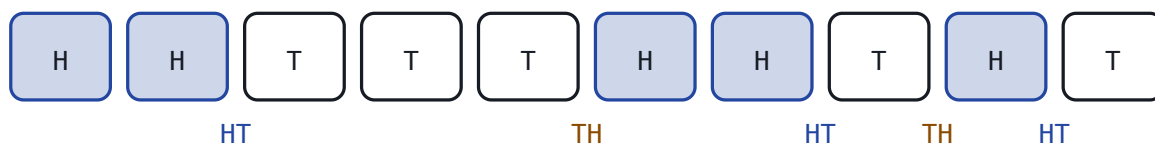
Key idea

Switches alternate between $H \rightarrow T$ and $T \rightarrow H$, so $\#HT - \#TH = [\text{first} = H] - [\text{last} = H]$.

Picture

Switches alternate: HT, TH, HT, ... So the two counts differ by at most 1, and the difference is decided by the first and last tosses alone. More HT means: starts with H and ends with T.

Click tosses to flip them. Runs of equal letters are separated by switches, which alternate HT, TH, HT, ... So $\#HT - \#TH$ depends only on the first and last toss.



$\#HT = 3$, $\#TH = 2$, difference **1** = [first is H] - [last is H] = $1 - 0$.

Solution

Setup. 10 fair tosses. $\#HT$ = the number of positions i where toss i is H and toss $i+1$ is T; $\#TH$ likewise.

1. Split the sequence into maximal runs of equal letters (e.g. HH | TTT | HH | T). At each boundary between two runs there is a switch, which is either HT or TH.

Why: a boundary is exactly where the letter changes.



switches alternate HT, TH, HT, ...

$$\#HT - \#TH = 1 \Leftrightarrow \text{starts H and ends T: } \frac{1}{2} \cdot \frac{1}{2}$$

2. The switches alternate between HT and TH.

Why: after an HT switch you are in a T-run, so the next switch must be TH, and vice versa.

3. So $\#HT - \#TH$ is $+1$ if the sequence starts with H and ends with T, -1 if it starts with T and ends with H, and 0 otherwise.

Why: an alternating list starting with HT and ending with HT has one more HT; if it starts and ends with different kinds, the counts are equal.

4. $\#HT > \#TH$ exactly when toss 1 is H and toss 10 is T: probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Why: step 3; the two tosses are independent. (Brute-force checked.)

Conclusion. More HT than TH happens with probability $\frac{1}{4}$. ■

△ Course claims “equal $\iff$ first = last” without proof. Steps 1–3 are the reason.

$n + 1$ coins vs n coins

Alice tosses $n + 1$ fair coins and Bob tosses n fair coins.

Question: what is the probability that Alice gets strictly more heads than Bob?

Key idea

Compare the first n coins of each. Alice's extra coin only matters when they are tied.

Picture

Compare the first n coins of each. Alice ahead or Bob ahead are equally likely. If they are tied, Alice's extra coin decides it, 50–50. So it is exactly $\frac{1}{2}$.

Run once

Run 100,000

Solution

Setup. Alice tosses $n + 1$ fair coins and Bob tosses n . Let A = Alice's heads among her first n coins, B = Bob's heads, and $E \in \{0, 1\}$ = Alice's extra coin.

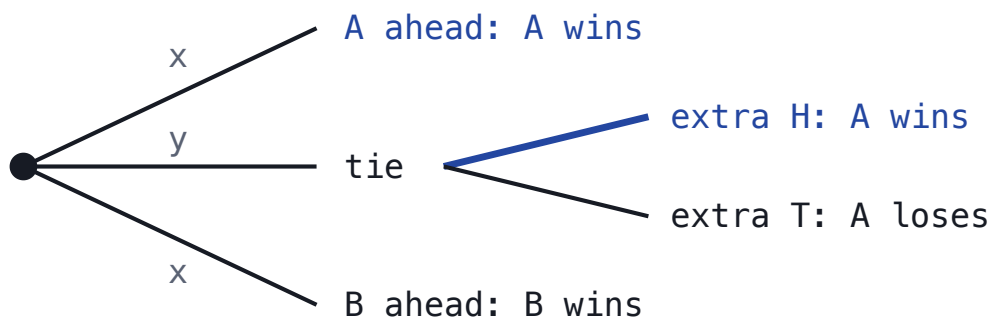
1. $P(A > B) = P(B > A) =: x$, and let $y = P(A = B)$.
Then $2x + y = 1$.

Why: A and B have the same distribution and are independent, so swapping them doesn't change probabilities; the three cases cover everything.

2. Alice (total $A + E$) has strictly more heads than Bob exactly when $A > B$, or $A = B$ and $E = 1$.

Why: if $A > B$ then $A + E \geq A > B$. If $A = B$, she needs the extra head. If $A < B$, then $A + E \leq A + 1 \leq B$.

first n coins each



$$P(\text{A wins}) = x + y/2 = (2x + y)/2 = \frac{1}{2}$$

3. $P = x + \frac{1}{2}y = \frac{2x+y}{2} = \frac{1}{2}$.

Why: E is independent of A, B with $P(E = 1) = \frac{1}{2}$; then use step 1.

Conclusion. Alice has strictly more heads with probability exactly $\frac{1}{2}$. ■

14 · Symmetry II: fewer states

Merge the states that a symmetry of the problem maps onto each other. They have equal values, so there are fewer unknowns.

Tetrahedron walk

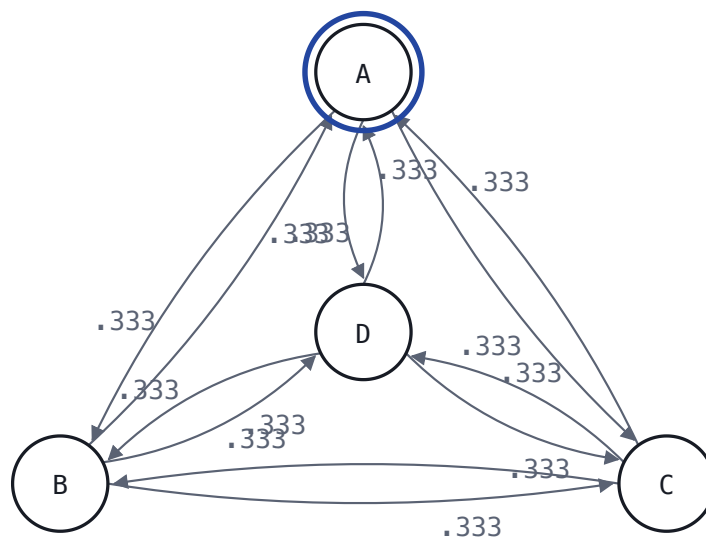
A tetrahedron has 4 corners, and every corner is joined by an edge to each of the other 3. A bug starts at corner A. At each move it walks along one of the 3 edges at its corner, each with probability $\frac{1}{3}$.

Question: what is the probability that the bug is back at A after exactly 6 moves?

Picture

Merge the three non-A corners into one state "N". From A you always go to N; from N you return to A with $\frac{1}{3}$. Follow the two numbers for 6 moves.

Press Step 6 times: the A bar reads $61/243 \approx 0.251$.



$P(X_n = j)$ for $n = 0$

0

0

0

Step

Run until stuck

Reset

click a state to start there

Step 0: at A.

Solution

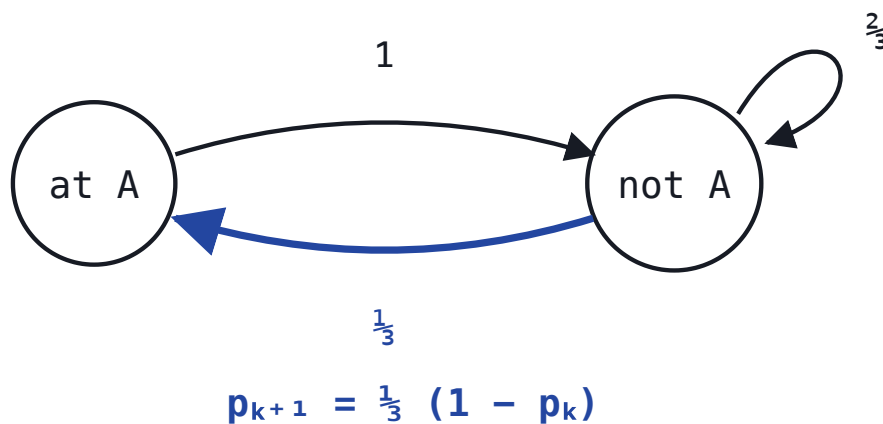
Form 1 ($i \rightarrow \text{end}$). **Turn** = one move. State s = at A, or at one of the other 3 corners (N). Fixed deadline (6 moves), so MEMORYLESS keeps i .

1. **LEVEL.** $\text{rec}(i, s) = P(\text{the bug is at A after move } 6), \text{ given that after } i \text{ moves it is at } s \in \{A, N\}.$ The answer is $\text{rec}(0, A).$

Why: the 3 non-A corners behave alike: each has 1 edge to A and 2 to the other non-A corners.

2. **CHOICE.** From A: all 3 edges go to N. From N: 1 of 3 edges goes to A ($\frac{1}{3}$), 2 go to N ($\frac{2}{3}$).

Why: every corner is joined to the other 3.



3. **CHECK.** $i = 6$: the walk is over, value 1 if $s = A$, else 0. $i < 6$: continue to $(i+1, s')$.

Why: the question asks about the position after exactly 6 moves.

4. **MOVE.** $\text{rec}(i, A) = \text{rec}(i+1, N).$ $\text{rec}(i, N) = \frac{1}{3} \text{rec}(i+1, A) + \frac{2}{3} \text{rec}(i+1, N).$

Why: law of total probability over the move.

5. **MEMORYLESS (check it).** No: 'after exactly 6 moves' is a fixed deadline, so keep i . i only grows, so this is plain recursion down to $i = 6$.

Why: a fixed deadline makes the turn number part of the state.

6. **SOLVE.** Start at the deadline and apply MOVE once per row:

i	rec(i, A)	rec(i, N)
6	1	0
5	0	$\frac{1}{3}$
4	$\frac{1}{3}$	$\frac{2}{9}$
3	$\frac{2}{9}$	$\frac{7}{27}$
2	$\frac{7}{27}$	$\frac{20}{81}$
1	$\frac{20}{81}$	$\frac{61}{243}$
0	$\frac{61}{243}$	$\frac{182}{729}$

So $\text{rec}(0, A) = \frac{61}{243} \approx 0.25103$.

Why: e.g. $\text{rec}(3, N) = \frac{1}{3} \cdot \text{rec}(4, A) + \frac{2}{3} \cdot \text{rec}(4, N) = \frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2}{9} = \frac{1}{9} + \frac{4}{27} = \frac{7}{27}$; and $\text{rec}(i, A)$ is always $\text{rec}(i+1, N)$ from the row above.

Answer. The bug is back at A after 6 moves with probability $\frac{61}{243}$. ■

Mouse on a cube

A mouse walks on the 8 corners of a cube. Label the corners by coordinates (x, y, z) , each coordinate 0 or 1.

At each move the mouse goes along one of the 3 edges at its corner, chosen at random. It starts at $(0, 0, 0)$.

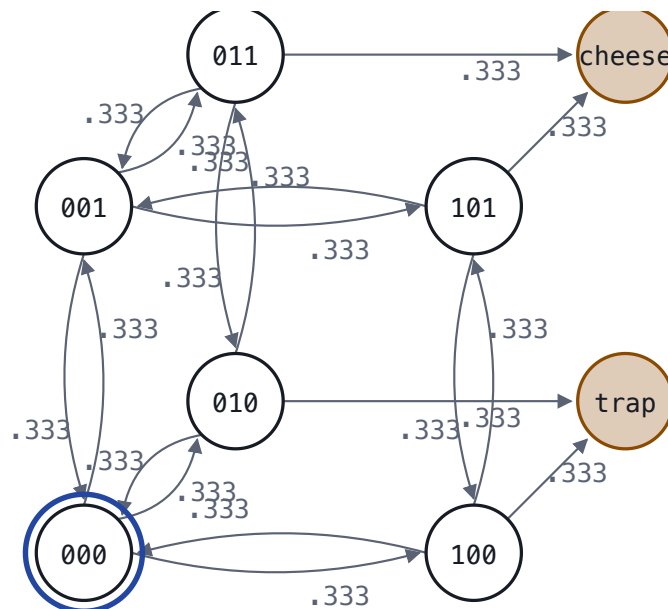
A trap is at $(1, 1, 0)$ and cheese is at $(1, 1, 1)$; the walk stops at whichever it reaches first.

Question: what is the probability that the mouse reaches the cheese before the trap?

Picture

Swapping x and y leaves everything the same (start, trap and cheese), so corners $100/010$ share a value, as do $101/011$. That leaves 4 unknowns and 4 one-step equations.

The cube with trap and cheese shaded. "Run 10 000": cheese first about $3/7 = 0.429$.



Step

Run until stuck

Reset

Run 10 000

click a state to start there

Step 0: at 000.

Solution

Form 1 ($i \rightarrow \text{end}$) · Turn = one move. State v = the corner. Trap $\rightarrow$ ends (0). Cheese $\rightarrow$ ends (1). Otherwise next turn.

- LEVEL.** $\text{rec}(i, v) = P(\text{cheese before trap})$, given the mouse is at corner $v = (x, y, z)$ before move

i. The answer is $\text{rec}(1, 000)$.

Why: only the current corner matters.

2. **CHOICE.** Flip one of the 3 coordinates, each $\frac{1}{3}$.

Why: a move goes along one of the 3 edges.

3. **CHECK.** Land on 110 (trap): ends, value 0. Land on 111 (cheese): ends, value 1. Otherwise continue to $(i+1, \text{new corner})$.

Why: the rules.

4. **MOVE.** $\text{rec}(i, v) = \frac{1}{3} \sum_{u \text{ next to } v} \text{rec}(i+1, u)$, with trap = 0 and cheese = 1.

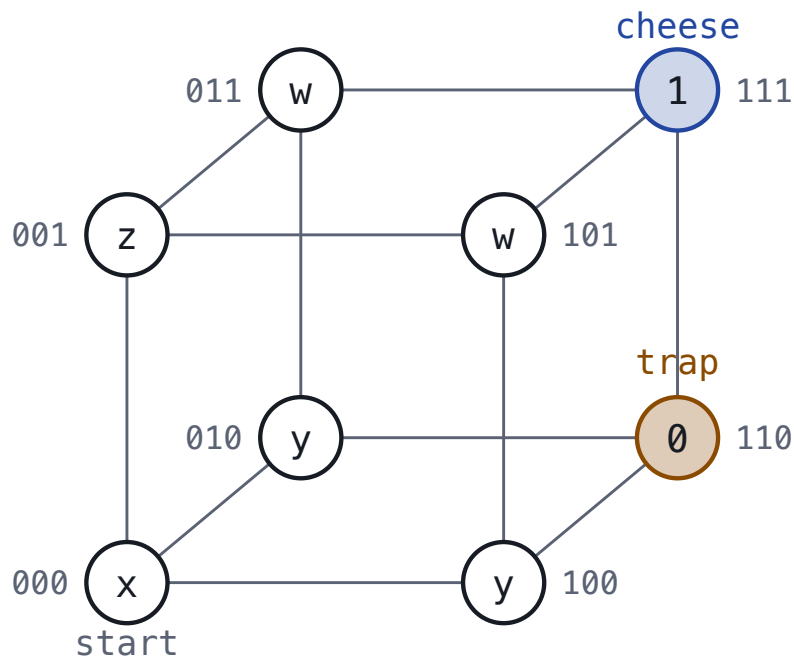
Why: law of total probability over the move.

5. **MEMORYLESS.** $\text{rec}(i+1, v) = \text{rec}(i, v) = \text{rec}(v)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **FEWER STATES.** Swapping x and y fixes the start, the trap and the cheese, so $\text{rec}(100) = \text{rec}(010)$ and $\text{rec}(101) = \text{rec}(011)$. Name $x = \text{rec}(000)$, $y = \text{rec}(100) = \text{rec}(010)$, $z = \text{rec}(001)$, $w = \text{rec}(101) = \text{rec}(011)$.

Why: a symmetry of the whole problem gives equal answers at matched corners.



7. SOLVE. MOVE at each corner:

$$3x = 2y + z \quad (\text{neighbours } 100, 010, 001)$$

$$3y = x + 0 + w \quad (000, \text{trap}, 101)$$

$$3z = x + 2w \quad (000, 101, 011)$$

$$3w = y + 1 + z \quad (100, \text{cheese}, 001)$$

Put $y = \frac{x+w}{3}$, $z = \frac{x+2w}{3}$ into $3x = 2y + z$: $9x = 2x + 2w + x + 2w = 3x + 4w$, so $w = \frac{3}{2}x$.

Into $3w = y + z + 1$: $9w = 2x + 3w + 3$, so $6w = 2x + 3$.

With $w = \frac{3}{2}x$: $9x = 2x + 3$, so $x = \frac{3}{7} \approx 0.42857$.

Why: multiply out; each line uses the one before.

8. **CHECK THE ANSWER.** $w = \frac{9}{14}$, $y = \frac{5}{14}$, $z = \frac{4}{7}$: $\frac{y+1+z}{3} = \frac{5/14+14/14+8/14}{3} = \frac{9}{14} = w$ ✓.

Why: substitute back.

Answer. The mouse reaches the cheese first with probability $\frac{3}{7}$. ■

15 · Symmetry III: continuous → discrete

If only the relative order of i.i.d. continuous points matters, replace them with a uniformly random ordering and count orderings.

Two random segments on a line

Four random points are chosen independently from the same continuous distribution on a line. The first two points are the ends of segment 1, and the last two are the ends of segment 2.

Question: what is the probability that the two segments overlap?

Key idea

Read the sorted points' owners as a word such as 1122. The 6 words are equally likely, and only 1122 and 2211 are disjoint.

Picture

Only the order of the 4 points matters. Write which segment each sorted point belongs to: one of 6 equally likely words like 1212. Only 1122 and 2211 are disjoint.

Run once

Run 100,000

Solution

Setup. Four i.i.d. points from a continuous distribution. Segment 1 joins points 1 and 2; segment 2 joins points 3 and 4.

1. Ties have probability 0, and all $4! = 24$ orders of the 4 points are equally likely.

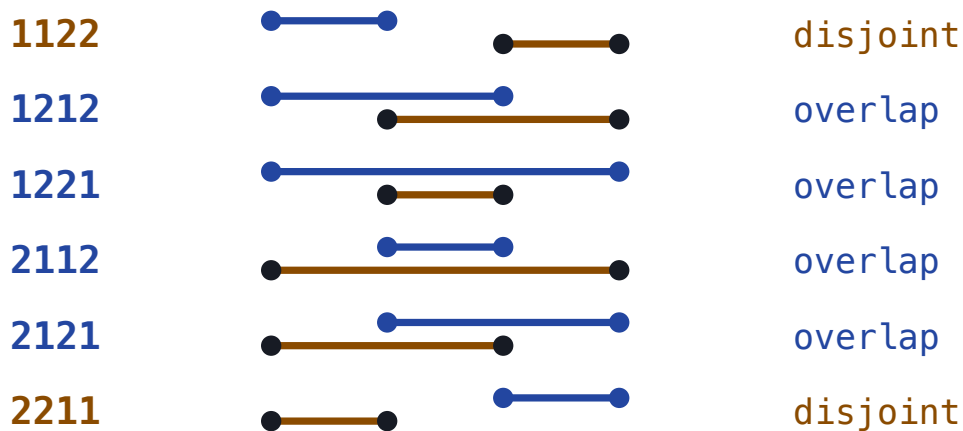
Why: i.i.d. continuous points are exchangeable.

2. Sort the points and write which segment owns each position: a word with two 1s and two 2s, such as 1212. There are $\binom{4}{2} = 6$ words, and each comes from $2! \cdot 2! = 4$ of the 24 orders, so each word has probability $\frac{1}{6}$.

Why: swapping the two points of a segment doesn't change the word.

3. The segments are disjoint exactly for the words 1122 and 2211.

Why: disjoint means both points of one segment are to the left of both points of the other. In every other word the points interleave (1212, 2121) or one segment sits inside the other (1221, 2112).



6 equally likely words, 2 disjoint $\Rightarrow P(\text{overlap}) = 4/6$

4. $P(\text{overlap}) = 1 - \frac{2}{6} = \frac{2}{3}$.

Why: complement of 2 of the 6 equally likely words.

Conclusion. The segments overlap with probability $\frac{2}{3}$, whatever the continuous distribution. ■

Two random chords

Four points are chosen independently and uniformly on a circle. Points 1 and 2 are joined by a chord (a straight segment), and points 3 and 4 by another chord.

Question: what is the probability that the two chords cross?

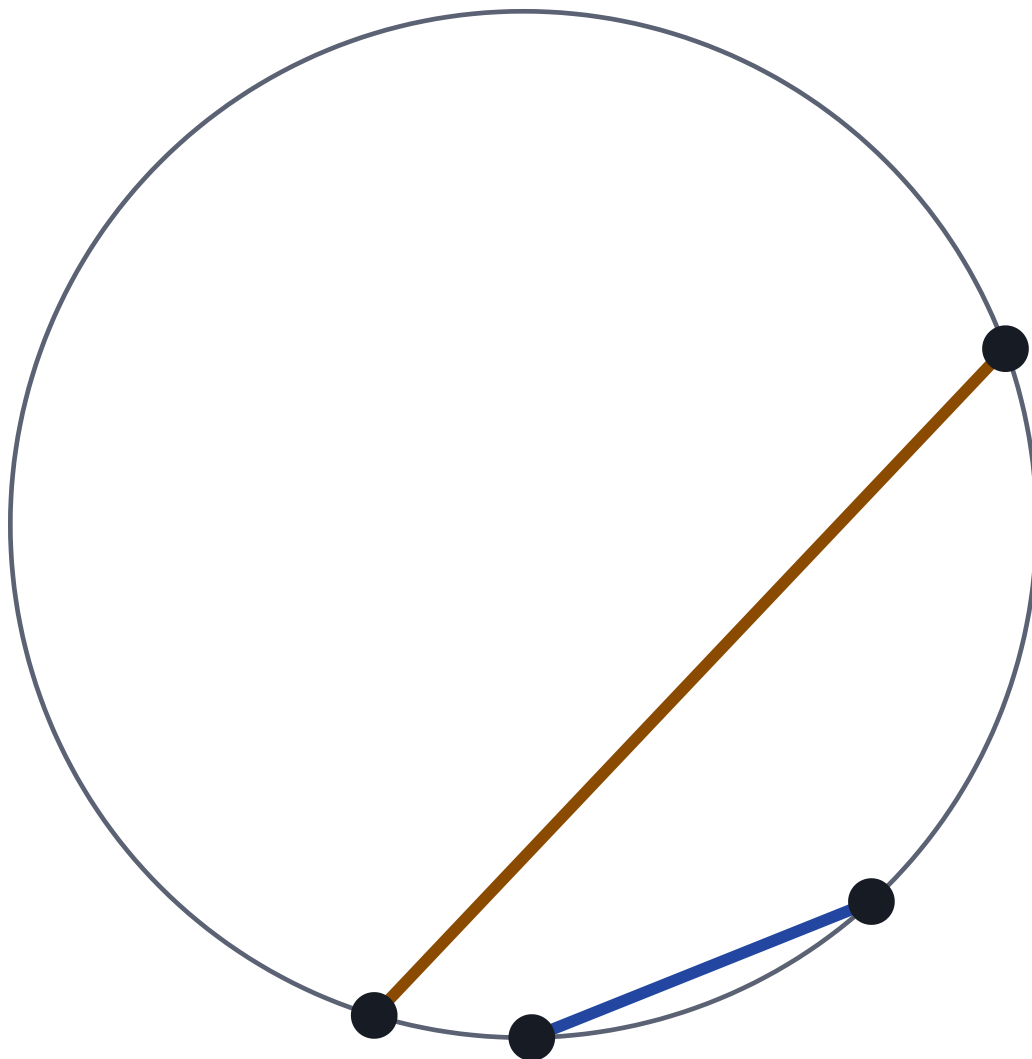
Key idea

Four points on a circle can be split into two pairs in 3 ways, and exactly one of those ways crosses.

Picture

Four points on a circle can be paired up in 3 ways, all equally likely here. Two pairings are opposite sides of the quadrilateral; only the diagonals cross.

Four random points on a circle, joined 1–2 and 3–4. Of the 3 ways to pair up 4 points, only one crosses.

[New chords](#)[Try 100 000](#)

They don't cross.

Solution

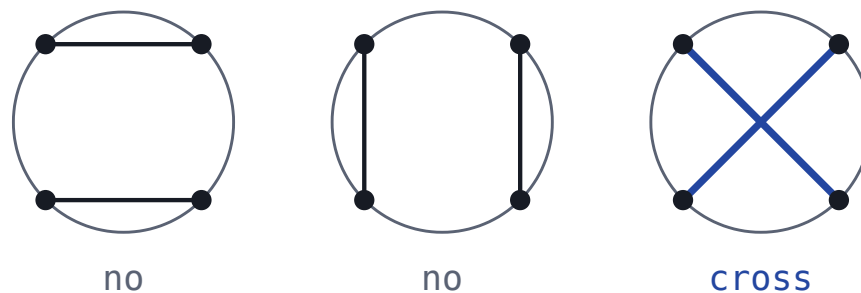
Setup. Four i.i.d. uniform points on a circle; chord 1 joins points 1 and 2, chord 2 joins points 3 and 4.

1. The 4 points are distinct with probability 1.
Given their positions, the way they are split into the two chords is uniform over the 3 possible pairings.

Why: the 24 ways to attach the labels 1–4 to the positions are equally likely, and each pairing arises from 8 of them.

2. Name the positions P_1, P_2, P_3, P_4 in order around the circle. The pairings $\{P_1P_2, P_3P_4\}$ and $\{P_1P_4, P_2P_3\}$ are pairs of opposite sides of the quadrilateral and don't cross. The pairing $\{P_1P_3, P_2P_4\}$ (the diagonals) crosses.

Why: see the figure below.



3. $P = \frac{1}{3}$.

Why: 1 of the 3 equally likely pairings crosses.

Conclusion. The chords cross with probability $\frac{1}{3}$. ■

Goalkeeper interval

A goalkeeper guards the interval between two random points x and y , chosen independently and uniformly on $[0, 1]$. Three shots land at independent uniform points on $[0, 1]$.

Question: what is the probability that exactly 2 of the 3 shots land inside the keeper's interval?

Key idea

5 i.i.d. points. The positions of the keeper's 2 points among the 5 sorted points are uniform over $\binom{5}{2} = 10$ choices.

Picture

5 random points, only their order matters. The keeper owns 2 of the 5 sorted positions, all 10 choices equally likely. Exactly 2 shots inside means the keeper's points are positions (1, 4) or (2, 5): 2 of 10.

Run once

Run 100,000

Solution

Setup. The keeper's two endpoints and the three shots are 5 i.i.d. uniform points on $[0,1]$. The keeper defends the interval between the two endpoints.

1. Sort the 5 points. The keeper's two points occupy sorted positions $i < j$, and all $\binom{5}{2} = 10$ position pairs are equally likely.

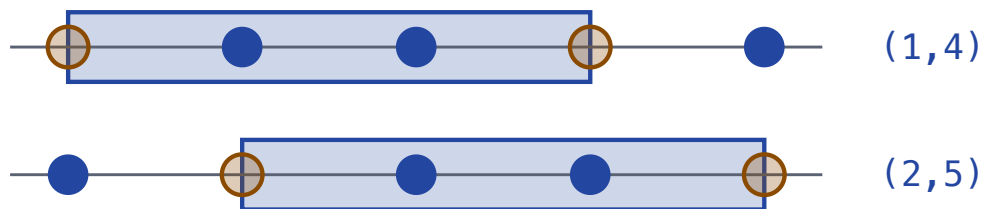
Why: all $5!$ orders are equally likely, and each position pair arises from $2! \cdot 3! = 12$ of the 120 orders.

2. The shots inside the interval are exactly the points between positions i and j : there are $j - i - 1$ of them.

Why: all other points are keeper points or outside.

3. $j - i - 1 = 2$ exactly for $(i, j) = (1, 4)$ or $(2, 5)$.

Why: $j = i + 3$ with $j \leq 5$.



keeper (orange): $C(5,2) = 10$ placements
 exactly 2 shots inside: 2 of the 10

4. $P = \frac{2}{10} = \frac{1}{5}$.

Why: 2 of the 10 equally likely position pairs.

Conclusion. Exactly two shots land inside with probability $\frac{1}{5}$. ■

16 · Random variables

1. A random variable is a function $X: \Omega \rightarrow \mathbb{R}$.
 Example: the number of heads in 5 tosses.
2. **Discrete:** takes countably many values. **Continuous** (in this course): has a density (§19).

17 · PMF and CDF (discrete)

1. PMF: $p(x) = P(X = x)$. Support = $\{x : p(x) > 0\}$.
2. CDF: $F(x) = P(X \leq x)$. It is non-decreasing and right-continuous, runs from 0 to 1, and jumps by $p(x_0)$ at x_0 .
3. **Difference trick** for integer X : $P(X = k) = F(k) - F(k - 1)$. Use it whenever " $\leq k$ " is easy

(maxima!).

Sum of two dice

Two fair dice are rolled, and S is their total.

Question: find $P(S = s)$ for every s from 2 to 12.

Key idea

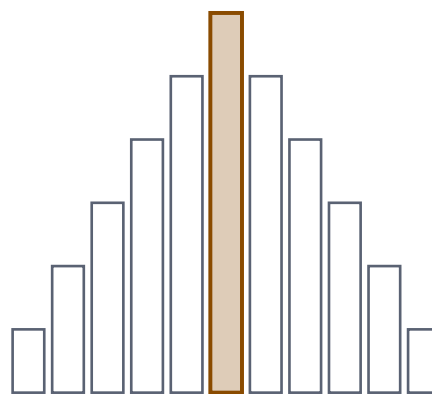
Count the pairs (i, j) with $i + j = s$.

Picture

Lay out all 36 pairs in a 6×6 grid. Pairs with the same total lie on a diagonal. The longest diagonal (total 7) has 6 cells; each step away from 7 loses one cell.

All 36 equally likely pairs. Slide the total s : its pairs lie on one diagonal, and diagonals get shorter away from 7.

2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12


 $P(S = s)$

total s  7

$s = 7$: 6 pairs $\rightarrow P = 6/36$.

Solution

Setup. Two fair dice; all 36 pairs (i, j) are equally likely. $S = i + j$.

1. For $2 \leq s \leq 7$: the pairs with $i + j = s$ are $i = 1, \dots, s - 1$, which is $s - 1$ pairs.

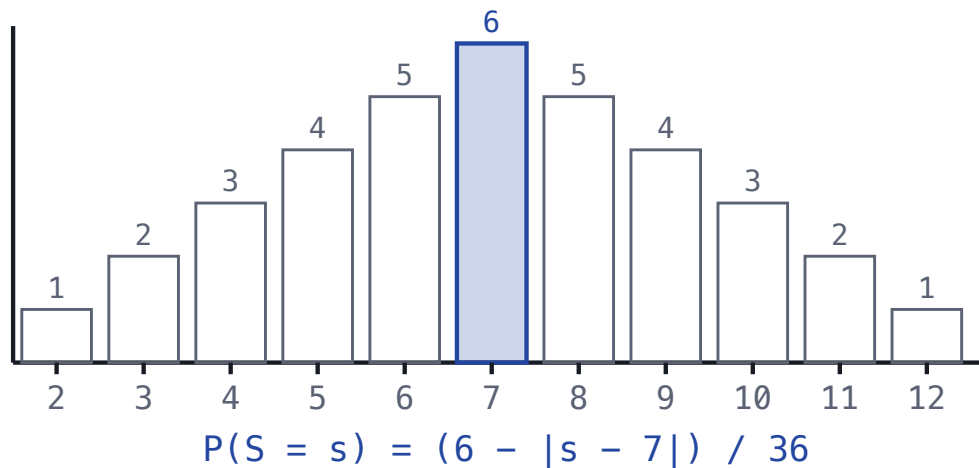
Why: then $j = s - i$ automatically lies between 1 and 6.

2. For $7 \leq s \leq 12$: i runs from $s - 6$ to 6, which is $13 - s$ pairs.

Why: we need $j = s - i \leq 6$.

3. Both cases equal $6 - |s - 7|$, so $P(S = s) = \frac{6 - |s - 7|}{36}$ for $s = 2, \dots, 12$.

Why: for $s \leq 7$: $6 - (7 - s) = s - 1$; for $s \geq 7$: $6 - (s - 7) = 13 - s$.



Conclusion. The distribution is the triangle pictured in step 3, peaking at 7. ■

CDF from a PMF

A random variable X equals 1 with probability 0.2, equals 3 with probability 0.5, and equals 5 with probability 0.3.

Question: write down its cumulative distribution function $F(x) = P(X \leq x)$ for every real number x .

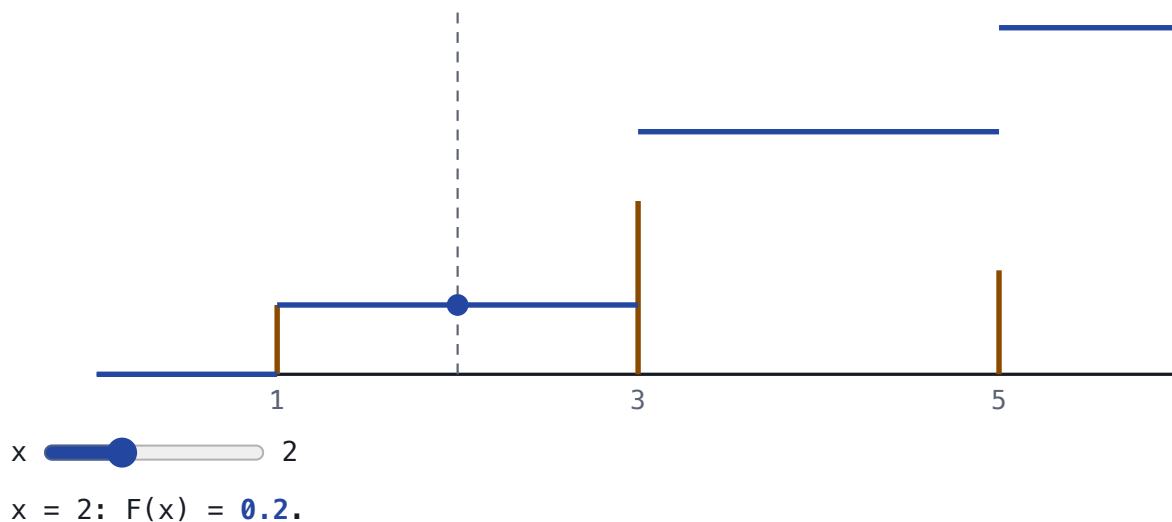
Key idea

Running sum, as a step function.

Picture

The CDF is a running total. Walk along the number line and add each probability as you pass its value: 0, then .2, then .7, then 1.

Spikes: the PMF. Staircase: the CDF $F(x)$ = total of the spikes at or left of x . Slide x .



Solution

Setup. $P(X = 1) = .2$, $P(X = 3) = .5$, $P(X = 5) = .3$.

$F(x) = P(X \leq x)$ = the sum of the probabilities of the values $\leq x$.

1. $x < 1$: no value is $\leq x$, so $F = 0$.

Why: definition.

2. $1 \leq x < 3$: only the value 1, so $F = .2$.

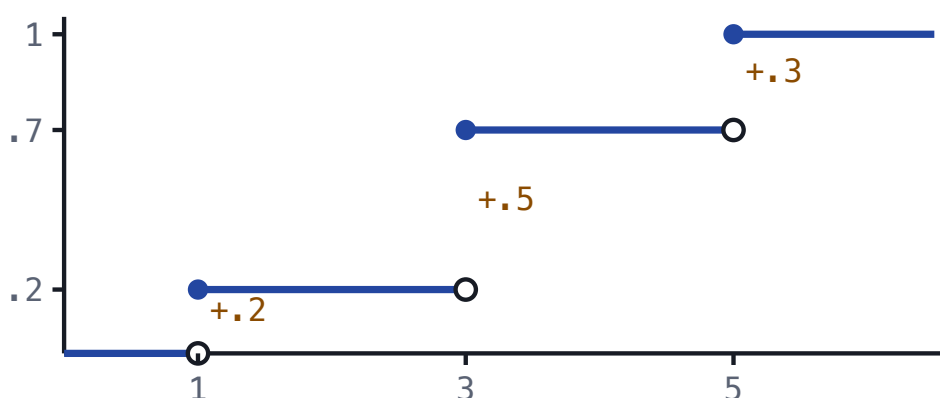
Why: definition.

3. $3 \leq x < 5$: the values 1 and 3, so $F = .2 + .5 = .7$.

Why: definition.

4. $x \geq 5$: all three values, so $F = 1$.

Why: definition.



jumps = the PMF values; filled dot = value at the jump

Conclusion. F is the step function pictured in step 4, jumping by the PMF value at each of 1, 3, 5. ■

Max of five d10

Five fair 10-sided dice (faces 1 to 10) are rolled, and X is the largest number shown.

Question: find $P(X = k)$ for each $k = 1, \dots, 10$.

Key idea

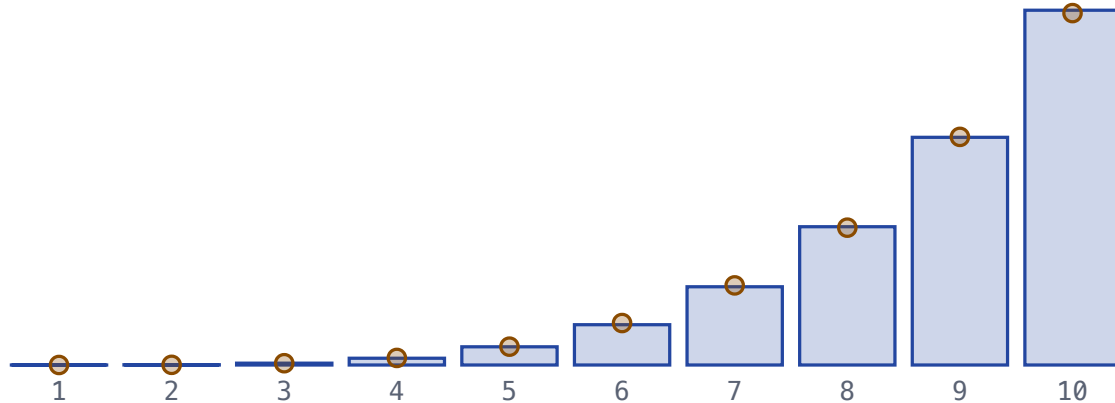
$\max \leq k \iff \text{every die} \leq k.$

Picture

"Max $\leq k$ " is easy: every die $\leq k$, probability $(k/10)^5$. Then "max = k " is "max $\leq k$ " minus "max $\leq k$

– 1".

The max is $\leq k$ exactly when all five dice are $\leq k$: $F(k) = (k/10)^5$. Each bar $P(X = k)$ is the step $F(k) - F(k - 1)$. Blue = formula, dots = simulation.



Simulate 50 000

$P(X = 10) = 1 - 0.9^5 = 0.4095$; simulated 0.4062.

Solution

Setup. X = the largest of 5 independent rolls of a fair 10-sided die (faces 1–10).

1. $X \leq k$ exactly when every die shows at most k .

Why: the maximum is at most k iff all of them are.

2. $F(k) = P(X \leq k) = \left(\frac{k}{10}\right)^5$.

Why: each die is $\leq k$ with probability $\frac{k}{10}$, independently.

3. $P(X = k) = F(k) - F(k - 1) = \left(\frac{k}{10}\right)^5 - \left(\frac{k-1}{10}\right)^5$, for $k = 1, \dots, 10$.

Why: $\{X \leq k\}$ is the disjoint union of $\{X \leq k - 1\}$ and $\{X = k\}$ (the difference trick).

Conclusion. $P(X = k) = (k/10)^5 - ((k - 1)/10)^5$. ■

18 · Common discrete distributions

Name	Models	$P(X = k)$
Uniform $\{x_1 \dots x_n\}$	equally likely values	$1/n$
Bernoulli(p)	one trial	$p^k(1-p)^{1-k}, k \in \{0, 1\}$
Geometric(p)	trials to the 1st success	$(1-p)^{k-1}p, k \geq 1$
Binomial(n, p)	successes in n trials	$\binom{n}{k}p^k(1-p)^{n-k}$
NegBin(r, p)	trials to the r -th success	$\binom{k-1}{r-1}p^r(1-p)^{k-r}, k \geq r$
Poisson(λ)	rare events at rate λ	$e^{-\lambda}\lambda^k/k!$
Hypergeometric	k successes in n draws, no replacement	$\binom{K}{k}\binom{N-K}{n-k}/\binom{N}{n}$

Why each formula holds: Geometric: $k-1$ failures, then a success. Binomial: choose which k trials succeed. NegBin: exactly $r-1$ successes in the first $k-1$ trials, then a success. Poisson: Binomial($n, \lambda/n$) as $n \rightarrow \infty$, because $\binom{n}{k}(\lambda/n)^k \rightarrow \lambda^k/k!$ and $(1 - \lambda/n)^{n-k} \rightarrow e^{-\lambda}$.

Identify the distribution

Question: for each situation, name the distribution and give $P(X = k)$.

- (a) X = the result of a fair die;
- (b) $X = 1$ if one flip of a coin with $P(\text{heads}) = 0.7$ is heads, else 0;
- (c) X = the number of items inspected up to and including the first good one, when each item is good with probability 0.9;
- (d) X = the number of heads in 5 fair flips;

(e) X = the number of shots needed to make 3 baskets, when each shot goes in with probability 0.6;

(f) X = the number of phone calls in a minute, when calls arrive at random at an average rate of 3 per minute;

(g) X = the number of red balls when 4 balls are drawn without replacement from 7 red and 5 blue.

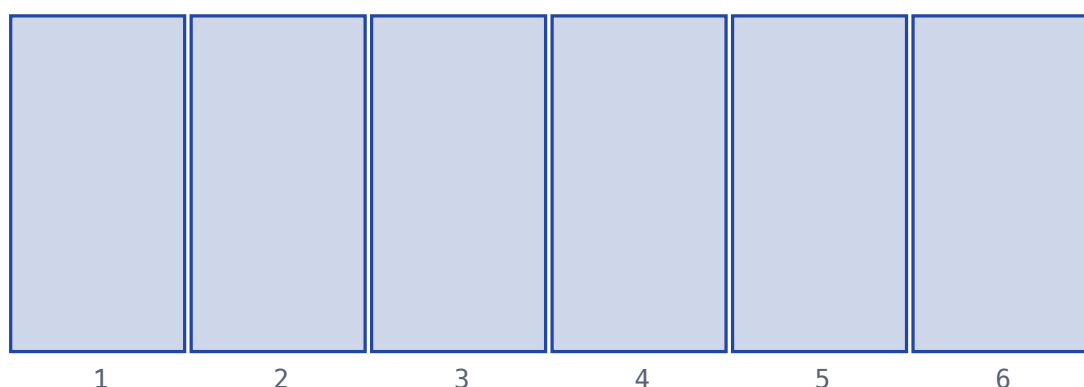
Key idea

Match the story to the table.

Picture

Match the story. One trial: Bernoulli. Count successes in fixed n : Binomial. Wait for the first success: Geometric; for the r -th: Negative Binomial. Random arrivals at a rate: Poisson. Draws without replacement: Hypergeometric.

Pick a story; the bars show its distribution $P(X = k)$.



a · Uniform die ✓

Mean = 3.500, total probability shown 1.000.

Solution

1. (a) A fair die: Uniform on $\{1, \dots, 6\}$, each value $\frac{1}{6}$.

Why: 6 equally likely faces.

2. (b) One flip with $P(H) = .7$: Bernoulli(.7).

Why: a single trial with success probability .7.

3. (c) The first good item, $p = .9$: Geometric, $P(X = k) = .1^{k-1} \cdot .9$.

Why: the first $k-1$ items are bad (probability .1 each) and item k is good.

4. (d) Heads in 5 fair flips: Binomial($5, \frac{1}{2}$).

Why: the number of successes in 5 independent trials.

5. (e) Shots until the 3rd make, $p = .6$: NegBin,
 $P(X = k) = \binom{k-1}{2} (.6)^3 (.4)^{k-3}$ for $k \geq 3$.

Why: exactly 2 makes among the first $k-1$ shots (choose which 2), then a make on shot k .

6. (f) Calls per minute with mean 3: Poisson(3).

Why: a count of rare independent events at rate 3.

7. (g) Reds in 4 draws from 7R + 5B without replacement: Hypergeometric, $P(X = k) = \frac{\binom{7}{k} \binom{5}{4-k}}{\binom{12}{4}}$.

Why: choose which k reds and which $4-k$ blues, out of all 4-subsets of the 12 balls.

Conclusion. Each story matches one row of the table.

■

19 • PDF and CDF (continuous)

1. X has density $f \geq 0$ if $P(a \leq X \leq b) = \int_a^b f$ for all $a \leq b$. Then $\int f = 1$, $P(X = x) = 0$ for every x , and f can be larger than 1.
2. $F(x) = \int_{-\infty}^x f$, and $F' = f$ wherever f is continuous.
3. To find a density, **find the CDF first, then differentiate**. This is usually easier than the " $f(x) dx = P(x \leq X \leq x + dx)$ " method.

Integrate a density

A continuous random variable with density f has $P(a \leq X \leq b) = \int_a^b f(x) dx$.

Question: (a) X has density $f(x) = \frac{3}{8}x^2$ for $0 \leq x \leq 2$ (and 0 elsewhere). Find $P(0.5 \leq X \leq 1.5)$. (b) X has density $f(x) = 3x^2$ for $0 \leq x \leq 1$. Find its CDF $F(x) = P(X \leq x)$.

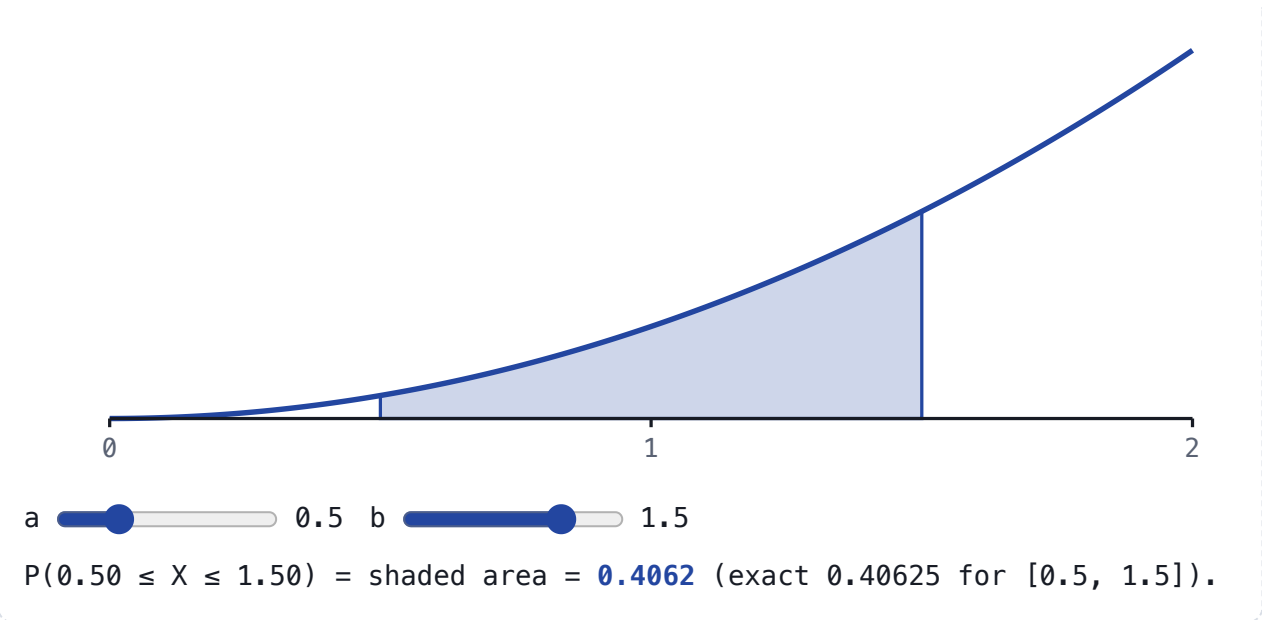
Key idea

The area under f .

Picture

Probability = area under the density. Shade the part between the two limits and integrate.

(a) $f(x) = 3x^2/8$ on $[0, 2]$. Move a and b .



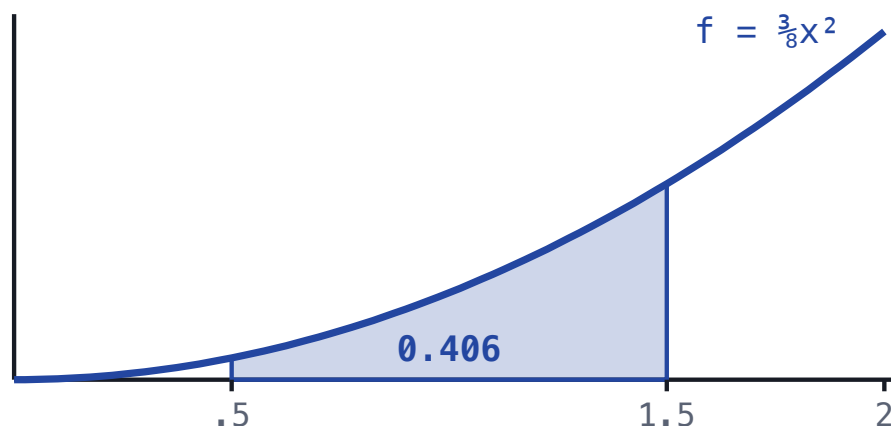
Solution

1. (a) $f(x) = \frac{3}{8}x^2$ on $[0, 2]$ is a density: $\int_0^2 \frac{3}{8}x^2 dx = \left. \frac{x^3}{8} \right|_0^2 = 1$.

Why: it is non-negative and integrates to 1.

2. $P(.5 \leq X \leq 1.5) = \int_{.5}^{1.5} \frac{3}{8}x^2 dx = \frac{1.5^3 - .5^3}{8} = \frac{3.375 - .125}{8} = 0.40625$.

Why: the probability of an interval is the area under the density.



3. (b) $f = 3x^2$ on $[0, 1]$: $F(x) = \int_0^x 3t^2 dt = x^3$ for $0 \leq x \leq 1$; $F = 0$ for $x < 0$ and $F = 1$ for $x > 1$.

Why: $F(x) = \int_{-\infty}^x f$, and $f = 0$ outside $[0, 1]$.

Conclusion. 0.40625, and $F(x) = x^3$ on $[0, 1]$. ■

Distance of a uniform point in the unit disc

A point is chosen uniformly at random inside a circle of radius 1. Let D be its distance from the centre.

Question: find the density of D .

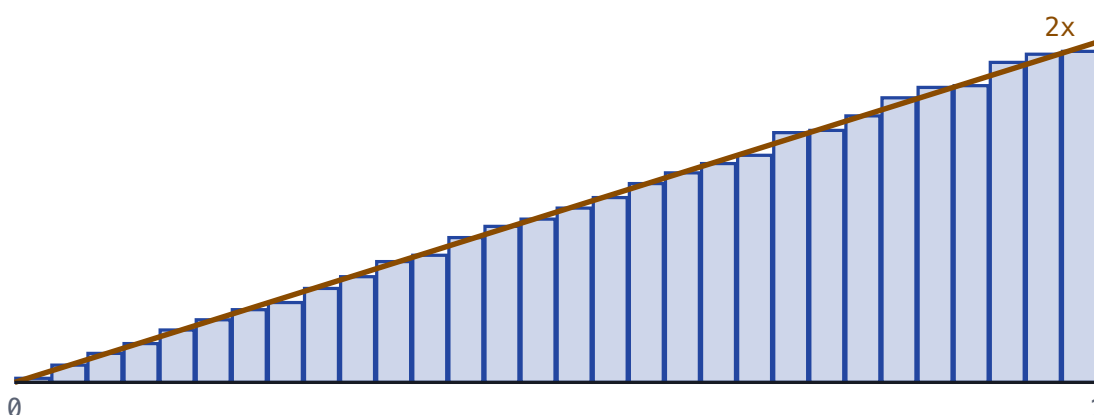
Key idea

$P(X \leq x)$ is the ratio of areas $\pi x^2 / \pi$.

Picture

Find $P(D \leq x)$ first: it is the small disc's share of the big disc, x^2 . Differentiate: density $2x$. Far-out distances are more likely because there is more room out there.

Distances of 50 000 random points in the disc.



Sample 50 000

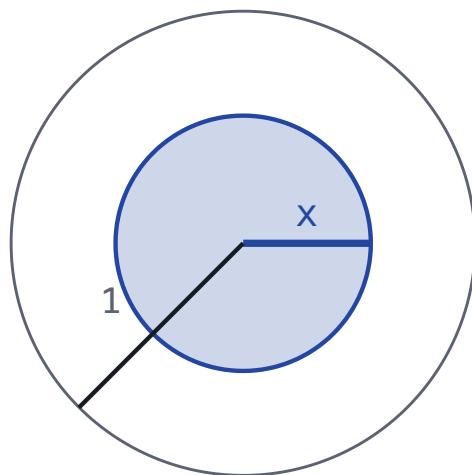
Blue bars: 50 000 samples. Orange: the formula. Sample mean **0.668** (exact $2/3$).

Solution

Setup. A point is uniform in the unit disc; D = its distance from the centre.

1. For $0 \leq x \leq 1$, $\{D \leq x\}$ is the disc of radius x , so $P(D \leq x) = \frac{\pi x^2}{\pi} = x^2$.

Why: for a uniform point, probability = area ratio.



$$P(\text{dist} \leq x) = \pi x^2 / \pi = x^2$$

$$f(x) = 2x$$

2. The density is $f(x) = F'(x) = 2x$ on $[0, 1]$.

Why: differentiate the CDF.

Conclusion. $f(x) = 2x$: points are more likely far from the centre, because there is more area there. ■

Minimum of 10 uniforms

Ten numbers are chosen independently and uniformly from $[0, 1]$.

Question: find the density of the smallest of them.

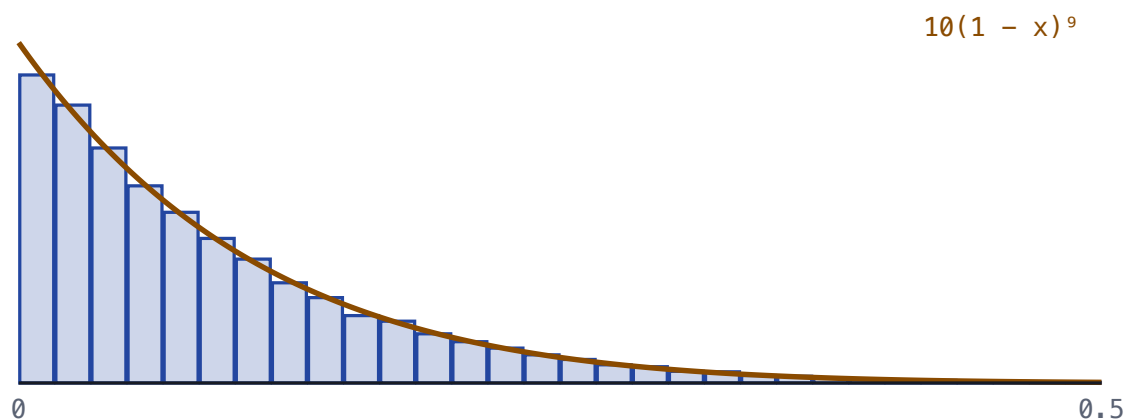
Key idea

$$\min > x \iff \text{all ten are } > x.$$

Picture

"Min > x" is easy: all ten > x, probability $(1 - x)^{10}$. So $P(\min \leq x) = 1 - (1 - x)^{10}$; differentiate.

The minimum of 10 random numbers, 50 000 times.



Sample 50 000

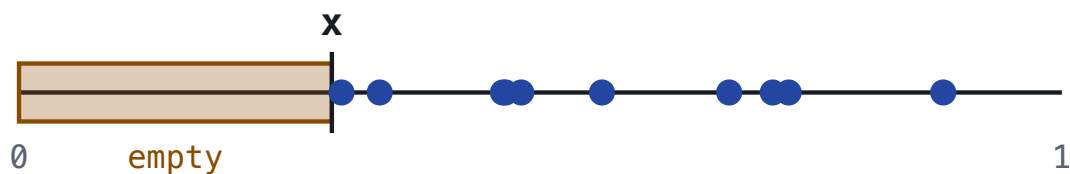
Blue bars: 50 000 samples. Orange: the formula. Sample mean **0.091** (exact $1/11 \approx 0.091$).

Solution

Setup. M = the smallest of 10 independent uniforms on $[0, 1]$.

1. $M > x$ exactly when all ten are $> x$, so $P(M > x) = (1 - x)^{10}$.

Why: each uniform is $> x$ with probability $1 - x$, independently.



$\min > x \Leftrightarrow$ all ten land right of x : $(1 - x)^{10}$

2. $F(x) = P(M \leq x) = 1 - (1 - x)^{10}$.

Why: complement.

3. $f(x) = F'(x) = 10(1 - x)^9$.

Why: chain rule.

Conclusion. The minimum has density $10(1-x)^9$. ■

20 · Common continuous distributions

Name	Models	density
$U[a, b]$	a uniform point	$\frac{1}{b-a}$ on $[a, b]$
$\text{Exp}(\lambda)$	wait for the 1st event at rate λ	$\lambda e^{-\lambda x}, x \geq 0$
$N(\mu, \sigma^2)$	sums of many small effects	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$
$\text{Gamma}(\alpha, \lambda)$	wait for the α -th event	$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$
$\text{Beta}(\alpha, \beta)$	an unknown proportion in $[0, 1]$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$

Exp: $P(T > t) = e^{-\lambda t}$. **Memoryless:**

$$P(T > s + t \mid T > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = P(T > t).$$

$\text{Gamma}(1, \lambda) = \text{Exp}(\lambda)$.

70% failure time

A machine's lifetime T (in years) is exponential with rate 0.1, so $P(T > t) = e^{-0.1t}$.

Question: by what time t have 70% of machines failed? That is, find t with $P(T < t) = 0.7$.

Key idea

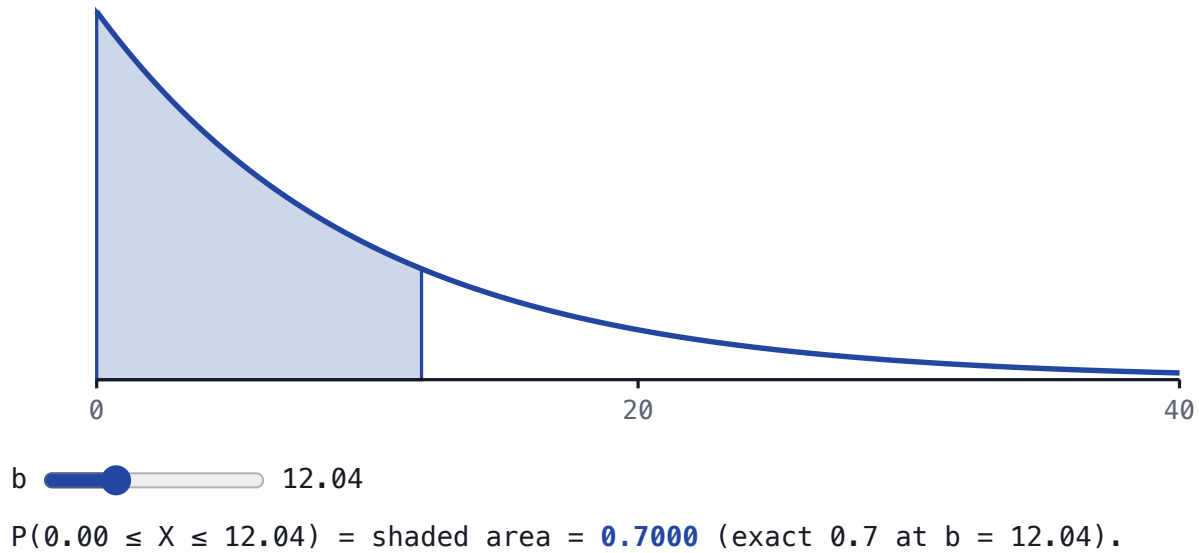
Invert the CDF $1 - e^{-\lambda t}$.

Picture

Undo the CDF. 70% failed means 30% still working:

$$e^{(-0.1t)} = 0.3, \text{ so } t = -10 \ln 0.3.$$

Exponential density with rate 0.1. Slide b until the shaded area is 0.7.



Solution

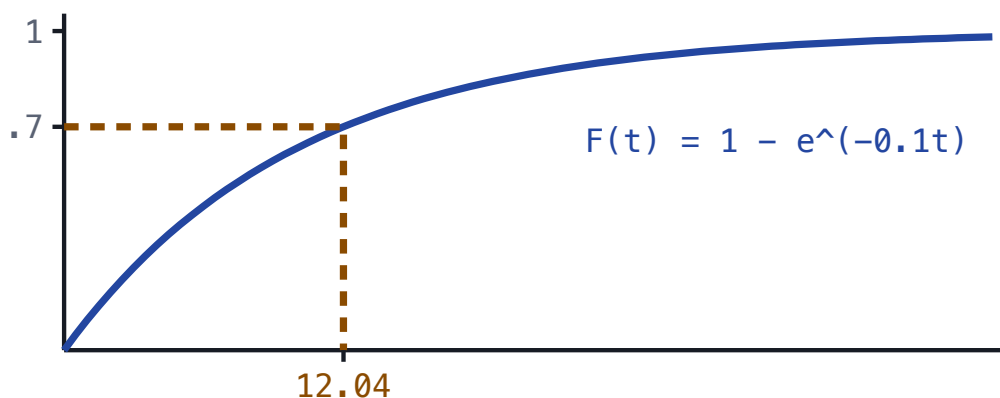
Setup. $T \sim \text{Exp}(0.1)$, so $P(T < t) = 1 - e^{-0.1t}$. We want t with $P(T < t) = 0.7$.

$$1. \quad 1 - e^{-0.1t} = 0.7 \iff e^{-0.1t} = 0.3.$$

Why: rearrange.

$$2. \quad -0.1t = \ln 0.3, \text{ so } t = -10 \ln 0.3 = 10 \ln \frac{10}{3} \approx 10 \cdot 1.20397 = 12.03973.$$

Why: take logs; $-\ln 0.3 = \ln \frac{1}{0.3} = \ln \frac{10}{3}$.



Conclusion. 70% of items have failed by about 12.04.

■

21 · The normal distribution

1. $\varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$ and $\Phi(x) = \int_{-\infty}^x \varphi$.
2. $\Phi(x) + \Phi(-x) = 1$. Since φ is even, $\Phi(-x) = \int_x^{\infty} \varphi = 1 - \Phi(x)$.
3. If $X \sim N(\mu, \sigma^2)$ and $a \neq 0$, then $aX + b \sim N(a\mu + b, a^2\sigma^2)$. So $Z = (X - \mu)/\sigma \sim N(0, 1)$.
4. The z-score $(x - \mu)/\sigma$ counts standard deviations from the mean. It lets you compare values from different normals.

$X \sim N(100, 36)$: $P(94 \leq X \leq 112)$

X is normally distributed with mean 100 and variance 36 (so standard deviation 6).

Question: find $P(94 \leq X \leq 112)$.

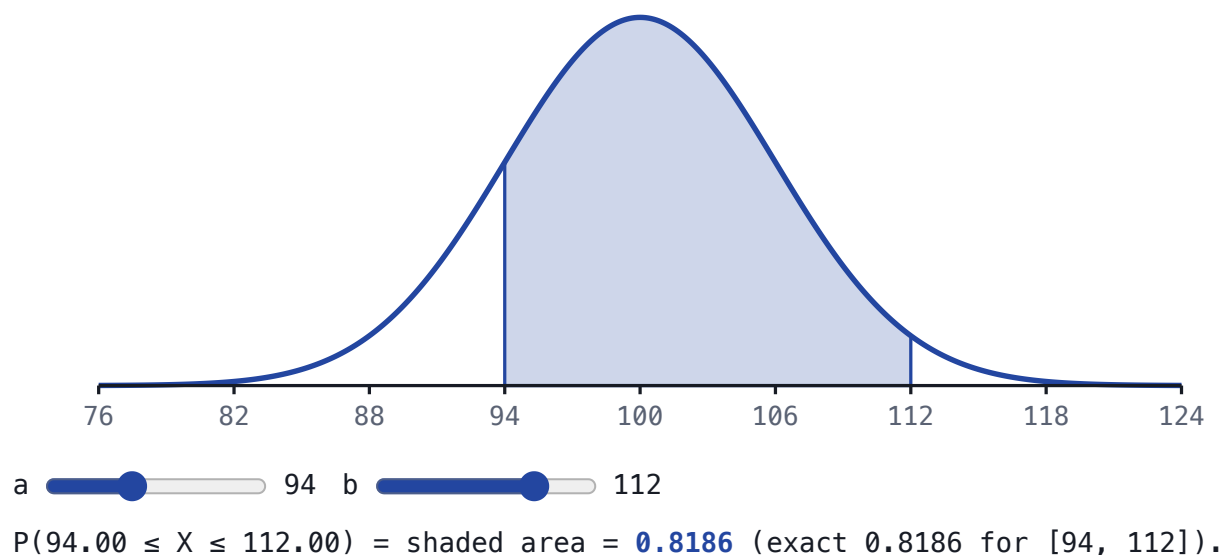
Key idea

Standardise with $\sigma = 6$.

Picture

Measure in standard deviations. 94 is one σ below the mean, 112 is two σ above. So the answer is $P(-1 \leq Z \leq 2)$ for a standard normal.

$N(100, 36)$. Ticks every 6 = one standard deviation.



Solution

Setup. $X \sim N(100, 36)$, so $\mu = 100$ and $\sigma = 6$. $Z = (X - 100)/6$ is standard normal.

$$1. \quad 94 \leq X \leq 112 \iff \frac{94-100}{6} \leq Z \leq \frac{112-100}{6} \iff -1 \leq Z \leq 2$$

.

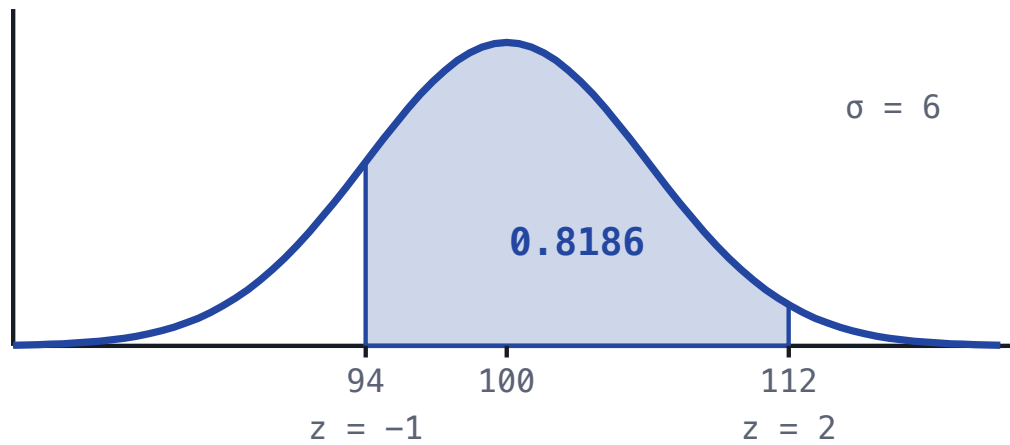
Why: subtracting 100 and dividing by 6 keeps the inequalities.

$$2. \quad P = \Phi(2) - \Phi(-1) = \Phi(2) - (1 - \Phi(1)).$$

Why: $\Phi(-x) = 1 - \Phi(x)$ by the symmetry of the bell curve.

$$3. \quad = 0.97725 - 0.15866 \approx 0.8186.$$

Why: table values $\Phi(2) \approx 0.97725$ and $\Phi(1) \approx 0.84134$.



Conclusion. $P(94 \leq X \leq 112) \approx 0.8186$. ■

22 · Total probability, continuous case

$$P(A) = \int P(A | X = x) f_X(x) dx.$$

Condition on the continuous quantity that the event depends on.

Rain and humidity

The humidity H on a given day is random with density $2h$ for $0 \leq h \leq 1$. If the humidity is h , it rains with probability h^2 .

Question: what is the probability that it rains?

Key idea

Average the conditional probability over the density.

Picture

Average the rain chance over all humidities, weighted by how likely each humidity is: $\int h^2 \cdot 2h \, dh = \frac{1}{2}$.

Run once

Run 100,000

Solution

Setup. Humidity H has density $2h$ on $[0,1]$, and $P(\text{rain} \mid H = h) = h^2$.

$$1. \quad P(\text{rain}) = \int_0^1 P(\text{rain} \mid H = h) f_H(h) \, dh = \int_0^1 h^2 \cdot 2h \, dh.$$

Why: continuous law of total probability.

$$2. \quad = \int_0^1 2h^3 \, dh = \frac{2}{4} = \frac{1}{2}.$$

Why: $\int_0^1 h^3 \, dh = \frac{1}{4}.$

Conclusion. It rains with probability $\frac{1}{2}$. ■

Nested points in the disc

A first point is placed in the unit disc so that its distance R_1 from the centre is uniform on $[0,1]$. Then a second point is placed uniformly at random (by area) in the ring between radius R_1 and radius 1.

Question: what is the probability that the second point is more than $\frac{1}{2}$ away from the centre?

Key idea

Given r_1 , $P(R_2 > \frac{1}{2})$ is an area ratio. Integrate it over r_1 .

Picture

Condition on where the first point lands. If it is beyond $\frac{1}{2}$, the second point surely is too. If not, the chance is an area ratio: (ring from $\frac{1}{2}$ to 1) $\div$ (ring from r_1 to 1). Average over r_1 .

Run once

Run 100,000

Solution

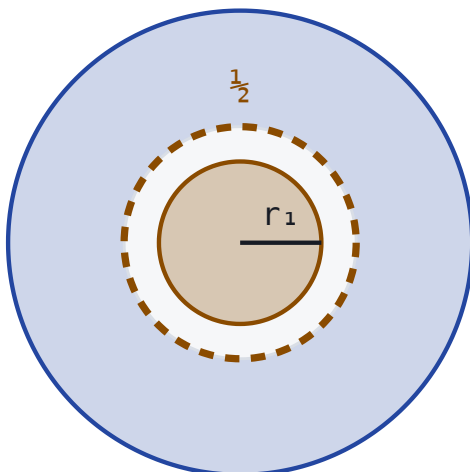
Setup. The first point's distance from the centre is $R_1 \sim U[0, 1]$. Given $R_1 = r_1$, the second point is uniform (by area) in the annulus $r_1 < r \leq 1$. We want $P(R_2 > \frac{1}{2})$.

1. If $r_1 \geq \frac{1}{2}$: every point of the annulus is at distance $> r_1 \geq \frac{1}{2}$, so $P(R_2 > \frac{1}{2} \mid r_1) = 1$.

Why: the annulus starts beyond $\frac{1}{2}$.

2. If $r_1 < \frac{1}{2}$: the favourable part is the ring from $\frac{1}{2}$ to 1, of area $\pi(1 - \frac{1}{4})$, out of the annulus area $\pi(1 - r_1^2)$. So $P(R_2 > \frac{1}{2} \mid r_1) = \frac{3/4}{1 - r_1^2}$.

Why: uniform by area.



second point: uniform
in the annulus $r > r_1$

$$P(R_2 > \frac{1}{2} \mid r_1 < \frac{1}{2}) \\ = (1 - \frac{1}{4}) / (1 - r_1^2)$$

(outside dashed $\frac{1}{2}$)

$$3. \quad P = \int_0^{1/2} \frac{3/4}{1-r^2} dr + \int_{1/2}^1 1 dr.$$

Why: continuous total probability over R_1 , whose density is 1 on $[0,1]$.

$$4. \quad \int_0^{1/2} \frac{dr}{1-r^2} = \frac{1}{2} \ln \frac{1+r}{1-r} \Big|_0^{1/2} = \frac{1}{2} \ln 3.$$

Why: partial fractions: $\frac{1}{1-r^2} = \frac{1}{2} \left(\frac{1}{1-r} + \frac{1}{1+r} \right)$; at $r = \frac{1}{2}$, $\frac{1.5}{0.5} = 3$.

$$5. \quad P = \frac{3}{4} \cdot \frac{1}{2} \ln 3 + \frac{1}{2} = \frac{4 + 3 \ln 3}{8} \approx 0.91198.$$

Why: substitute step 4 into step 3.

Conclusion. $P(R_2 > \frac{1}{2}) = \frac{4+3 \ln 3}{8}$. ■

Three points, pairwise $\geq 90^\circ$ apart

Three points are chosen independently and uniformly on a circle.

Question: what is the probability that every two of them are at least 90° apart, measured the short way round the circle?

Key idea

Fix the first point at 0 . The second point is at $90^\circ + x$ (doubled by symmetry). The third point's allowed arc has length $90^\circ - x$.

Picture

Fix the first point at 0° . The second must be at least 90° away; say it is $90^\circ + x$. The third then has only an arc of $90^\circ - x$ left. Averaging gives $1/16$.

Run once

Run 200,000

Solution

Setup. Three independent uniform points on a circle, with angles measured in degrees (0 to 360). We want every pair to be at least 90° apart (measured the short way round).

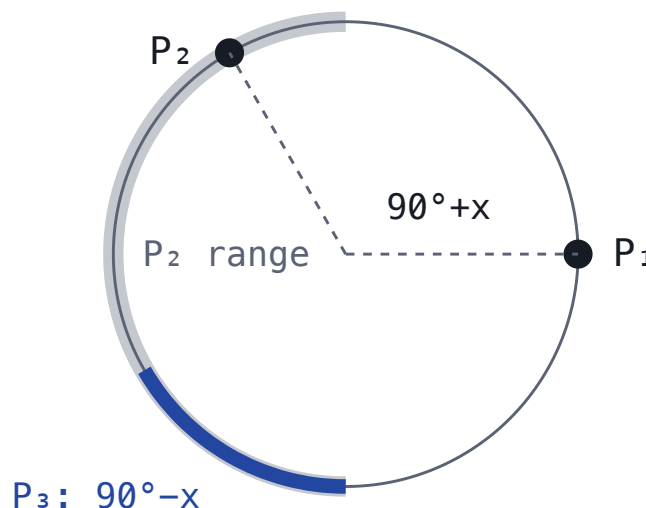
1. Rotate so that $P_1 = 0^\circ$.

Why: rotation doesn't change the problem, and P_2, P_3 stay independent and uniform.

2. P_2 must lie in $[90^\circ, 270^\circ]$. Reflecting $\theta \mapsto -\theta$ fixes P_1 and swaps $[90^\circ, 180^\circ]$ with $[180^\circ, 270^\circ]$, so compute with $P_2 = 90^\circ + x$, $x \in [0^\circ, 90^\circ]$, and double.

Why: the reflection preserves the uniform law.

3. Given $P_2 = 90^\circ + x$: P_3 must be $\geq 90^\circ$ from 0° , i.e. in $[90^\circ, 270^\circ]$, and $\geq 90^\circ$ from P_2 , i.e. outside $(x, 180^\circ + x)$. Together: the arc $[180^\circ + x, 270^\circ]$, of length $90^\circ - x$.



Why: intersect the two allowed sets.

$$4. \quad P = 2 \int_0^{90} \frac{1}{360} \cdot \frac{90-x}{360} dx = \frac{2 \cdot 4050}{360^2} = \frac{8100}{129600} = \frac{1}{16}.$$

Why: P_2 has density $\frac{1}{360}$ per degree; given P_2 , P_3 succeeds with probability $\frac{90-x}{360}$; $\int_0^{90} (90-x) dx = \frac{90^2}{2} = 4050$.

Conclusion. All three are pairwise at least 90° apart with probability $\frac{1}{16}$ (simulated 0.0622). ■

General: n points with every gap at least a fraction a of the circle: $(1-na)^{n-1}$.

23 · Expected value

$\mathbb{E}[X] = \sum x P(X=x)$ (discrete), or $\int x f(x) dx$ (continuous). It is the long-run average.

Die, and a d30 difference

Question: (a) What is the expected value of one roll of a fair six-sided die? (b) Two fair 30-sided dice (faces 1 to 30) are rolled. What is the expected value of the absolute difference between the two results?

Key idea

(b) $|x-y| = d$ occurs in $2(30-d)$ of the 900 outcomes.

Picture

Count how often each difference d happens: $2(30 - d)$ of the 900 pairs. Weight each d by that count.

Run once

Run 100,000

Solution

1. (a) $\mathbb{E}[\text{die}] = \frac{1+2+\dots+6}{6} = \frac{21}{6} = 3.5.$

Why: each face has weight $\frac{1}{6}.$

2. (b) Two independent d_{30} rolls x, y : 900 equally likely pairs. For $d \geq 1$, $|x - y| = d$ happens for $2(30 - d)$ pairs.

Why: $(x, x + d)$ with $x = 1, \dots, 30 - d$, and the same pairs reversed.

3. $\mathbb{E}|x - y| = \frac{1}{900} \sum_{d=1}^{29} d \cdot 2(30 - d) = \frac{2}{900} \left(30 \sum d - \sum d^2 \right).$

Why: $d = 0$ contributes 0.

4. $\sum_{d=1}^{29} d = 435$ and $\sum_{d=1}^{29} d^2 = \frac{29 \cdot 30 \cdot 59}{6} = 8555$, so the value is $\frac{2}{900} (13050 - 8555) = \frac{8990}{900} \approx 9.99.$

Why: standard sums; in general the answer is $\frac{n^2-1}{3n}.$

Conclusion. 3.5 and about 9.99. ■

First repeat on a d4

A fair 4-sided die (faces 1 to 4) is rolled over and over until some number comes up for the second time.

Question: on average, on which roll does this first repeat happen? (Give the expected roll number.)

Picture

Keep track of how many values you have seen. With k seen, the next roll repeats with chance $k/4$. Stepping k up from 0 to 4 gives a short chain of "1 + ..." equations.

Run once

Run 100,000

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one roll. State k = different values seen so far. A seen value $\rightarrow$ ends (this roll is the repeat). A new value $\rightarrow$ next turn with $k+1$.

1. **LEVEL.** $\text{rec}(i, k)$ = expected rolls from roll i onward (roll i included) until the first repeat, given rolls 1 to $i-1$ show k different values and no repeat. The answer is $\text{rec}(1, 0)$, which is the roll number of the repeat.

Why: only how many values are 'used up' matters.

2. **CHOICE.** Roll i shows an already-seen value with probability $\frac{k}{4}$, a new value with $\frac{4-k}{4}$.

Why: k of the 4 faces are seen.

3. **CHECK.** Seen: ends; this roll counts, cost 1. New: cost 1, continue to $(i+1, k+1)$.

Why: the rules.

4. **MOVE.** $\text{rec}(i, k) = \frac{k}{4} \cdot 1 + \frac{4-k}{4} (1 + \text{rec}(i+1, k+1)) = 1 + \frac{4-k}{4} \text{rec}(i+1, k+1)$.

Why: law of total expectation; $\frac{k}{4} + \frac{4-k}{4} = 1$.

5. **MEMORYLESS.** $\text{rec}(i+1, k) = \text{rec}(i, k) = \text{rec}(k)$. k only goes up, so plain recursion; base $\text{rec}(4) = 1$

(the next roll must repeat).

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** $\text{rec}(3) = 1 + \frac{1}{4} \cdot 1 = \frac{5}{4}$. $\text{rec}(2) = 1 + \frac{2}{4} \cdot \frac{5}{4} = \frac{13}{8}$. $\text{rec}(1) = 1 + \frac{3}{4} \cdot \frac{13}{8} = 1 + \frac{39}{32} = \frac{71}{32}$. $\text{rec}(0) = 1 + 1 \cdot \frac{71}{32} = \frac{103}{32} \approx 3.22$.

```
double dp[5]; // all -1
double rec(int k) {
    if (k == 4) return 1; // base case: the next roll is 4
    if (dp[k] != -1) return dp[k]; // cache check
    double ans = 1 + (4.0 - k) / 4 * rec(k + 1);
    return dp[k] = ans; // save and return
} // rec(0)
```

Why: MOVE with $k = 3, 2, 1, 0$ in turn.

Answer. The repeat happens on roll $\frac{103}{32}$ on average. ■

△ The course's original wording is "rolls before the first repetition", which would be $\frac{103}{32} - 1$. The course answers the roll index.

Densities

Question: (a) X has density $3x^2$ on $[0, 1]$. Find $\mathbb{E}[X]$.

(b) A point is chosen uniformly inside a circle of radius 1. What is its expected distance from the centre?

Key idea

$\int x f(x) dx$. The disc distance has $f = 2r$.

Picture

Average = $\int x f(x) dx$. For the disc, use the distance density $2r$ from above: $\int r \cdot 2r dr = \frac{2}{3}$.

Run once

Run 100,000

Solution

1. (a) $\mathbb{E}[X] = \int_0^1 x \cdot 3x^2 dx = \int_0^1 3x^3 dx = \frac{3}{4}$.

Why: $\mathbb{E}[X] = \int x f(x) dx$.

2. (b) The distance D of a uniform point in the unit disc has density $2r$ (§19), so $\mathbb{E}[D] = \int_0^1 r \cdot 2r dr = \frac{2}{3}$.

Why: same formula.

Conclusion. $\frac{3}{4}$ and $\frac{2}{3}$. ■

Ferris wait

A Ferris wheel has radius 1 m, its centre is 1 m above the ground, and it turns once per minute. You can only get on when the boarding car is at most 0.5 m above the ground. You arrive at a random time.

Question: what is your expected waiting time before you can get on?

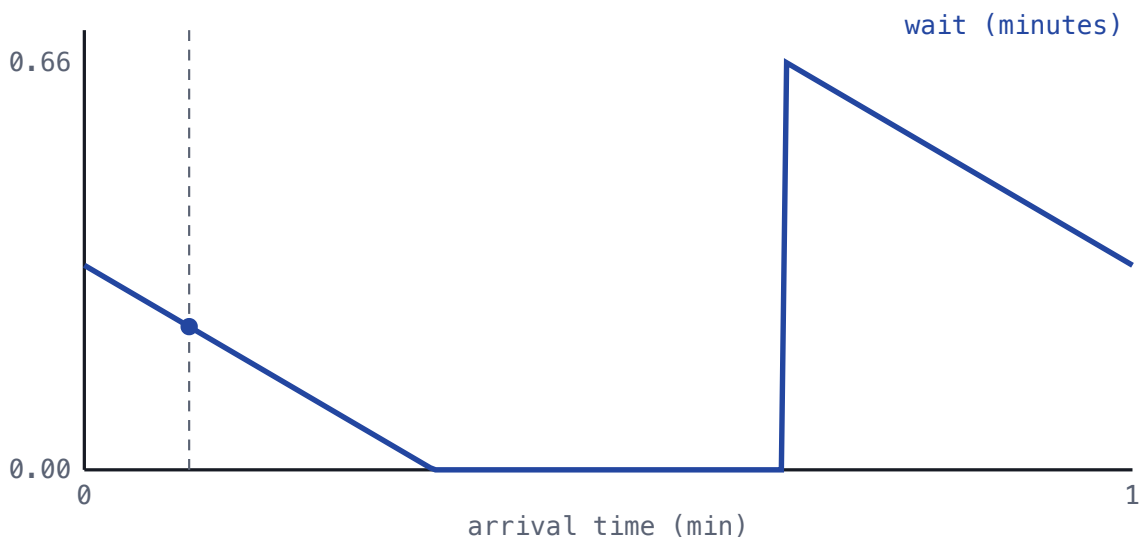
Key idea

Boarding is possible for $\frac{1}{3}$ of each cycle. The wait is 0 inside that window, and otherwise the time to the window's next start.

Picture

Draw the wait against your arrival time. It is 0 during the boarding third of each minute, and otherwise counts down to the next window. The average wait is the area under that sawtooth.

Wait as a function of arrival time. Average = area = $2/9$.



arrival time (min) 0.1

arrival time (min) = 0.1: wait (minutes) 0.233

Run once

Run 100,000

Solution

Setup. A Ferris wheel of radius 1 m has its centre 1 m above the ground and turns once per minute. At time t the boarding car is at height $h(t) = 1 + \cos 2\pi t$. You can board when $h \leq 0.5$. You arrive at a uniform random time.

$$1. \quad h \leq \frac{1}{2} \iff \cos 2\pi t \leq -\frac{1}{2} \iff 2\pi t \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right] \iff t \in \left[\frac{1}{3}, \frac{2}{3}\right] \pmod{1}.$$

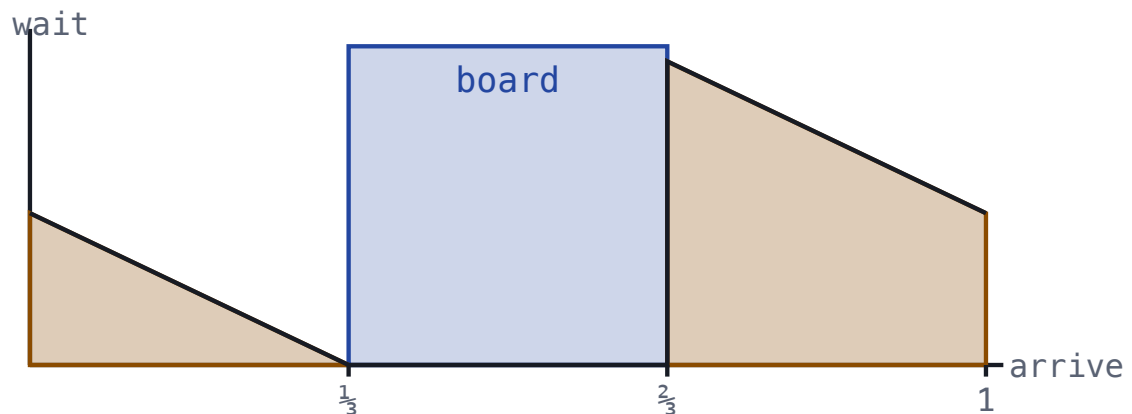
Why: cosine is at most $-\frac{1}{2}$ exactly between 120° and 240° .

2. So each minute has a boarding window of length $\frac{1}{3}$, and an unavailable stretch of length $L = \frac{2}{3}$ (from $\frac{2}{3}$ to $\frac{4}{3}$).

Why: $1 - \frac{1}{3} = \frac{2}{3}$.

3. If you arrive inside the window (probability $\frac{1}{3}$), the wait is 0. If you arrive in the stretch (probability $\frac{2}{3}$), the wait is the time left until the stretch ends, which is uniform on $[0, L]$, with mean $\frac{L}{2} = \frac{1}{3}$.

Why: a uniform arrival inside an interval of length L leaves a uniform remaining time.



$$E[\text{wait}] = \text{shaded area} = \frac{1}{2} \cdot \left(\frac{2}{3}\right)^2 = 2/9$$

4. $\mathbb{E}[W] = \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9} \approx 0.22222$ min.

Why: law of total expectation. The sawtooth figure above shows the same thing as an area.

Conclusion. The expected wait is $\frac{2}{9}$ of a minute. ■

24 · Law of total expectation

$\mathbb{E}[X] = \sum_y \mathbb{E}[X | Y = y] P(Y = y)$, or $\int \mathbb{E}[X | Y = y] f_Y(y) dy$. Compactly, the tower property: $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$.

Mixture and nested uniform

Question: (a) One of three dice is picked at random: a 6-sided, a 7-sided or a 9-sided die (faces numbered from 1 up to the number of sides). It is rolled. What is the expected result? (b) Y is uniform on $[0,1]$, and then, given Y , X is uniform on $[0,Y]$. Find $\mathbb{E}[X]$.

Key idea

Condition on which die, or on Y .

Picture

Average in two stages. First the average given the die (3.5, 4, 5), then average those. For (b): given Y the average is $Y/2$, and Y averages $\frac{1}{2}$.

Run once

Run 100,000

Solution

- (a) Given the die, the mean roll is $\frac{k+1}{2}$: 3.5 for d6, 4 for d7, 5 for d9. Each die is picked with probability $\frac{1}{3}$, so $\mathbb{E}[X] = \frac{3.5+4+5}{3} = \frac{12.5}{3} \approx 4.167$.

Why: law of total expectation over which die.

- (b) Given Y , X is uniform on $[0,Y]$, so $\mathbb{E}[X | Y] = \frac{Y}{2}$. Then $\mathbb{E}[X] = \mathbb{E}[\frac{Y}{2}] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Why: tower property; $\mathbb{E}[Y] = \frac{1}{2}$.

Conclusion. About 4.167 and $\frac{1}{4}$. ■

25 · LOTUS

$\mathbb{E}[g(X)] = \sum g(x) P(X = x)$, or $\int g(x) f_X(x) dx$. You never need the distribution of $g(X)$.

$\mathbb{E}[X^2]$

Question: find $\mathbb{E}[X^2]$ when (a) X is the roll of a fair die; (b) X is uniform on $[0,1]$.

Key idea

Weight $g(x) = x^2$ by the law of X .

Picture

Square the values, keep the probabilities. You never need the distribution of X^2 .

Solution

1. (a) Die: $\mathbb{E}[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx 15.16667$.

Why: LOTUS: weight x^2 by $P(X = x) = \frac{1}{6}$.

2. (b) $U[0,1]$: $\mathbb{E}[X^2] = \int_0^1 x^2 dx = \frac{1}{3}$.

Why: LOTUS with density 1.

Conclusion. $\frac{91}{6}$ and $\frac{1}{3}$. ■

26 · Tail sum (discrete)

For $X \in \{0, 1, 2, \dots\}$:

$$\mathbb{E}[X] = \sum_{k \geq 1} P(X \geq k).$$

Proof: $X = \sum_{k \geq 1} 1[X \geq k]$; take expectations. Use it when " $\geq k$ " is easy (minima: all $\geq k$).

Min of three d100

Three fair 100-sided dice (faces 1 to 100) are rolled.

Question: what is the expected value of the smallest number shown?

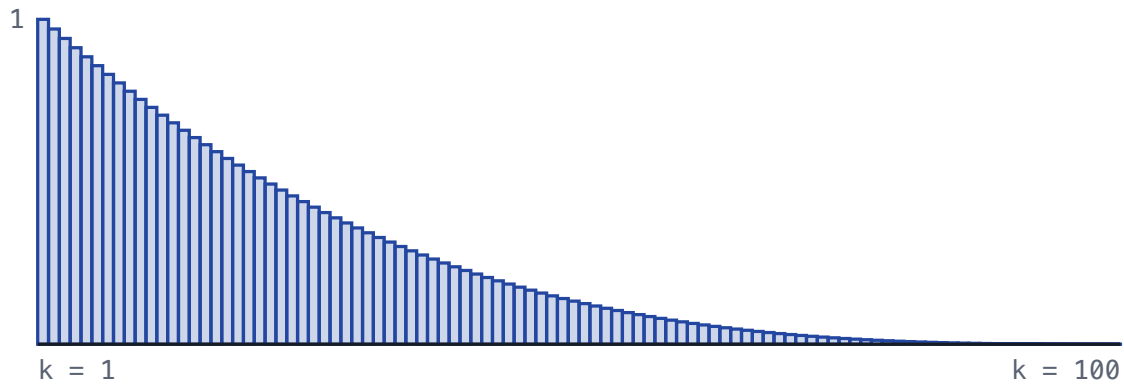
Key idea

$\min \geq k \iff \text{all three} \geq k.$

Picture

Add up the chances $P(\min \geq k)$. "Min $\geq k$ " means all three are $\geq k$: $((101 - k)/100)^3$. The sum of those bars is the average.

Bars $P(\min \geq k)$ for $k = 1 \dots 100$.



Each bar is $P(X \geq k)$. Their total is $E[X] = 25.5025$.

Solution

Setup. X = the smallest of three independent rolls of a 100-sided die (faces 1–100).

1. $X \geq k$ exactly when all three rolls are $\geq k$, so $P(X \geq k) = \left(\frac{101-k}{100}\right)^3$.

Why: a roll is $\geq k$ for the $101 - k$ faces $k, \dots, 100$; independence.

2. $\mathbb{E}[X] = \sum_{k=1}^{100} P(X \geq k) = \sum_{j=1}^{100} \frac{j^3}{100^3}$.

Why: tail sum; substitute $j = 101 - k$.

3. $\sum_{j=1}^{100} j^3 = \left(\sum j\right)^2 = 5050^2 = 25\,502\,500$, so $\mathbb{E}[X] = \frac{25\,502\,500}{1\,000\,000} = 25.5025$.

Why: the sum of the first n cubes is the square of the sum of the first n integers.

Conclusion. The expected minimum is 25.5025. ■

Smallest ball in ball 10's box

Balls numbered 1 to 10 are each put into one of 5 boxes, chosen independently and uniformly at random. Look at the box that contains ball 10.

Question: what is the expected value of the smallest-numbered ball in that box? (It might be ball 10 itself.)

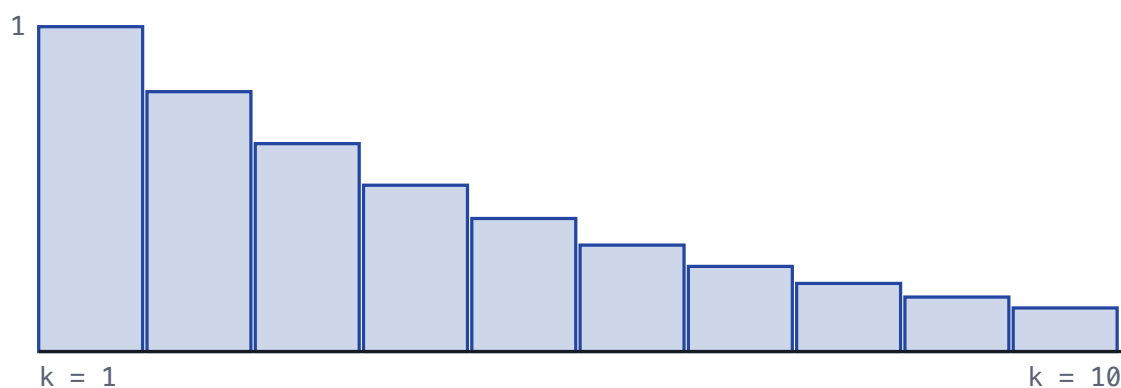
Key idea

$Y \geq k \iff$ none of balls $1..k-1$ shares ball 10's box.

Picture

"Smallest $\geq k$ " means none of balls $1 \dots k-1$ joined ball 10's box: $(4/5)^{(k-1)}$. Add these up.

Bars $P(Y \geq k)$ for $k = 1 \dots 10$.



Each bar is $P(X \geq k)$. Their total is $E[X] = 4.4631$.

Solution

Setup. Balls 1–10 each go independently into one of 5 boxes, uniformly. Y = the smallest label in the box that holds ball 10 (possibly 10 itself).

- For $1 \leq k \leq 10$: $Y \geq k$ exactly when none of balls $1, \dots, k-1$ is in ball 10's box.

Why: definition of the smallest label.

- $P(Y \geq k) = \left(\frac{4}{5}\right)^{k-1}$.

Why: each of those $k-1$ balls independently avoids that box with probability $\frac{4}{5}$.

$$3. \mathbb{E}[Y] = \sum_{k=1}^{10} \left(\frac{4}{5}\right)^{k-1} = \frac{1 - 0.8^{10}}{1 - 0.8} = 5(1 - 0.8^{10}) \approx 4.46313.$$

Why: tail sum ($Y \leq 10$, so larger k contribute 0); finite geometric series; $0.8^{10} \approx 0.10737$.

Conclusion. $\mathbb{E}[Y] = 5(1 - 0.8^{10})$. ■

27 · Tail integral

For $X \geq 0$:

$$\mathbb{E}[X] = \int_0^{\infty} P(X > t) dt.$$

Proof: $X = \int_0^{\infty} 1[X > t] dt$; take expectations (swapping is fine because the integrand is ≥ 0).

Distance to the base of a triangle

A point is chosen uniformly at random inside a triangle of height h .

Question: what is its expected distance from the base of the triangle?

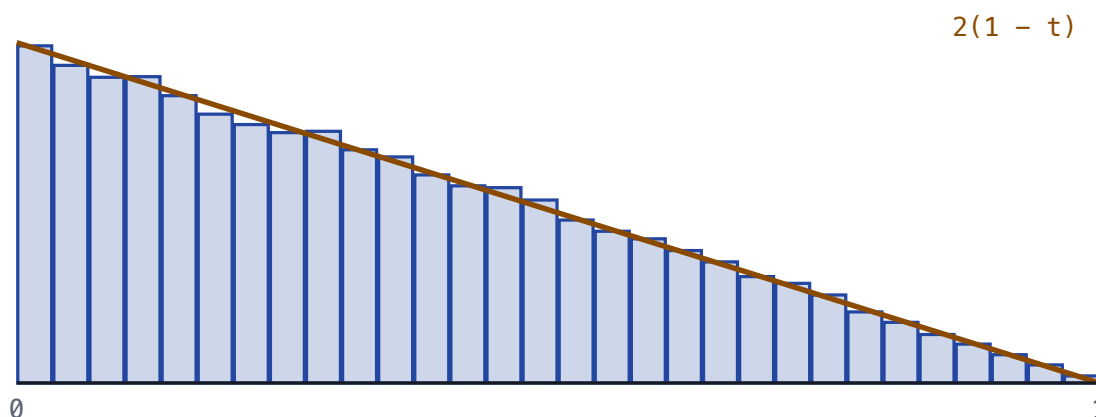
Key idea

$\{D > t\}$ is a similar triangle of height $h - t$, so its area ratio is $\left(\frac{h-t}{h}\right)^2$.

Picture

"Distance $> t$ " is a smaller copy of the triangle, scaled by $(h - t)/h$, so its area share is $((h - t)/h)^2$. Integrate over t : $h/3$.

Heights of 50 000 random points in a triangle of height 1.



Sample 50 000

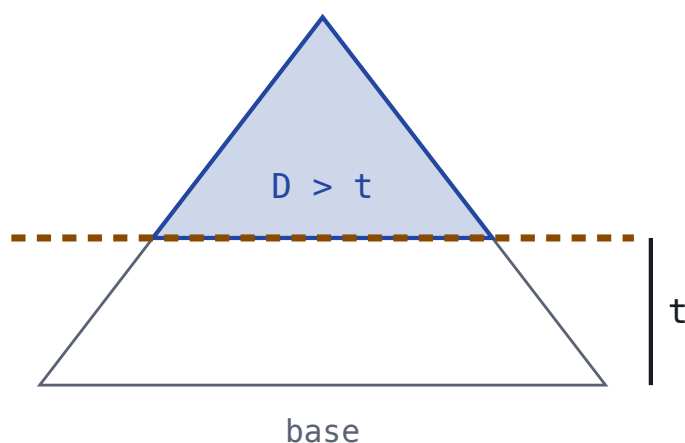
Blue bars: 50 000 samples. Orange: the formula. Sample mean **0.335** (exact $1/3$ (height 1)).

Solution

Setup. A point is uniform in a triangle of height h ; D = its distance to the base.

1. $\{D > t\}$ is the part above the line parallel to the base at height t : a smaller triangle, similar to the whole one, with height $h - t$.

Why: it shares the apex and has parallel sides.



similar triangle,
height $h - t$

$$P(D > t) = (1 - t/h)^2$$

$$2. P(D > t) = \left(\frac{h-t}{h}\right)^2 = (1 - t/h)^2.$$

Why: similar figures with ratio $\frac{h-t}{h}$ have area ratio $\left(\frac{h-t}{h}\right)^2$; for a uniform point, probability = area ratio.

$$3. \mathbb{E}[D] = \int_0^h (1 - t/h)^2 dt = h \int_0^1 u^2 du = \frac{h}{3}.$$

Why: tail integral (§27); substitute $u = 1 - t/h$, $dt = -h du$.

Conclusion. The average distance to the base is a third of the height. ■

Max of n uniforms

n numbers are chosen independently and uniformly from $[0, 1]$.

Question: what is the expected value of the largest of them?

Key idea

$$P(M > t) = 1 - t^n.$$

Picture

$P(\max \leq t) = t^n$, so $P(\max > t) = 1 - t^n$. Integrate: $n/(n+1)$. Picture: n points cut $[0, 1]$ into $n+1$ pieces of equal average size.

n random numbers in $[0, 1]$: the largest averages $n/(n+1)$ – the n points split $[0, 1]$ into $n+1$ gaps of equal average size.



$n = 4$: average max over 100 000 tries **0.801** ($n/(n+1) = 0.800$).

Solution

Setup. M = the largest of n independent uniforms on $[0, 1]$.

1. $P(M \leq t) = t^n$, so $P(M > t) = 1 - t^n$.

Why: the maximum is $\leq t$ iff every one is, each with probability t , independently.

2. $\mathbb{E}[M] = \int_0^1 (1 - t^n) dt = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

Why: tail integral.

Conclusion. $\mathbb{E}[\max] = \frac{n}{n+1}$. ■

28 · Linearity of expectation

1. $\mathbb{E}[\sum c_i X_i] = \sum c_i \mathbb{E}[X_i]$. This holds **always, even for dependent variables**.
2. The method: name X , then write $X = X_1 + \dots + X_n$ explicitly, using either (a) indicators or (b) "the extra amount added by step k ".
3. A geometric wait with success probability p has mean $1/p$.

Sum of three dice

Question: three fair dice are rolled. What is the expected total?

Key idea

$$X = X_1 + X_2 + X_3.$$

Picture

Average of a sum = sum of the averages. Each die averages 3.5, so three dice average 10.5. No need to list the 216 outcomes.

Run once

Run 100,000

Solution

Setup. $X = X_1 + X_2 + X_3$, where X_i is the value of die i .

1. $\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}[X_3].$

Why: linearity of expectation.

2. Each $\mathbb{E}[X_i] = 3.5$, so $\mathbb{E}[X] = 3 \cdot 3.5 = 10.5.$

Why: $\frac{1+\dots+6}{6} = 3.5.$

Conclusion. 10.5. ■

Letters in the right envelopes

n letters are put into n addressed envelopes completely at random, one letter per envelope.

Question: what is the expected number of letters that end up in their correct envelope?

Key idea

$I_i = 1$ if letter i is correct, and $P(I_i = 1) = 1/n.$

Picture

Count with indicators. Letter i is right with chance $1/n$. Adding n of these gives exactly 1, even though the letters depend on each other: linearity doesn't care.

Letters into envelopes at random. Each letter is in the right place with chance $1/n$, and there are n letters: on average exactly 1 correct, whatever n is.



n  8 [Shuffle](#)

This shuffle: 0 correct (blue). Average over 20 000 shuffles: **1.005**.

Solution

Setup. n letters go into n envelopes in a uniformly random way (a random permutation). $I_i = 1$ if letter i is in its own envelope, else 0. The number of correct letters is $X = I_1 + \dots + I_n$.

1. $P(I_i = 1) = \frac{1}{n}$.

Why: letter i is equally likely to be in each of the n envelopes.

2. $\mathbb{E}[X] = \sum_i \mathbb{E}[I_i] = n \cdot \frac{1}{n} = 1$, for every n .

Why: linearity, which holds even though the I_i are dependent; $\mathbb{E}[I_i] = P(I_i = 1)$.

Conclusion. On average exactly one letter is in the right envelope. ■

HT count in 10 flips

A fair coin is flipped 10 times.

Question: what is the expected number of times an H is immediately followed by a T?

Key idea

9 adjacent pairs, each HT with probability $\frac{1}{4}$.

Picture

One indicator per neighbouring pair. 9 pairs, each HT with chance $\frac{1}{4}$: $9/4$.

Run once

Run 100,000

Solution

Setup. 10 fair flips. For each of the 9 adjacent pairs $(i, i+1)$, let $I_i = 1$ if flip i is H and flip $i+1$ is T.

$$1. P(I_i = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Why: two independent fair flips.

$$2. \mathbb{E}[\#HT] = \sum_{i=1}^9 \mathbb{E}[I_i] = \frac{9}{4} = 2.25.$$

Why: linearity.

Conclusion. 2.25. ■

Coupon collector (d6)

A fair die is rolled until every face from 1 to 6 has appeared at least once.

Question: what is the expected number of rolls?

Picture

Split the wait into stages. With k faces seen, each roll finds a new one with chance $(6 - k)/6$, so that stage takes $6/(6 - k)$ rolls on average. Add the six stages: $6/6 + 6/5 + 6/4 + 6/3 + 6/2 + 6/1 = 14.7$.

Rolls coloured by stage: with k faces already seen, a new face has chance $(6 - k)/6$, so the stage lasts $6/(6 - k)$ rolls on average: $1 + 1.2 + 1.5 + 2 + 3 + 6 = 14.7$.



Roll until all six

Average of 50 000

20 rolls. Ringed = a new face; colour = stage (faces seen before that roll).

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one roll. State k = faces seen so far. $k = 6 \rightarrow \text{ends}$. A seen face $\rightarrow$ same state (self-loop). A new face $\rightarrow k+1$.

1. **LEVEL.** $\text{rec}(i, k)$ = expected rolls from roll i onward until all 6 faces have appeared, given k different faces among rolls 1 to $i-1$. The answer is $\text{rec}(1, 0)$.

Why: only the number of faces seen matters.

2. **CHOICE.** Roll i shows a new face with probability $\frac{6-k}{6}$, a seen face with $\frac{k}{6}$.

Why: k of the 6 faces are seen.

3. **CHECK.** $k = 6$: ends, 0 more rolls. Otherwise the roll costs 1; new $\rightarrow (i+1, k+1)$; seen $\rightarrow (i+1, k)$.

Why: the rules.

4. **MOVE.** $\text{rec}(i, k) = 1 + \frac{6-k}{6} \text{rec}(i+1, k+1) + \frac{k}{6} \text{rec}(i+1, k)$.

Why: law of total expectation over roll i .

5. **MEMORYLESS.** $\text{rec}(i+1, k) = \text{rec}(i, k) = \text{rec}(k)$. The 'seen' case is a self-loop: $\text{rec}(k) - \frac{k}{6}\text{rec}(k) = 1 + \frac{6-k}{6}\text{rec}(k+1)$, i.e. $\frac{6-k}{6}\text{rec}(k) = 1 + \frac{6-k}{6}\text{rec}(k+1)$, so $\text{rec}(k) = \frac{6}{6-k} + \text{rec}(k+1)$.

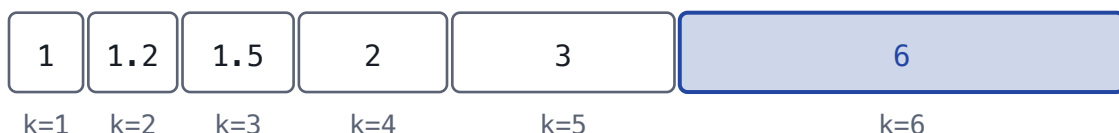
Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) . Solve the self-loop by algebra: multiply by $\frac{6}{6-k}$.

6. **SOLVE.** $\text{rec}(6) = 0$; $\text{rec}(5) = 6$; $\text{rec}(4) = 3 + 6 = 9$; $\text{rec}(3) = 2 + 9 = 11$; $\text{rec}(2) = 1.5 + 11 = 12.5$; $\text{rec}(1) = 1.2 + 12.5 = 13.7$; $\text{rec}(0) = 1 + 13.7 = 14.7$.

```
double dp[7];                                // all -1
double rec(ll k) {
    if (k == 6) return 0;                     // base case: every face
    if (dp[k] != -1) return dp[k];           // cache check
    double ans = 6.0 / (6 - k) + rec(k + 1);  // self-loop solved
    return dp[k] = ans;                       // save and return
} // rec(0)
```

Why: $\frac{6}{6-k}$ for $k = 5, 4, 3, 2, 1, 0$ is 6, 3, 2, 1.5, 1.2, 1.

wait for the k -th new face $\sim \text{Geometric}((7-k)/6)$



$$1 + 1.2 + 1.5 + 2 + 3 + 6 = 14.7$$

the last face alone costs 6 rolls on average

Answer. It takes 14.7 rolls on average to see every face. ■

Rectangles from 20 random lines

20 straight lines are drawn one after another. Each is horizontal or vertical with probability $\frac{1}{2}$, independently, and no two lines coincide. Together they cut the plane into rectangular cells (counting the unbounded ones at the edges).

Question: what is the expected number of cells?

Key idea

The k -th line adds (#perpendicular lines so far + 1) cells, which is $\frac{k+1}{2}$ on average.

Picture

Count what each new line adds. A horizontal line crosses every column of cells and splits each one: it adds (vertical lines + 1). On average the k -th line adds $(k + 1)/2$. Start with 1 cell and add these up.

Run once

Run 100,000

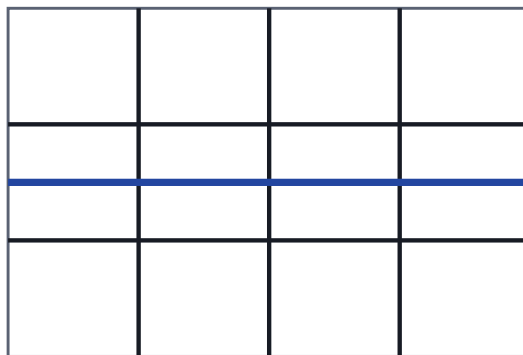
Solution

Setup. 20 lines are drawn one at a time, each horizontal or vertical with probability $\frac{1}{2}$,

independently. With h horizontal and v vertical lines, the plane is cut into $(h+1)(v+1)$ cells.

1. Adding a horizontal line when there are v vertical lines crosses $v+1$ columns of cells and splits each of them into two, so it adds $v+1$ cells. A vertical line adds $h+1$.

Why: a new line is cut into $v+1$ pieces by the v perpendicular lines, and each piece splits one cell.



new horizontal line
cuts $v + 1 = 4$ cells
 $\Rightarrow$ adds 4 cells

$$h = 2, v = 3: (h+1)(v+1) = 12$$

2. Before the k -th line there are $h + v = k - 1$ lines. The k -th line adds $\frac{1}{2}(v+1) + \frac{1}{2}(h+1) = \frac{h+v+2}{2} = \frac{k+1}{2}$ cells on average.

Why: it is horizontal or vertical with probability $\frac{1}{2}$ each; the average depends only on $h+v$, which is fixed.

3. $\mathbb{E}[\text{cells}] = 1 + \sum_{k=1}^{20} \frac{k+1}{2} = 1 + \frac{230}{2} = 116$.

Why: start with 1 cell (the whole plane) and add the expected gains (linearity); $\sum_{k=1}^{20} (k+1) = 210 + 20 = 230$.

4. Check: $h \sim \text{Bin}(20, \frac{1}{2})$, so $\mathbb{E}[(h+1)(21-h)] = \mathbb{E}[-h^2 + 20h + 21] = -105 + 200 + 21 = 116 \checkmark$.

Why: $\mathbb{E}[h] = 10$, $\mathbb{E}[h^2] = \text{Var} + 10^2 = 5 + 100 = 105$.

Conclusion. The 20 lines make 116 cells on average.

■

△ Course says “smaller rectangles” but its count includes the unbounded cells. Counting bounded ones only, $(h-1)(v-1)$, gives 76.

29 · Indicator variables

$1_A = 1$ if A happens, 0 otherwise. $\mathbb{E}[1_A] = P(A)$. To count something, write $\text{count} = \sum 1$ and take expectations term by term.

GB pairs; colour components

Question:

(a) 5 boys and 5 girls stand in a row in random order. What is the expected number of places where a girl stands immediately to the left of a boy?

(b) 10 squares in a row are each coloured red or blue at random ($\frac{1}{2}$ each, independently). A **run** is a maximal block of neighbouring squares of the same colour. What is the expected number of runs?

Key idea

(a) One indicator per adjacent pair. (b) $\# \text{runs} = 1 + \# \text{colour changes}$.

Picture

(a) One indicator per neighbouring pair of seats: each is (girl, boy) with chance $\frac{5}{10} \times \frac{5}{9}$. **(b)** Runs = 1 + colour changes, and each of the 9 boundaries is a change with chance $\frac{1}{2}$.

Run once

Run 100,000

Run once

Run 100,000

Solution

1. (a) 5 boys and 5 girls in a random row. For seats $i, i+1$ ($i = 1, \dots, 9$), $P(\text{seat } i = G, \text{ seat } i+1 = B) = \frac{5}{10} \cdot \frac{5}{9} = \frac{5}{18}$.

Why: seat i is a girl with probability $\frac{5}{10}$; given that, 9 people remain, 5 of them boys.

2. $\mathbb{E}[\#GB] = 9 \cdot \frac{5}{18} = 2.5$.

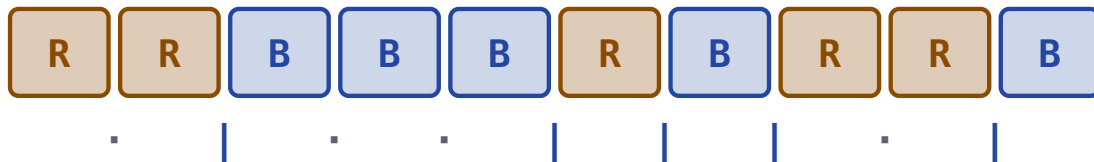
Why: one indicator per adjacent pair of seats; linearity.

3. (b) 10 squares, each red or blue independently with probability $\frac{1}{2}$. The number of runs is 1 + # colour changes.

Why: each change starts a new run.

4. Each of the 9 boundaries is a change with probability $\frac{1}{2}$, so $\mathbb{E}[\#\text{runs}] = 1 + \frac{9}{2} = 5.5$.

Why: two independent fair colours differ with probability $\frac{1}{2}$; linearity.



9 boundaries, each a change w.p. $\frac{1}{2}$
 $\#runs = 1 + \#changes \Rightarrow 1 + 4.5$

Conclusion. 2.5 and 5.5. ■

Records among 10 i.i.d. values

Ten random numbers $X_1, \dots, X_{10}$ are drawn independently from the same continuous distribution. X_i is a **record** if it is larger than every number before it (X_1 always counts as a record).

Question: what is the expected number of records?

Key idea

X_i is a record $\iff$ it is the largest of the first i , which has probability $1/i$.

Picture

X_i is a record exactly when it is the largest of the first i , chance $1/i$. Add: $1 + \frac{1}{2} + \frac{1}{3} + \dots + 1/10$.

Run once

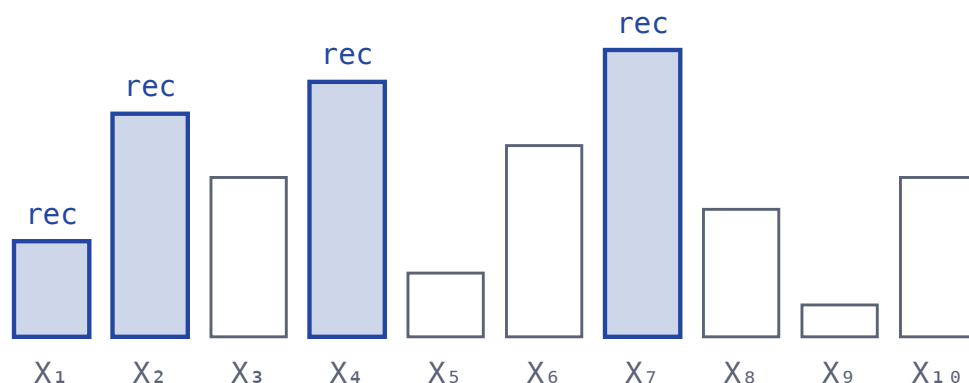
Run 100,000

Solution

Setup. $X_1, \dots, X_{10}$ are i.i.d. continuous. X_i is a **record** if it is larger than all of $X_1, \dots, X_{i-1}$ (X_1 always is). Let I_i indicate that X_i is a record.

$$1. P(I_i = 1) = \frac{1}{i}.$$

Why: X_i is a record iff it is the largest of the first i values, and each of those i values is equally likely to be the largest (i.i.d., no ties).



$$P(X_i \text{ is a record}) = P(\text{largest of first } i) = 1/i$$

$$2. \mathbb{E}[\# \text{records}] = \sum_{i=1}^{10} \frac{1}{i} = H_{10} \approx 2.92897.$$

Why: linearity.

Conclusion. About 2.93 records on average. ■

2-cycles in a random permutation

A permutation (rearrangement) of $1, \dots, n$ is chosen uniformly at random. A **2-cycle** is a pair of numbers that swap places: i goes to position j and j goes to position i .

Question: what is the expected number of 2-cycles?

Key idea

One indicator per pair $i < j$. $P(\pi(i) = j, \pi(j) = i) = \frac{(n-2)!}{n!}$.

Picture

One indicator per pair $\{i, j\}$. They swap with chance $1/(n(n-1))$, and there are $n(n-1)/2$ pairs: $\frac{1}{2}$ on average, for any n .

Run once

Run 100,000

Solution

Setup. A uniformly random permutation π of $\{1, \dots, n\}$. A 2-cycle is a pair $i < j$ with $\pi(i) = j$ and $\pi(j) = i$.

1. For a fixed pair, $P(\pi(i) = j, \pi(j) = i) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$.

Why: fixing those two values leaves $(n-2)!$ permutations of the rest, out of $n!$.

2. There are $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs, so $\mathbb{E}[\#2\text{-cycles}] = \frac{n(n-1)}{2} \cdot \frac{1}{n(n-1)} = \frac{1}{2}$.

Why: one indicator per pair; linearity.

Conclusion. $\frac{1}{2}$, for every $n \geq 2$. ■

Divisor-erasing moves on 1..20

The numbers 1 to 20 are written on a board. Each move: pick one of the numbers still on the board uniformly at random, and erase it together with all of its divisors that are still on the board. This continues until the board is empty.

Question: what is the expected number of moves?

Key idea

#moves = #numbers that get picked. k is picked $\iff$ k comes first among its multiples in a uniformly random order.

Picture

A number is picked exactly when it comes before all its other multiples in a random order (otherwise a multiple erases it first). Number k has $\lfloor 20/k \rfloor$ multiples, so it is picked with chance $1/\lfloor 20/k \rfloor$. Add over k .

Run once

Run 100,000

Solution

Setup. The numbers 1–20 are on the board. Each move picks a uniformly random remaining number and erases it together with all its divisors that are still on the board. The number of moves is the number of numbers that get *picked*.

1. Equivalent model: take a uniformly random order of 1–20 and scan it left to right; each time you reach a number that is still on the board, pick it.

Why: the unscanned part of a random order is itself in random order, and every number still on the board is unscanned, so the next one reached is uniform among those on the board, exactly like a uniform pick.

2. Let A_k = the multiples of k in 1–20, so $|A_k| = \lfloor 20/k \rfloor$. Claim: k is picked exactly when k comes

first among the numbers of A_k in the order.

Why: proved in the next two steps.

3. If k is first in A_k , it is still on the board when reached, so it is picked.

Why: k can only be erased when some multiple of k is picked, and every multiple of k comes later in the order.

4. If some other multiple m of k comes before k , then k is erased before it is reached, so it is not picked.

Why: when the scan reaches m , either m is picked (erasing its divisor k), or m was already erased by an earlier pick m' that is a multiple of m ; then $k \mid m \mid m'$, so that pick erased k too.

5. $P(k \text{ is picked}) = \frac{1}{\lfloor 20/k \rfloor}$.

Why: in a random order, each element of A_k is equally likely to come first.



$A_5 = \{5, 10, 15, 20\}$: whichever comes first is picked
 $5 \text{ is picked} \Leftrightarrow 5 \text{ is first of the 4} \Rightarrow 1/4$

6. $\mathbb{E}[\text{\#moves}] = \sum_{k=1}^{20} \frac{1}{\lfloor 20/k \rfloor}$. The terms are $\frac{1}{20}, \frac{1}{10}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3}$ for $k = 1..6$, $\frac{1}{2}$ for $k = 7..10$, and 1 for $k = 11..20$.

Sum = $1.1 + 2 + 10 = 13.1$.

Why: linearity; $\frac{1}{20} + \frac{1}{10} + \frac{1}{6} + \frac{1}{5} + \frac{1}{4} + \frac{1}{3} = 1.1$. (Simulated 13.10.)

Conclusion. The game lasts 13.1 moves on average. ■

△ Course asserts “the order within A_k is random” without step 1 (why sequential uniform picks behave like a random permutation).

30 · Products of independent variables

If $X \perp Y$, then $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$; likewise for n independent variables. This **needs** independence. Trick: expand a square or product, and every cross term containing a mean-zero independent factor vanishes. Centre coordinates first when you are allowed to (e.g. area doesn't change under translation).

Cut a 1×2 rectangle

A 1×2 rectangle is cut by a straight line. Pick the corner where a short side and a long side meet. The line joins the point on the short side at distance X from that corner (X uniform on $[0,1]$) to the point on the long side at distance Y from it (Y uniform on $[0,2]$), independently.

Question: what is the expected area of the larger of the two pieces?

Key idea

The corner triangle has area $XY/2 \leq 1$, so the other piece is always the larger.

Picture

The corner triangle is never more than half (area $XY/2 \leq 1$), so the larger piece is $2 - XY/2$. X and Y are independent: $E[XY] = \frac{1}{2} \times 1$.

Run once

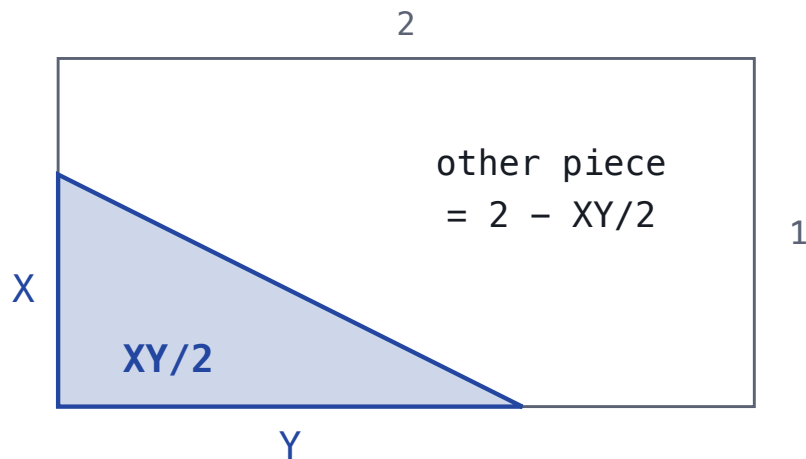
Run 100,000

Solution

Setup. A 1×2 rectangle (area 2). From one corner, $X \sim U(0,1)$ is marked along the short side and $Y \sim U(0,2)$ along the long side, independently; the cut joins the two marks.

1. The corner piece is a right triangle with legs X and Y , of area $\frac{XY}{2} \leq \frac{1 \cdot 2}{2} = 1$.

Why: $X \leq 1$ and $Y \leq 2$.



2. So the other piece, of area $2 - \frac{XY}{2} \geq 1$, is always the larger one.

Why: the two areas add to 2 and the triangle has at most half.

3. $E[\text{larger}] = 2 - \frac{1}{2}E[XY] = 2 - \frac{1}{2}E[X]E[Y] = 2 - \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{7}{4}$.

Why: linearity; independence gives $E[XY] = E[X]E[Y]$.

Conclusion. The larger piece has expected area $\frac{7}{4}$. ■

Stock after 10 days

A stock starts at price 100. Each day, independently, its price is multiplied by 1.05 with probability $\frac{3}{5}$ or by 0.97 with probability $\frac{2}{5}$.

Question: what is the expected price after 10 days?

Key idea

The expectation of a product of independent factors is the product of their expectations.

Picture

Independent factors: the average of the product is the product of the averages. Each day multiplies the price by 1.018 on average, so 100×1.018^{10} .

Run once

Run 100,000

Solution

Setup. Start at 100. Each day the price is multiplied by a factor F_i : 1.05 with probability $\frac{3}{5}$, 0.97 otherwise, independently.

$$1. \quad \mathbb{E}[F_i] = 0.6 \cdot 1.05 + 0.4 \cdot 0.97 = 0.63 + 0.388 = 1.018.$$

Why: definition of expectation.

$$2. \quad \text{The price after 10 days is } 100 F_1 F_2 \cdots F_{10}, \text{ so its mean is } 100 \cdot \mathbb{E}[F_1] \cdots \mathbb{E}[F_{10}] = 100 \cdot 1.018^{10} \approx 119.53024.$$

Why: the expectation of a product of independent variables is the product of the expectations.

Conclusion. The expected price is about 119.53. ■

Random jumps, $\mathbb{E}[\text{distance}^2]$

A point starts at the origin and makes 10 jumps. Each jump goes in a random direction (angle uniform on $[0, 2\pi)$) with a random length (uniform on $[0, 1]$); all of these are independent.

Question: what is the expected value of the squared distance from the origin at the end?

Key idea

Cross terms contain $\mathbb{E}[\cos \theta] = \mathbb{E}[\sin \theta] = 0$.

Picture

Expand the squared distance. Cross terms between different jumps average to 0 (random directions cancel). What is left is the sum of the squared jump lengths: $10 \times \frac{1}{3}$.

Run once

Run 100,000

Solution

Setup. 10 jumps; jump i has length $r_i \sim U[0, 1]$ and direction $\theta_i \sim U[0, 2\pi)$, all independent. The final position is $(X, Y) = \sum_i (r_i \cos \theta_i, r_i \sin \theta_i)$.

1. $X^2 = \sum_i r_i^2 \cos^2 \theta_i + \sum_{i \neq j} r_i r_j \cos \theta_i \cos \theta_j$, and similarly for Y^2 with sines.

Why: expand the square of a sum.

2. Every cross term has mean 0: $\mathbb{E}[r_i \cos \theta_i \cdot r_j \cos \theta_j] = \mathbb{E}[r_i \cos \theta_i] \mathbb{E}[r_j \cos \theta_j] = 0$.

Why: different jumps are independent, and $\mathbb{E}[\cos \theta] = \mathbb{E}[\sin \theta] = 0$ for a uniform angle.

3. The squared terms add to $r_i^2(\cos^2 \theta_i + \sin^2 \theta_i) = r_i^2$.

Why: $\cos^2 + \sin^2 = 1$.

4. $\mathbb{E}[X^2 + Y^2] = \sum_{i=1}^{10} \mathbb{E}[r_i^2] = 10 \cdot \frac{1}{3} \approx 3.33333$.

Why: $\mathbb{E}[r^2] = \int_0^1 r^2 dr = \frac{1}{3}$.

Conclusion. $\mathbb{E}[\text{distance}^2] = \frac{10}{3}$. ■

$\mathbb{E}[\text{area}^2]$ for 3 grid points (9×9)

Three points are chosen independently and uniformly from the 81 lattice points of a 9×9 grid.

Question: what is the expected value of the *square* of the area of the triangle they form?

Key idea

Shift the coordinates to -4.4 (area is unchanged) so every coordinate has mean 0. Then only the 6 squared terms survive.

Picture

Centre the grid at 0 first (moving the triangle doesn't change its area). Then every coordinate

averages 0, all cross terms vanish, and only 6 squared terms remain: $6 \times (20/3)^2 \div 4$.

Run once

Run 200,000

Solution

Setup. Three points are chosen independently and uniformly from a 9×9 grid of lattice points; every coordinate is independent and uniform over 9 consecutive integers. Shift the grid so the coordinates run over $-4, \dots, 4$; this doesn't change the area.

1. After the shift, each coordinate has mean 0 and $\mathbb{E}[x^2] = \frac{2(1+4+9+16)}{9} = \frac{60}{9} = \frac{20}{3}$.

Why: the values $-4, \dots, 4$ are symmetric about 0.

2. The area is $A = \frac{1}{2}|D|$ with $D = x_1y_2 + x_2y_3 + x_3y_1 - x_2y_1 - x_3y_2 - x_1y_3$, so $A^2 = \frac{1}{4}D^2$.

Why: the shoelace formula.

3. D is a sum of the 6 terms $\pm x_a y_b$ with $a \neq b$. In D^2 , a cross term multiplies two different terms $x_a y_b$ and $x_c y_d$.

If $a \neq c$: x_a appears exactly once.

If $a = c$: then $b \neq d$, and y_b appears exactly once.

Either way the cross term has mean 0.

Why: a coordinate appearing to the first power is independent of the others and has mean 0 (step 1).

4. So $\mathbb{E}[D^2] =$ the sum of the 6 squared terms $= 6 \mathbb{E}[x^2] \mathbb{E}[y^2] = 6 \cdot \left(\frac{20}{3}\right)^2 = \frac{800}{3}$.

Why: each squared term is $x_a^2 y_b^2$ with $a \neq b$, a product of independent factors.

$$5. \mathbb{E}[A^2] = \frac{1}{4} \cdot \frac{800}{3} = \frac{200}{3} \approx 66.66667.$$

Why: step 2. (Exact brute force agrees.)

Conclusion. $\mathbb{E}[\text{area}^2] = \frac{200}{3}$. ■

31 · Means of common distributions

Distribution	$\mathbb{E}[X]$	Distribution	$\mathbb{E}[X]$
Bernoulli(p)	p	$U(a, b)$	$\frac{a+b}{2}$
Binomial(n, p)	np	$\text{Exp}(\lambda)$	$1/\lambda$
Geometric(p)	$1/p$	$N(\mu, \sigma^2)$	μ
NegBin(r, p)	r/p	$\text{Gamma}(\alpha, \text{rate } \lambda)$	α/λ
Poisson(λ)	λ	$\text{Beta}(\alpha, \beta)$	$\frac{\alpha}{\alpha+\beta}$
Uniform{1.. n }	$\frac{n+1}{2}$	$\text{Hypergeo}(N, K, n)$	nK/N

Short proofs. Binomial and hypergeometric: a sum of n indicators, each with mean p (or K/N). NegBin: a sum of r geometrics. Geometric by first-step analysis: $E = 1 + (1 - p)E$, so $E = 1/p$. Exponential: tail integral $\int_0^\infty e^{-\lambda t} dt = 1/\lambda$.

32 · Recursion in expectation

The [§5 template](#) again, with a cost: a continuing outcome contributes $\text{prob} \times (\text{cost} + \text{rec}(i+1, s'))$, and

$\text{rec} = 0$ once the process has ended. A self-loop $\text{rec} = c + q\text{rec}$ solves to $\text{rec} = \frac{c}{1-q}$.

Sum of rolls until a 6

A fair die is rolled until the first 6 appears.

Question: what is the expected sum of all the rolls, including the final 6?

Picture

Each roll adds its value, then either stops (a 6) or repeats the same game. $x = \frac{1}{6} \cdot 6 + \sum_{k=1}^5 \frac{1}{6}(k + x)$, so $x = 21$. (Also: 6 rolls on average $\times 3.5$ per roll.)

Run once

Run 100,000

Solution

Form 1 (i → end) · Turn = one roll. 6 → adds 6 and ends. 1–5 → adds that value, next turn.

1. **LEVEL.** $\text{rec}(i)$ = the expected sum of the rolls from turn i onward (turn i included), given no 6 in turns 1 to $i-1$. The answer is $\text{rec}(1)$.

Why: 'from turn i onward' is the amount still to come.

2. **CHOICE.** At turn i the roll is k , for $k = 1, \dots, 6$, each with probability $\frac{1}{6}$.

Why: a fair die.

3. **CHECK.** $k = 6$: adds 6, ends. $k \leq 5$: adds k , continue to turn $i+1$.

Why: the rules.

4. **MOVE.** $\text{rec}(i) = \frac{1}{6} \cdot 6 + \sum_{k=1}^5 \frac{1}{6} (k + \text{rec}(i+1))$.

Why: law of total expectation over turn i : a continuing outcome contributes $\text{prob} \times (\text{this roll's value} + \text{what is still to come})$.

5. **MEMORYLESS.** At turn $i+1$ the roll is again uniform on 1–6 with the same stopping rule, so $\text{rec}(i+1) = \text{rec}(i)$.

Why: each turn is independent of earlier turns, has the same probabilities and the same rules, and nothing from earlier turns is carried forward.

6. **SOLVE.** Let $x = \text{rec}(i)$. $x = 1 + \frac{1}{6}(1 + 2 + 3 + 4 + 5) + 5 \cdot \frac{1}{6}x = 1 + \frac{15}{6} + \frac{5}{6}x = \frac{21}{6} + \frac{5}{6}x$. So $\frac{1}{6}x = \frac{21}{6}$, so $x = 21$.

Why: $\frac{1}{6} \cdot 6 = 1$; five rec terms each carry $\frac{1}{6}$; move x terms left; multiply by 6.

Answer. The expected sum is 21. (Check: 6 rolls on average times a mean of 3.5, by Wald's identity on the Martingales page.) ■

Bob's routes

Bob is lost. At the start he picks one of three routes at random:

Route A (probability 0.2): 4 hours, back to the start.

Route B (probability 0.5): 2 hours, back to the start.

Route C (probability 0.3): 3 hours, home.

Every time he is back at the start he picks again in the same way.

Question: what is the expected time until Bob gets home?

Picture

Every return to the start is a fresh start. $x = 0.2(4 + x) + 0.5(2 + x) + 0.3 \cdot 3$, so $0.3x = 2.7$ and $x = 9$.

Run once

Run 100,000

Solution

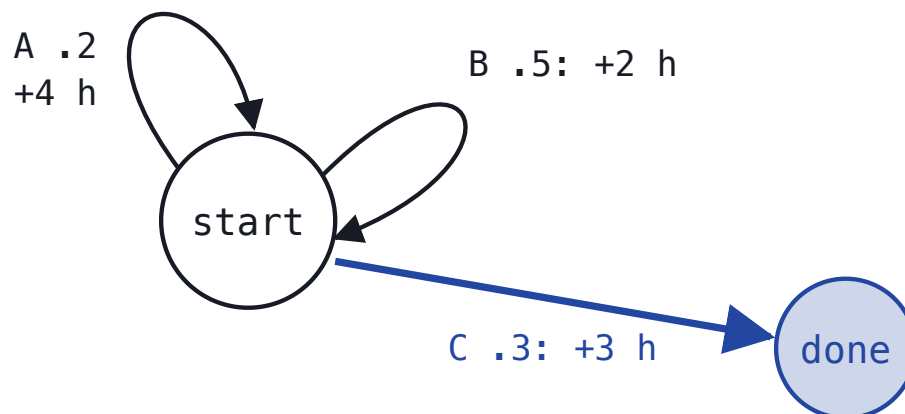
Form 1 (i → end) · **Turn** = one route choice. C → adds 3 h and ends. A or B → adds its hours, back at the start, next turn.

1. **LEVEL.** $\text{rec}(i)$ = the expected hours from Bob's i -th route choice until he is home. The answer is $\text{rec}(1)$.

Why: the time still to come from turn i .

2. **CHOICE.** At turn i : route A (0.2), B (0.5) or C (0.3).

Why: given.



$$\text{rec} = .2(4 + \text{rec}) + .5(2 + \text{rec}) + .3 \cdot 3 \Rightarrow 9$$

3. **CHECK.** C: adds 3 h, ends. A: adds 4 h, continue to turn $i+1$. B: adds 2 h, continue to turn $i+1$.

Why: the rules.

4. **MOVE.** $\text{rec}(i) = 0.2(4 + \text{rec}(i + 1)) + 0.5(2 + \text{rec}(i + 1)) + 0.3 \cdot 3$.

Why: law of total expectation over the route.

5. **MEMORYLESS.** Back at the start Bob picks with the same probabilities and the routes take the same times. So $\text{rec}(i + 1) = \text{rec}(i)$.

Why: each choice is independent of earlier ones; nothing is carried forward.

6. **SOLVE.** Let $x = \text{rec}(i)$. $x = 0.8 + 0.2x + 1 + 0.5x + 0.9 = 2.7 + 0.7x$. So $0.3x = 2.7$, so $x = 9$ hours.

Why: expand; move x terms left; divide by 0.3 .

Answer. Bob needs 9 hours on average. ■

Tosses until THH

A fair coin is tossed until the pattern T, H, H appears in three consecutive tosses.

Question: what is the expected number of tosses?

Picture

Track the useful ending: nothing, T, or TH. Each toss costs 1. Solve the three "1 + average of next states" equations.

Run once

Run 100,000

Solution

Form 1 ($i \rightarrow \text{end}$) · **Turn** = one toss. State s = the useful ending: $\emptyset$, T or TH . TH then $H \rightarrow \text{ends}$. Each toss costs 1.

1. **LEVEL**. $\text{rec}(i, s)$ = expected tosses from toss i onward until THH , given the useful ending before toss i is $s \in \{\emptyset, T, TH\}$. The answer is $\text{rec}(1, \emptyset)$.

Why: a T is always the start of a new attempt; an H with no T before it is useless.

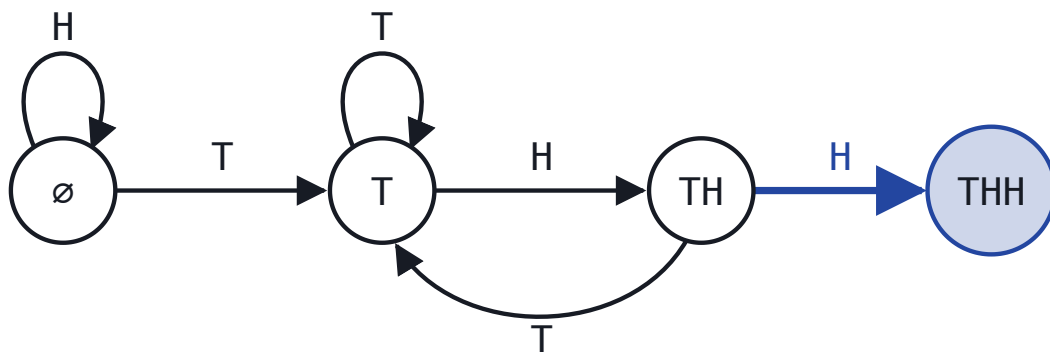
2. **CHOICE**. Toss i is H or T , each $\frac{1}{2}$; each toss costs 1.

Why: a fair coin.

3. **CHECK**. $\emptyset$: $T \rightarrow T$, $H \rightarrow \emptyset$. T : $H \rightarrow TH$, $T \rightarrow T$. TH : $H \rightarrow THH$, ends (0 more), $T \rightarrow T$.

Why: read off the pattern.

every edge $\frac{1}{2}$



4. **MOVE**. $\text{rec}(i, \emptyset) = 1 + \frac{1}{2}\text{rec}(i+1, T) + \frac{1}{2}\text{rec}(i+1, \emptyset)$.
 $\text{rec}(i, T) = 1 + \frac{1}{2}\text{rec}(i+1, TH) + \frac{1}{2}\text{rec}(i+1, T)$.
 $\text{rec}(i, TH) = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}\text{rec}(i+1, T)$.

Why: law of total expectation over toss i .

5. **MEMORYLESS**. $\text{rec}(i+1, s) = \text{rec}(i, s) = \text{rec}(s)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has

no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. **SOLVE.** $\text{rec}(\emptyset) = 1 + \frac{1}{2}\text{rec}(T) + \frac{1}{2}\text{rec}(\emptyset)$, so $\frac{1}{2}\text{rec}(\emptyset) = 1 + \frac{1}{2}\text{rec}(T)$, $\text{rec}(\emptyset) = 2 + \text{rec}(T)$. Same algebra: $\text{rec}(T) = 2 + \text{rec}(TH)$. $\text{rec}(TH) = 1 + \frac{1}{2}\text{rec}(T) = 1 + \frac{1}{2}(2 + \text{rec}(TH)) = 2 + \frac{1}{2}\text{rec}(TH)$, so $\text{rec}(TH) = 4$. Then $\text{rec}(T) = 6$ and $\text{rec}(\emptyset) = 8$.

Why: move each self-loop term left and multiply by 2; substitute upwards.

Answer. THH takes 8 tosses on average. ■

At least 3 H and 3 T

A fair coin is tossed until at least 3 heads and at least 3 tails have appeared in total (not necessarily in a row).

Question: what is the expected number of tosses?

Picture

Fill a small grid of (heads so far, tails so far).

Once one side reaches 3, you are just waiting for the other: 2 tosses per missing one on average.

Rows h = heads so far, columns t = tails so far (capped at 3). Click a cell.

$h \setminus t$	0	1	2	3
0	$\frac{63}{8}$	$\frac{55}{8}$	$\frac{25}{4}$	6
1	$\frac{55}{8}$	$\frac{11}{2}$	$\frac{9}{2}$	4
2	$\frac{25}{4}$	$\frac{9}{2}$	3	2
3	6	4	2	0

$$\text{rec}(0,0) = 1 + \frac{1}{2} \cdot \text{rec}(1,0) + \frac{1}{2} \cdot \text{rec}(0,1) = 1 + \frac{1}{2} \cdot 55/8 + \frac{1}{2} \cdot 55/8 = 63/8$$

Solution

Form 1 (i → end) · **Turn** = one toss. State (h, t) = heads and tails so far, each capped at 3. (3, 3) → ends. Each toss costs 1.

1. **LEVEL.** $\text{rec}(i, h, t)$ = expected tosses from toss i onward, given h heads and t tails before toss i (each capped at 3). The answer is $\text{rec}(1, 0, 0)$.

Why: only whether each count has reached 3 affects the future.

2. **CHOICE.** Toss i is H or T, each $\frac{1}{2}$; each toss costs 1.

Why: a fair coin.

3. **CHECK.** (3, 3): ends, 0 more. H → ($i+1$, $\min(h+1, 3)$, t). T → ($i+1$, h , $\min(t+1, 3)$).

Why: the cap: a 4th head changes nothing.

4. **MOVE.** $\text{rec}(i, h, t) = 1 + \frac{1}{2} \text{rec}(i+1, h+1, t) + \frac{1}{2} \text{rec}(i+1, h, t+1)$ (counts capped at 3).

Why: law of total expectation over toss i .

5. **MEMORYLESS.** $\text{rec}(i+1, h, t) = \text{rec}(i, h, t) = \text{rec}(h, t)$. On an edge the capped count stays put, a self-loop: $\text{rec}(3, t) = 1 + \frac{1}{2} \text{rec}(3, t) + \frac{1}{2} \text{rec}(3, t+1)$, so $\frac{1}{2} \text{rec}(3, t) = 1 + \frac{1}{2} \text{rec}(3, t+1)$, i.e. $\text{rec}(3, t) = 2 + \text{rec}(3, t+1)$. Same for $\text{rec}(h, 3)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from ($i+1, s$) is exactly the

future from (i, s). Solve the self-loop by algebra instead of recursing into it.

6. SOLVE.

```
double dp[4][4]; // all -1
double rec(ll h, ll t) {
    if (h == 3 && t == 3) return 0; // base case
    if (dp[h][t] != -1) return dp[h][t]; // cache check
    double ans;
    if (h == 3) ans = 2 + rec(3, t + 1); // self-loop solved
    else if (t == 3) ans = 2 + rec(h + 1, 3);
    else ans = 1 + 0.5 * rec(h + 1, t) + 0.5 * rec(h, t + 1);
    return dp[h][t] = ans; // save and return
} // rec(0, 0) cost: 16 states x 2 transitions
```

Filling the grid from (3, 3) back to (0, 0) gives $\text{rec}(0, 0) = \frac{63}{8} = 7.875$.

Why: counts only go up (apart from the solved self-loops), so fill from the corner (3, 3).

	tails j →				
	7.875	6.875	6.25	6	
	6.875	5.5	4.5	4	
	6.25	4.5	3	2	
	6	4	2	0	
heads ↓					

move right: T, down: H

edges (orange): one side capped, a self-loop: 2 + next

start: 63/8

Answer. It takes $\frac{63}{8} = 7.875$ tosses on average. ■

33 · Symmetry in expectation I

If a transformation T keeps the law of the randomness unchanged and sends X to $c - X$ (or $-X$), then $\mathbb{E}[X] = \mathbb{E}[c - X]$, so $\mathbb{E}[X] = c/2$ (or 0). For random walks, pair the moves m and $-m$: $\mathbb{E}|v + m|^2 = |v|^2 + \mathbb{E}|m|^2$.

Median of five d100

Five fair 100-sided dice (faces 1 to 100) are rolled. Their **median** is the third smallest of the five results.

Question: what is the expected value of the median?

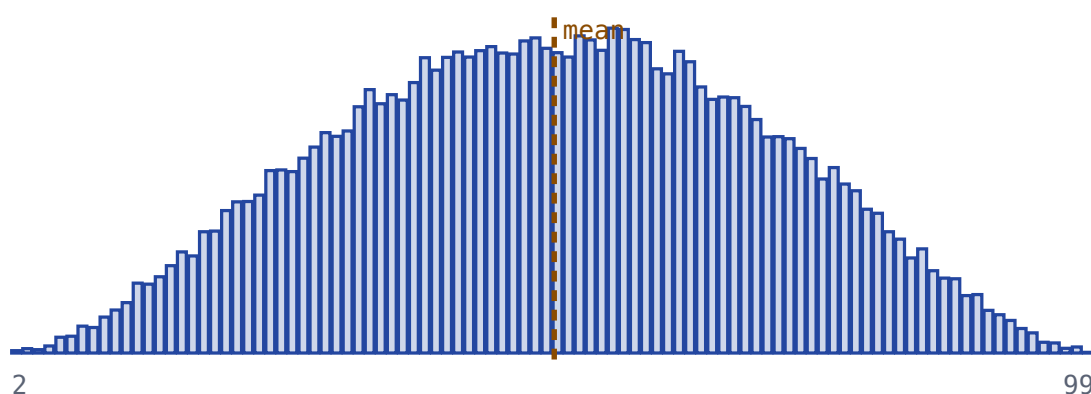
Key idea

Replacing each roll r by $101 - r$ keeps the law and sends the median z to $101 - z$.

Picture

Flip every roll: $r \rightarrow 101 - r$. The rolls look just as random, and the median becomes $101 - \text{median}$. So the median and $101 - \text{median}$ have the same average: 50.5.

The median of five d100 rolls: a symmetric histogram centred at 50.5.



Sample 50,000

Mean **50.604**, variance **358.997** (exact mean 50.5).**Solution**

Setup. Five independent rolls of a 100-sided die; z = their median (the third smallest).

1. Replacing every roll r by $101 - r$ keeps the rolls independent and uniform on $\{1, \dots, 100\}$.

Why: $r \mapsto 101 - r$ is a bijection of $\{1..100\}$ onto itself.

2. It reverses the sorted order, so the median z becomes $101 - z$.

Why: the third smallest of the new values is 101 minus the third largest of the old ones, which is also the old third smallest (5 values).

3. So z and $101 - z$ have the same distribution: $\mathbb{E}[z] = \mathbb{E}[101 - z] = 101 - \mathbb{E}[z]$, which gives $\mathbb{E}[z] = 50.5$.

Why: steps 1–2; solve $2\mathbb{E}[z] = 101$.

Conclusion. The expected median is 50.5. ■

 $\mathbb{E}[\det]$ of a random 5×5 matrix

Each entry of a 5×5 matrix is chosen independently and uniformly from $\{1, 2, \dots, 25\}$.

Question: what is the expected value of the matrix's determinant?

Key idea

Swapping two rows keeps the law and flips the sign of the determinant.

Picture

Swap two rows. The new matrix is just as likely as the old one, and its determinant is the negative. So the average equals minus itself: 0.

A random 5×5 matrix. Swapping two rows gives another equally likely matrix with the opposite determinant, so positive and negative values cancel on average.

New matrix

Swap rows 1 and 2

23	9	14	4	17
24	4	22	16	22
24	7	3	4	23
1	18	17	8	16
16	8	7	13	23

det = **-273,156**

Solution

Setup. A is a 5×5 matrix with i.i.d. entries uniform on $\{1, \dots, 25\}$. The determinant is bounded, so its expectation exists.

1. Swapping rows 1 and 2 gives a matrix A' with the same distribution as A .

Why: the entries are i.i.d., so rearranging them doesn't change the joint law.

2. $\det A' = -\det A$.

Why: swapping two rows flips the sign of the determinant.

3. $\mathbb{E}[\det A] = \mathbb{E}[\det A'] = -\mathbb{E}[\det A]$, so $\mathbb{E}[\det A] = 0$.

Why: step 1 for the first equality, step 2 for the second; a number equal to its negative is 0.

Conclusion. The expected determinant is 0. ■

Knight's walk, $\mathbb{E}[\text{distance}^2]$ after 100 moves

A knight starts on some square of an infinite chessboard and makes 100 moves. Each move is chosen uniformly from its 8 possible knight moves (2 squares one way and 1 square the other).

Question: what is the expected value of the squared distance between its starting square and its final square?

Key idea

The move m is independent of the position v , and $\mathbb{E}[m] = 0$.

Picture

Each move adds exactly 5 to the average squared distance. The move is independent of where you are and averages to zero, so the cross term vanishes; every knight move has $\text{length}^2 = 1 + 4 = 5$.

Run once

Run 100,000

Solution

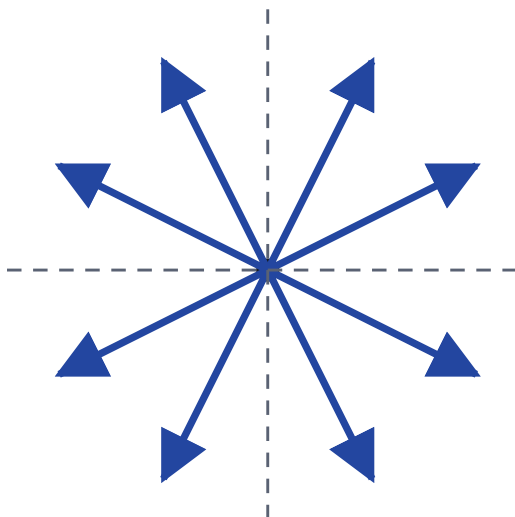
Setup. A knight on an infinite board makes 100 moves, each chosen uniformly from its 8 moves $(\pm 1, \pm 2), (\pm 2, \pm 1)$, independently. v_k = its position after k moves (starting at the origin), and m = the next move.

$$1. |v + m|^2 = |v|^2 + 2v \cdot m + |m|^2.$$

Why: expand the dot product $(v + m) \cdot (v + m)$.

$$2. \mathbb{E}[v \cdot m] = 0.$$

Why: m is independent of v , so $\mathbb{E}[v_x m_x + v_y m_y] = \mathbb{E}[v_x] \mathbb{E}[m_x] + \mathbb{E}[v_y] \mathbb{E}[m_y]$, and $\mathbb{E}[m] = 0$ because the 8 moves come in $\pm$ pairs.



8 moves cancel in $\pm$ pairs
 $\Rightarrow \mathbb{E}[m] = 0$

each $|m|^2 = 2^2 + 1^2 = 5$

100 moves $\Rightarrow$ 500

$$3. |m|^2 = 2^2 + 1^2 = 5 \text{ for every move.}$$

Why: all 8 moves have this length.

$$4. \text{ So each move adds exactly 5 to } \mathbb{E}|v|^2, \text{ and after 100 moves } \mathbb{E}|v_{100}|^2 = 500.$$

Why: steps 1–3, starting from $|v_0|^2 = 0$.

Conclusion. The expected squared distance after 100 moves is 500. ■

34 · Symmetry in expectation

II: fewer states

Ant to the opposite corner of a cube

An ant walks on the corners of a cube. At each step it moves along one of the 3 edges at its corner, chosen at random.

Question: what is the expected number of steps for the ant to get from one corner to the opposite corner?

Picture

Only the distance to the target (0–3) matters. From distance d you step closer with chance $d/3$. Solve three "1 + average" equations.

Distance to the opposite corner. "Run 10 000": about 10 steps on average.



Step

Run until stuck

Reset

Run 10 000

click a state to start there

Step 0: at **d=3**.

Solution

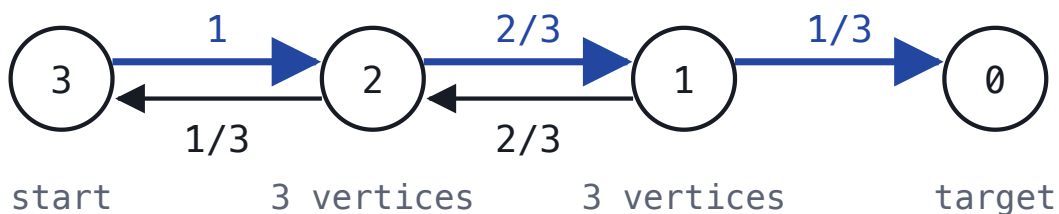
Form 1 ($i \rightarrow \text{end}$) · Turn = one step. State d = distance to the target (coordinates still different). $d = 0 \rightarrow \text{ends}$. Each step costs 1.

1. **LEVEL.** $\text{rec}(i, d)$ = expected steps from step i onward until the target, given distance d before step i . The answer is $\text{rec}(1, 3)$.

Why: by symmetry only the distance matters.

2. **CHOICE.** From distance d , the step flips one of the 3 coordinates: closer ($d \rightarrow d-1$) with probability $\frac{d}{3}$, farther ($d \rightarrow d+1$) with $\frac{3-d}{3}$.

Why: flipping one of the d differing coordinates brings the ant closer.



3. **CHECK.** $d = 0$: ends, 0 more steps. Otherwise the step costs 1 and we continue to $(i+1, d \pm 1)$.

Why: the rules.

4. **MOVE.** $\text{rec}(i, d) = 1 + \frac{d}{3} \text{rec}(i+1, d-1) + \frac{3-d}{3} \text{rec}(i+1, d+1)$.

Why: law of total expectation over the step.

5. **MEMORYLESS.** $\text{rec}(i+1, d) = \text{rec}(i, d) = \text{rec}(d)$.

Why: the rules at a turn depend only on the state s , never on the turn number i , and the coin/die/walk has no memory. So the future from $(i+1, s)$ is exactly the future from (i, s) .

6. SOLVE.

$$d = 3: \text{rec}(3) = 1 + \text{rec}(2).$$

$$d = 1: \text{rec}(1) = 1 + \frac{2}{3}\text{rec}(2) \quad (\text{rec}(0) = 0).$$

$$d = 2: \text{rec}(2) = 1 + \frac{2}{3}\text{rec}(1) + \frac{1}{3}\text{rec}(3) = 1 + \frac{2}{3}(1 + \frac{2}{3}\text{rec}(2)) + \frac{1}{3}(1 + \text{rec}(2)) = 2 + \frac{7}{9}\text{rec}(2).$$

So $\frac{2}{9}\text{rec}(2) = 2$, **$\text{rec}(2) = 9$** . Then $\text{rec}(3) = 10$ (and $\text{rec}(1) = 7$).

Why: $1 + \frac{2}{3} + \frac{1}{3} = 2$ and $\frac{4}{9} + \frac{1}{3} = \frac{7}{9}$.

Answer. The ant needs 10 steps on average. ■

35 · Random points on a segment

n i.i.d. uniform points cut $[0, L]$ into $n + 1$ gaps that are **exchangeable** (every gap has the same law). So each has mean $\frac{L}{n+1}$. The same holds for the gaps between randomly shuffled special items.

Blue end of a 9-inch rod

A rod 9 inches long has one end painted blue. It is broken at two points chosen independently and uniformly along it, giving 3 pieces.

Question: what is the expected length of the piece with the blue end?

Key idea

3 exchangeable pieces.

Picture

The three pieces are interchangeable: each has the same average, and they add to 9. So each averages 3, including the blue one.

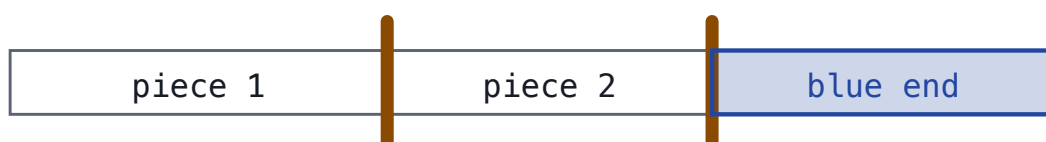
Run onceRun 100,000

Solution

Setup. A 9-inch rod is broken at two independent uniform points, giving 3 pieces; one end piece is painted blue.

1. The 3 pieces are exchangeable: each has the same distribution.

Why: the gaps between n i.i.d. uniform points on a segment are exchangeable (§35).



3 exchangeable pieces $\Rightarrow$ each has mean $9/3 = 3$

2. Their lengths add to 9, so each has mean 3. The blue end is one of them: 3 in.

Why: linearity: 3 equal means summing to 9.

Conclusion. The blue end is 3 inches long on average. ■

Smaller piece of a broken metre stick

A metre stick is broken at a point chosen uniformly at random along it.

Question: what is the expected length of the shorter piece?

Key idea

By symmetry, look at a break in the left half. It is uniform on $[0, \frac{1}{2}]$, and the smaller piece is exactly that length.

Picture

The shorter piece is uniform on $[0, \frac{1}{2}]$ (fold the stick in half), so it averages $\frac{1}{4}$.

Run once

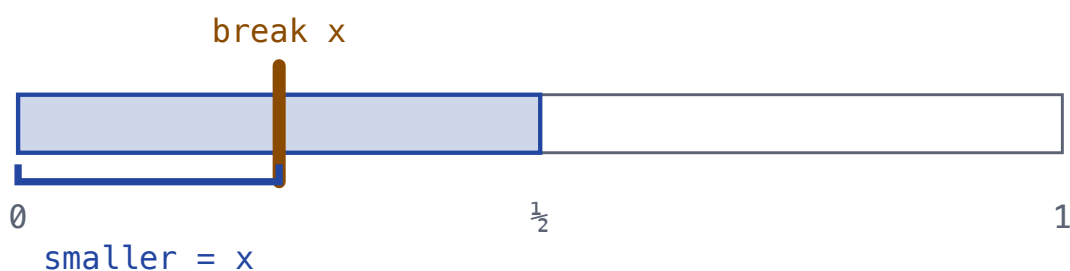
Run 100,000

Solution

Setup. A metre stick is broken at $U \sim U[0,1]$. The smaller piece has length $\min(U, 1-U)$.

1. If $U \leq \frac{1}{2}$, the smaller piece is U ; given $U \leq \frac{1}{2}$, U is uniform on $[0, \frac{1}{2}]$, with mean $\frac{1}{4}$.

Why: a uniform variable conditioned on a subinterval stays uniform there.



break in left half (by symmetry): $x \sim U[0, \frac{1}{2}] \Rightarrow \text{mean } \frac{1}{4}$

2. If $U > \frac{1}{2}$, the smaller piece is $1 - U$, which is again uniform on $[0, \frac{1}{2}]$, with mean $\frac{1}{4}$.

Why: symmetry $U \mapsto 1 - U$.

3. $\mathbb{E}[\text{smaller}] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$ m.

Why: law of total expectation.

Conclusion. The smaller piece is 25 cm on average. ■

Cards up to and including the first ace

A shuffled 52-card deck is dealt face up one card at a time.

Question: what is the expected number of cards dealt up to and including the first ace?

Key idea

Each of the 48 non-aces lies before all 4 aces with probability $\frac{1}{5}$.

Picture

Each non-ace comes before all 4 aces with chance $\frac{1}{5}$ (it is equally likely to be anywhere among those 5 cards). Add 1 for the ace itself: $1 + 48/5$.

Run once

Run 100,000

Solution

Setup. A shuffled 52-card deck. X = the number of cards up to and including the first ace = $1 +$ (the number of non-aces before the first ace).

1. For a non-ace c , let $I_c = 1$ if c comes before all 4 aces.

Why: then the number of non-aces before the first ace is $\sum_c I_c$.

2. $P(I_c = 1) = \frac{1}{5}$.

Why: the relative order of the 5 cards $\{c$ and the 4 aces $\}$ is uniform, so c is equally likely to be in any of the 5 places, and it is first with probability $\frac{1}{5}$.

4 aces cut the deck into 5 slots



each non-ace: slot 1 with probability $\frac{1}{5}$

$$48/5 + 1 \text{ (the ace)} = 10.6$$

3. $\mathbb{E}[X] = 1 + 48 \cdot \frac{1}{5} = 1 + 9.6 = 10.6$.

Why: linearity over the 48 non-aces, plus the ace itself.

Conclusion. On average 10.6 cards are turned up to reach the first ace. ■

△ Course says the 48 non-aces are “equally distributed among 5 segments” without justifying it. Step 2 is the justification.

36 · Conditional expectation

$$\mathbb{E}[X \mid A] = \sum_x x P(X = x \mid A) = \frac{\mathbb{E}[X 1_A]}{P(A)}.$$

Always compute the conditional distribution;

intuition fails here. For "repeat until" processes, don't condition inside the recursion: run the §5 template twice, for $p = P(A)$ and $g = \mathbb{E}[X 1_A]$, then divide.

Rolls until 6, given no 5 came first

A fair die is rolled until a 6 appears.

Question: given that no 5 was rolled before that 6, what is the expected number of rolls?

Picture

The game is really decided at the first roll that is a 5 or a 6 (chance $\frac{1}{3}$ per roll, so 3 rolls on average). "No 5 first" just says that deciding roll was a 6.

Whether it is a 5 or a 6 doesn't depend on *when* it comes, so the condition doesn't change the length: 3. The template below confirms it: $g \div p = (3/2) \div (1/2)$.

Roll until 6. Keep only games with no 5 before the 6. Average their length.

Run once

Run 100 000

Solution

Form 1 (i → end) · Turn = one roll. 6 → ends (A happens). 5 → ends (A fails). 1–4 → next turn. Run the template twice: for $p = P(A)$ and for $g = \mathbb{E}[N \cdot 1_A]$.

1. **SPLIT.** N = the number of rolls until the first 6.
 A = 'no 5 before that 6'. $1_A = 1$ if A happens,
 else 0. $\mathbb{E}[N \mid A] = \frac{\mathbb{E}[N 1_A]}{P(A)}$. On A the game ends at
 the first 5-or-6, so we only follow it until then.

Why: definition of conditional expectation. Each of
 the two pieces is an ordinary 'repeat until' quantity,
 so the template applies to each.

2. **LEVEL.** Both given that rolls 1 to $i-1$ were all in
 $\{1, 2, 3, 4\}$: $p(i) = P(A, \text{looking from turn } i)$.
 $g(i) = \mathbb{E}[R_i 1_A]$, where R_i = rolls from turn i
 onward (turn i included). We need $p(1)$ and $g(1)$,
 since $R_1 = N$.

Why: definitions.

3. **CHOICE.** At turn i : 6 ($\frac{1}{6}$), 5 ($\frac{1}{6}$), or 1-4 ($\frac{4}{6}$).

Why: a fair die.

4. **CHECK.** 6: ends, A happens, $R_i = 1$. 5: ends, A
 fails, $1_A = 0$. 1-4: continue to turn $i+1$, and $R_i =$
 $1 + R_{i+1}$.

Why: the rules.

5. **MOVE.** $p(i) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 0 + \frac{4}{6} p(i+1)$. $g(i) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 0 +$
 $\frac{4}{6} \mathbb{E}[(1 + R_{i+1})1_A] = \frac{1}{6} + \frac{4}{6} (p(i+1) + g(i+1))$.

Why: total probability / total expectation over turn
 i . $(1 + R_{i+1})1_A = 1_A + R_{i+1}1_A$, with expectations $p(i+1)$
 and $g(i+1)$: this roll's 1 only counts if A happens.

6. **MEMORYLESS.** At turn $i+1$ the roll is again 6, 5 or
 1-4 with the same probabilities and rules, so
 $p(i+1) = p(i)$ and $g(i+1) = g(i)$.

Why: each turn is independent of earlier turns, has
 the same probabilities and the same rules, and nothing

from earlier turns is carried forward.

7. **SOLVE.** $p = \frac{1}{6} + \frac{4}{6}p$, so $\frac{2}{6}p = \frac{1}{6}$, so $p = \frac{1}{2}$. Then $g = \frac{1}{6} + \frac{4}{6}(\frac{1}{2} + g) = \frac{1}{6} + \frac{2}{6} + \frac{4}{6}g = \frac{1}{2} + \frac{2}{3}g$, so $\frac{1}{3}g = \frac{1}{2}$, so $g = \frac{3}{2}$.
 $\mathbb{E}[N | A] = \frac{g}{p} = \frac{3/2}{1/2} = 3$.

Why: collect terms; multiply by 3; then SPLIT.

Answer. $\mathbb{E}[N | A] = 3$. ■

△ The tempting answer 5 (treat the rolls as uniform on {1, 2, 3, 4, 6}) is wrong: conditioning on A is not the same as deleting the face 5. The template avoids the trap because it never conditions; it only computes p and g.

Alice's tosses, given she beats Bob

Alice and Bob each toss their own fair coin until they get heads. Suppose Alice needed strictly fewer tosses than Bob.

Question: given this, what is the expected number of tosses Alice needed?

Picture

Think in rounds where both toss. The race is decided at the first round that isn't (T, T), which happens with chance $\frac{3}{4}$ per round. Alice wins if that round is (H, T).

Which kind of round decides it doesn't depend on *when* it comes. So even given that Alice wins, the race is still a " $\frac{3}{4}$ per round" wait: $\frac{4}{3}$ rounds on average. The template below confirms it: $g \div p = (4/9) \div (1/3)$.

Both toss until heads. Keep runs where Alice needed fewer tosses; average Alice's count.

Run once

Run 100 000

Solution

Form 1 (*i* → end) · **Turn** = one round where both toss. (H, T) → ends (Alice first). (T, T) → next turn. (T, H) or (H, H) → ends (fails). Run the template for $p = P(X < Y)$ and $g = E[X \cdot 1_{X < Y}]$.

1. **SPLIT.** X, Y = Alice's and Bob's tosses until their first head. $E[X \mid X < Y] = \frac{E[X \cdot 1_{X < Y}]}{P(X < Y)}$. Think of them tossing together, one round at a time.

Why: definition of conditional expectation.

2. **LEVEL.** Both given rounds 1 to $i-1$ were (T, T): $p(i) = P(X < Y, \text{ looking from round } i)$. $g(i) = E[R_i \cdot 1_{X < Y}]$, R_i = Alice's tosses from round i onward. We need $p(1)$ and $g(1)$, since $R_1 = X$.

Why: definitions.

3. **CHOICE.** Round i , (Alice, Bob): (H, T), (T, T), (T, H), (H, H), each $\frac{1}{4}$.

Why: two independent fair coins.

4. **CHECK.** (H, T): ends, $1_{X < Y} = 1$, $R_i = 1$. (T, T): continue, $R_i = 1 + R_{i+1}$. (T, H) or (H, H): ends, $1_{X < Y} = 0$.

Why: definition of $X < Y$.

5. **MOVE.** $p(i) = \frac{1}{4} + \frac{1}{4} p(i+1)$. $g(i) = \frac{1}{4} \cdot 1 + \frac{1}{4} (p(i+1) + g(i+1))$.

Why: total probability / expectation over round i ;
 $(1 + R_{i+1})1_{X < Y}$ has expectation $p(i+1) + g(i+1)$.

6. **MEMORYLESS.** After (T, T) neither has a head and the coins are the same, so $p(i+1) = p(i)$, $g(i+1) = g(i)$.

Why: each turn is independent of earlier turns, has the same probabilities and the same rules, and nothing from earlier turns is carried forward.

7. **SOLVE.** $p = \frac{1}{4} + \frac{1}{4}p$, so $\frac{3}{4}p = \frac{1}{4}$, $p = \frac{1}{3}$. $g = \frac{1}{4} + \frac{1}{4}(\frac{1}{3} + g) = \frac{1}{4} + \frac{1}{12} + \frac{1}{4}g = \frac{1}{3} + \frac{1}{4}g$, so $\frac{3}{4}g = \frac{1}{3}$, $g = \frac{4}{9}$. $\mathbb{E}[X \mid X < Y] = \frac{4/9}{1/3} = \frac{4}{3} \approx 1.33333$.

Why: collect terms; multiply by $\frac{4}{3}$; then **SPLIT**.

Answer. Given that she beats Bob, Alice needed $\frac{4}{3}$ tosses on average. ■

$\mathbb{E}[X \mid X < Y]$ for two uniforms

X and Y are independent and uniform on $[0,1]$.

Question: find the expected value of X given that $X < Y$.

Key idea

The conditional density is proportional to $P(Y > x) = 1 - x$.

Picture

Given $X < Y$, small X is favoured: the chance Y beats x is $1 - x$. So the conditional density is $2(1 - x)$, which averages $\frac{1}{3}$.

Two random numbers. Keep pairs with $X < Y$; average X .

Run once

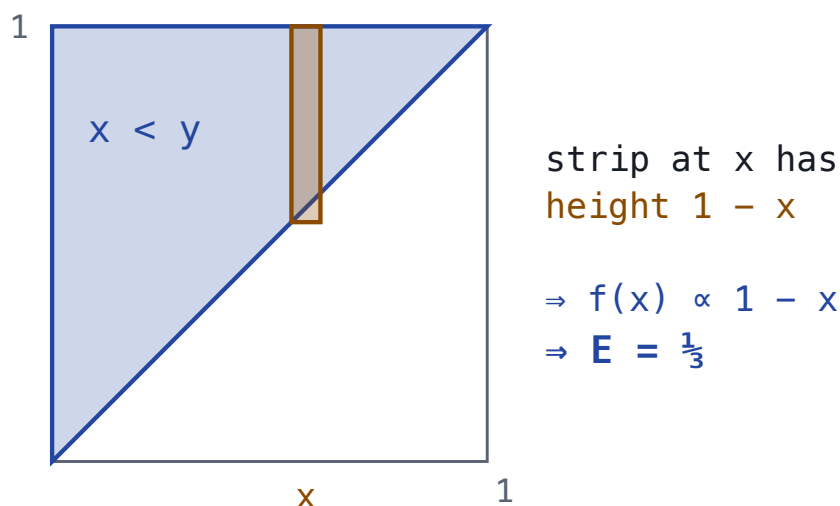
Run 100 000

Solution

Setup. X, Y are independent uniforms on $[0, 1]$. We want $\mathbb{E}[X \mid X < Y]$.

1. The conditional density of X given $X < Y$ is proportional to $f_X(x) P(Y > x) = 1 \cdot (1 - x)$.

Why: Bayes for densities: weight each x by the chance of the event given $X = x$.



2. $\int_0^1 (1 - x) dx = \frac{1}{2}$, so the density is $f(x) = 2(1 - x)$.

Why: normalise to total 1.

3. $\mathbb{E} = \int_0^1 x \cdot 2(1 - x) dx = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}$.

Why: integrate. This is also the mean of the minimum of two uniforms.

Conclusion. $\mathbb{E}[X \mid X < Y] = \frac{1}{3}$. ■

37 · Inclusion–exclusion for waiting times

Each trial produces at most one of the target events i , with probability p_i (the events are **mutually exclusive within a trial**), independently across trials.

Let T be the time until every event has occurred. Then

$$\mathbb{E}[T] = \sum_{\emptyset \neq J} \frac{(-1)^{|J|+1}}{\sum_{j \in J} p_j}.$$

Proof: tail sum plus PIE on $\{T \geq t\} = \bigcup_i \{i \text{ unseen in } t - 1 \text{ trials}\}$.

$P(\text{no } j \in J \text{ in one trial}) = 1 - \sum_J p_j$ (this uses exclusivity).

Each term is a geometric series summing to $1/\sum_J p_j$.

△ Course omits the "mutually exclusive within a trial" condition. Without it, $1 - \sum_J p_j$ is wrong.

See both 5 and 6; see both H and T ($P(H) = \frac{1}{3}$)

Question: (a) A fair die is rolled until both a 5 and a 6 have appeared (in any order, on different rolls). What is the expected number of rolls? (b) A coin with $P(\text{heads}) = \frac{1}{3}$ is tossed until both a head and a tail have appeared. What is the expected number of tosses?

Key idea

Two events: $\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p_1 + p_2}$.

Picture

Wait for both = wait for the first + wait for the second – wait for either. $6 + 6 - 3 = 9$ for the die;
 $3 + 1.5 - 1 = 3.5$ for the coin.

Run once

Run 100,000

Solution

Setup. Two target outcomes that can't happen on the same trial, with probabilities p_1, p_2 per trial. T = the time until both have appeared, T_i = the time until outcome i first appears.

1. $T = \max(T_1, T_2)$ and $\max + \min = T_1 + T_2$, so $\mathbb{E}[T] = \mathbb{E}[T_1] + \mathbb{E}[T_2] - \mathbb{E}[\min(T_1, T_2)]$.

Why: for two numbers, $\max + \min =$ their sum; linearity.

2. T_i is geometric with mean $\frac{1}{p_i}$, and $\min(T_1, T_2)$ is the wait for 'either', geometric with mean $\frac{1}{p_1 + p_2}$.

Why: the outcomes are exclusive within a trial, so 'either' has probability $p_1 + p_2$.

3. Die, 5 and 6: $6 + 6 - 3 = 9$.

Why: $p_1 = p_2 = \frac{1}{6}$, $p_1 + p_2 = \frac{1}{3}$.

4. Coin with $P(H) = \frac{1}{3}$: $3 + \frac{3}{2} - 1 = 3.5$.

Why: $p_1 = \frac{1}{3}$, $p_2 = \frac{2}{3}$, $p_1 + p_2 = 1$.

Conclusion. 9 and 3.5. ■

d10: see 7, 8, 9 and one of {5,10}

A fair 10-sided die (faces 1 to 10) is rolled over and over.

Question: what is the expected number of rolls until each of 7, 8 and 9 has appeared at least once *and* at least one 5 or 10 has appeared?

Key idea

Four exclusive events with $p = .1, .1, .1, .2$.

Picture

Inclusion-exclusion on waiting times. Add the waits for each single target, subtract waits for "either of two", add back for "any of three", subtract for "any of four". Each wait is $1 \div (\text{total probability})$.

Run once

Run 100,000

Solution

Setup. A d10 is rolled until 7, 8, 9 and one of {5,10} have all appeared. Four exclusive events per roll with $p = 0.1, 0.1, 0.1, 0.2$. Formula: $\mathbb{E}[T] = \sum_{\emptyset \neq J} (-1)^{|J|+1} / p_J$, where p_J is the total probability of the events in J .

1. Singletons: $\frac{1}{0.1} \cdot 3 + \frac{1}{0.2} = 30 + 5 = 35$.

Why: one term per event.

2. Pairs (subtract): three pairs from {7,8,9} with $p = 0.2$ give $3 \cdot 5 = 15$; three pairs with {5,10} with $p = 0.3$ give $3 \cdot \frac{10}{3} = 10$. Total 25.

Why: $\binom{4}{2} = 6$ pairs.

3. Triples (add): $\{7, 8, 9\}$ with $p = 0.3$ gives $\frac{10}{3}$; three triples containing $\{5, 10\}$ with $p = 0.4$ give $3 \cdot 2.5 = 7.5$. Total $\frac{65}{6}$.

Why: $\binom{4}{3} = 4$ triples.

4. All four (subtract): $p = 0.5$, giving 2.

Why: one term.

5. $\mathbb{E}[T] = 35 - 25 + \frac{65}{6} - 2 = \frac{48}{6} + \frac{65}{6} = \frac{113}{6} \approx 18.83333$.

Why: alternating signs. (Exact; simulated 18.85.)

Conclusion. About 18.83 rolls on average. ■

38 · Summation via expectation

$\sum_{x \in K} x = |K| \cdot$ (the mean of a uniform element of K).

Write the number digit by digit, $N = \sum 10^i d_i$, and use linearity: $\mathbb{E}[N] = \sum 10^i \mathbb{E}[d_i]$.

Palindromes (third proof)

Same palindromes as before: all five-digit numbers $\overline{abcba}$ with $a \neq 0$.

Question: find their sum once more, this time as (number of palindromes) $\times$ (average palindrome).

Key idea

For a uniform palindrome: $\mathbb{E}[a] = 5$ and $\mathbb{E}[b] = \mathbb{E}[c] = 4.5$.

Picture

Sum = count × average. A random palindrome has average digits $a = 5$, $b = c = 4.5$, so it averages 55000. Times 900.

Solution

Setup. A uniformly random five-digit palindrome $\overline{abcba}$ corresponds to independent a uniform on $\{1..9\}$ and b, c uniform on $\{0..9\}$. Its value is $N = 10001a + 1010b + 100c$.

1. $\mathbb{E}[a] = 5$ and $\mathbb{E}[b] = \mathbb{E}[c] = 4.5$.

Why: means of uniform distributions on 1..9 and 0..9.

2. $\mathbb{E}[N] = 10001 \cdot 5 + 1010 \cdot 4.5 + 100 \cdot 4.5 = 50005 + 4545 + 450 = 55000$.

Why: linearity.

3. Sum of all palindromes $= 900 \cdot \mathbb{E}[N] = 900 \cdot 55000 = 49\,500\,000$.

Why: sum = (number of items) × (average item).

Conclusion. 49,500,000 again. ■

8-digit strings of 1s and 9s

X is an 8-digit number whose digits are each 1 or 9, for example 19911191.

Question: what is the sum of all possible values of X ?

Key idea

The set of possible X is all 2^8 strings over $\{1,9\}$. Averaging uniformly over that set, each digit has mean 5.

Picture

Sum = count $\times$ average. Each digit averages 5, so the average string is 55,555,555. Times $2^8 = 256$ strings.

Solution

Setup. The set K of possible X is all $2^8 = 256$ eight-digit strings over $\{1,9\}$.

1. For a uniform element of K , each digit is independently 1 or 9 with probability $\frac{1}{2}$, so each has mean 5.

Why: uniform on all strings = independent uniform digits.

2. The mean of X is $5 \cdot (10^7 + 10^6 + \dots + 1) = 5 \cdot 11111111 = 55555555$.

Why: linearity over the 8 place values.

3. Sum over $K = 256 \cdot 55555555 = 14222222080$.

Why: sum = count $\times$ mean.

Conclusion. The sum of all possible X is 14,222,222,080. ■

△ **Note:** the two-step random choice of k is a distraction. The question is about the set of possible values, so average uniformly over that set (step 1).

0..99999 with no "11" (as 5-digit strings)

Write each of the numbers 0 to 99999 with exactly 5 digits, adding leading zeros (so 417 becomes 00417). Keep only the numbers in which the digit 1 never appears twice in a row.

Question: what is the sum of the numbers you keep?

Key idea

Count the valid strings with $f_n = 9f_{n-1} + 9f_{n-2}$. For each position, find $P(\text{digit} = 1)$. A non-1 digit is uniform over the other 9 digits (mean $\frac{44}{9}$).

Picture

Sum = count × average, digit by digit. Count strings with digit i equal to 1 (its neighbours can't be 1). The other strings have digit i spread evenly over the 9 other digits (average $44/9$).

Brute-force check: add up every 5-digit string 00000–99999 with no "11".

Add them all up

Solution

Setup. We add up all 5-digit strings $d_1d_2d_3d_4d_5$ (leading zeros allowed) that contain no '11', read as numbers.

- Let f_n be the number of length- n strings with no '11': $f_n = 9f_{n-1} + 9f_{n-2}$, with $f_0 = 1$, $f_1 = 10$. So $f_2 = 99$, $f_3 = 981$, $f_4 = 9720$, $f_5 = 96309$.

Why: split on the first digit: not 1 (9 ways), then any valid string of length $n-1$; or 1, then a non-1 second digit (9 ways), then any valid string of length $n-2$.

2. Count the valid strings with $d_i = 1$. Its neighbours must be non-1 (9 ways each), and the parts further left and right are independent valid strings. $i = 1$ or 5: $9f_3 = 8829$. $i = 2$ or 4: $9 \cdot 9 \cdot f_2 = 8019$. $i = 3$: $9 \cdot 9 \cdot f_1 \cdot f_1 = 8100$.

Why: the only rule is 'no adjacent 1s', and non-1 neighbours cut all links between the pieces.

3. Among valid strings with $d_i \neq 1$, the digit d_i is equally likely to be each of the 9 digits 0, 2, 3, ..., 9, whose average is $\frac{44}{9}$.

Why: swapping two non-1 values at position i keeps a string valid (the rule only involves 1s).

4. So the sum of digit i over all valid strings is $S_i = \#(d_i = 1) \cdot 1 + (96309 - \#(d_i = 1)) \cdot \frac{44}{9}$: $S_1 = S_5 = 8829 + 9720 \cdot 44 = 436\,509$; $S_2 = S_4 = 8019 + 9810 \cdot 44 = 439\,659$; $S_3 = 8100 + 9801 \cdot 44 = 439\,344$.

Why: $\frac{87480}{9} = 9720$, $\frac{88290}{9} = 9810$, $\frac{88209}{9} = 9801$.

5. Total = $\sum_i 10^{5-i} S_i = 4\,365\,090\,000 + 439\,659\,000 + 43\,934\,400 + 4\,396\,590 + 436\,509 = 4\,853\,516\,499$.

Why: each position contributes its digit sum times its place value. (Brute-force checked.)

Conclusion. The sum is 4,853,516,499. ■

39 · Variance

1. $\text{Var } X = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. $\text{SD} = \sqrt{\text{Var}}$.
2. $\text{Var}(aX + b) = a^2 \text{Var } X$, and $\text{Var}(c) = 0$.
3. $\text{Var}(X + Y) = \text{Var } X + \text{Var } Y + 2 \text{Cov}(X, Y)$. For independent (or just uncorrelated) variables the covariance is 0 and **variances add**. They do not add in general.
4. For a count $X = \sum I_i$: $\mathbb{E}[X^2] = \sum P(I_i) + \sum_{i \neq j} P(I_i I_j)$.
5. **Law of total variance:** $\text{Var } X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$.

Dist.	Var	Dist.	Var
Bernoulli(p)	$p(1 - p)$	$U(a, b)$	$\frac{(b-a)^2}{12}$
Bin(n, p)	$np(1 - p)$	Exp(λ)	$1/\lambda^2$
Geo(p)	$\frac{1-p}{p^2}$	$N(\mu, \sigma^2)$	σ^2
NegBin(r, p)	$\frac{r(1-p)}{p^2}$	Gamma(α , rate λ)	α/λ^2
Poisson(λ)	λ	Beta(α, β)	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Unif{1.. n }	$\frac{n^2-1}{12}$	Hypergeo	$n \frac{K}{N} (1 - \frac{K}{N}) \frac{N-n}{N-1}$

Die; uniform{1.. n }; exponential

Question: (a) Find the variance and standard deviation of one roll of a fair die, and more generally the variance of a number chosen uniformly from $\{1, \dots, n\}$. (b) Find the variance of an exponential random variable with rate λ .

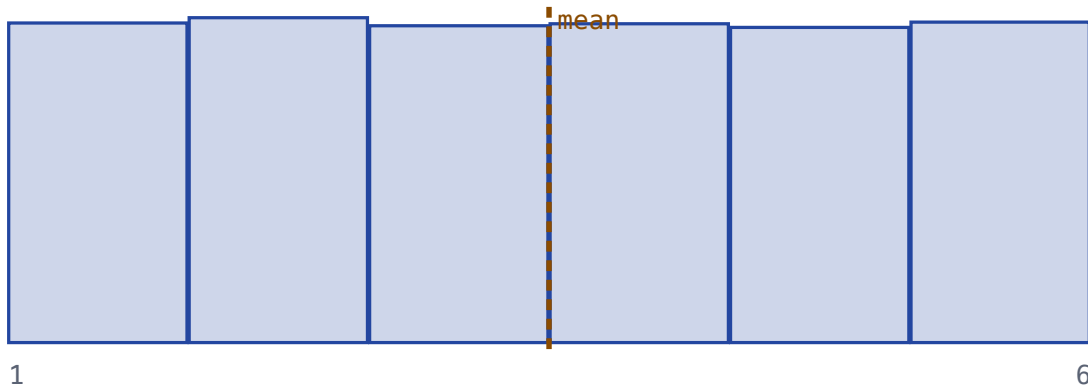
Key idea

$$\mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Picture

Variance = average of the square – square of the average. For a die: $91/6 - 3.5^2 = 35/12$.

One die, 50 000 times.



Sample 50,000

Mean **3.494**, variance **2.923** (exact 3.5 and $35/12 \approx 2.917$).

Solution

1. Uniform on $\{1..n\}$: $\mathbb{E}[X] = \frac{n+1}{2}$ and $\mathbb{E}[X^2] = \frac{1}{n} \sum k^2 = \frac{(n+1)(2n+1)}{6}$.

Why: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.

2. $\text{Var } X = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)[2(2n+1)-3(n+1)]}{12} = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}$.

Why: common denominator 12, then factor out $(n+1)$.

3. Die ($n = 6$): $\text{Var} = \frac{35}{12}$, $\text{SD} = \sqrt{35/12} \approx 1.70783$.

Why: substitute.

4. $\text{Exp}(\lambda)$: $\mathbb{E}[Y^2] = \int_0^\infty y^2 \lambda e^{-\lambda y} dy = \frac{2}{\lambda^2}$.

Why: integrate by parts: the boundary term vanishes and the rest is $\frac{2}{\lambda} \int_0^\infty y e^{-\lambda y} dy = \frac{2}{\lambda} \cdot \frac{1}{\lambda}$.

5. $\text{Var } Y = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$.

Why: $\mathbb{E}[Y] = \frac{1}{\lambda}$.

Conclusion. $\frac{n^2-1}{12}$ (SD ≈ 1.708 for a die) and $\frac{1}{\lambda^2}$. ■

x -coordinate of a uniform point on the sphere

A point (X, Y, Z) is chosen uniformly at random on the surface of a sphere of radius 1 centred at the origin.

Question: what is the variance of its x -coordinate X ?

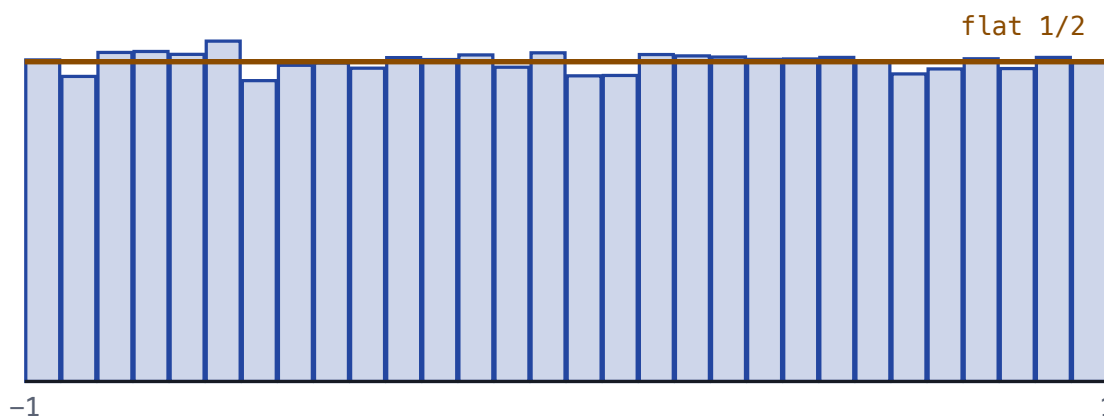
Key idea

By symmetry $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \mathbb{E}[Z^2]$, and they sum to 1.

Picture

x^2, y^2, z^2 are interchangeable and add to 1, so each averages $\frac{1}{3}$. (Surprise bonus: x is exactly uniform on $[-1, 1]$.)

x -coordinates of 50 000 random points on the sphere: perfectly flat.



Sample 50 000

Blue bars: 50 000 samples. Orange: the formula. Sample mean **-0.002** (exact 0 (variance 1/3)).

Solution

Setup. (X, Y, Z) is a uniform point on the unit sphere, so $X^2 + Y^2 + Z^2 = 1$.

1. $\mathbb{E}[X] = 0$.

Why: the reflection $x \mapsto -x$ preserves the uniform distribution on the sphere.

2. $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \mathbb{E}[Z^2]$.

Why: rotations that permute the axes preserve the uniform distribution.

3. Their sum is $\mathbb{E}[X^2 + Y^2 + Z^2] = 1$, so each is $\frac{1}{3}$.

Why: linearity.

4. $\text{Var } X = \mathbb{E}[X^2] - 0^2 = \frac{1}{3}$.

Why: steps 1 and 3.

Conclusion. $\text{Var } X = \frac{1}{3}$. ■

Fixed points of a random permutation ($n = 1000$)

A permutation of $1, \dots, 1000$ is chosen uniformly at random. A **fixed point** is a number that stays in its own place.

Question: what is the variance of the number of fixed points?

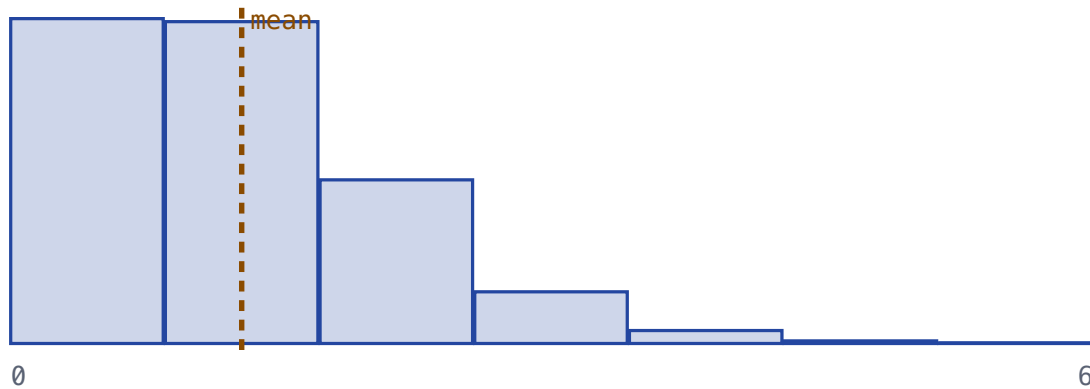
Key idea

$$X = \sum I_i, \text{ with } P(I_i I_j = 1) = \frac{1}{n(n-1)}.$$

Picture

Count with indicators and square. $E[X^2] = E[X] +$
 (pairs both fixed) $= 1 + 1$. So variance $= 2 - 1 = 1$.

Fixed points of 20 000 random permutations of 1..1000.



Sample 20,000

Mean **0.993**, variance **0.988** (exact mean 1, variance 1).

Solution

Setup. π is a uniform permutation of $\{1, \dots, n\}$, $n = 1000$. $I_i = 1$ if $\pi(i) = i$, and $X = \sum I_i$.

1. $\mathbb{E}[X] = n \cdot \frac{1}{n} = 1$.

Why: $P(\pi(i) = i) = \frac{1}{n}$; linearity.

2. $X^2 = \sum_i I_i^2 + \sum_{i \neq j} I_i I_j = X + \sum_{i \neq j} I_i I_j$.

Why: expand the square; $I_i^2 = I_i$ because $I_i \in \{0, 1\}$.

3. $\mathbb{E}[I_i I_j] = P(\pi(i) = i, \pi(j) = j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$, and there are $n(n-1)$ ordered pairs $i \neq j$, so the second sum has mean 1.

Why: fix two values; count ordered pairs.

4. $\mathbb{E}[X^2] = 1 + 1 = 2$, so $\text{Var } X = 2 - 1^2 = 1$ (for every $n \geq 2$).

Why: $\text{Var} = \mathbb{E}[X^2] - (\mathbb{E}X)^2$.

Conclusion. The number of fixed points has mean 1 and variance 1. ■

$|A \cap B|$ for random subsets of $\{1..25\}$

Two subsets A and B of $\{1, \dots, 25\}$ are chosen independently and uniformly at random (every subset equally likely).

Question: what is the variance of $|A \cap B|$, the number of elements they have in common?

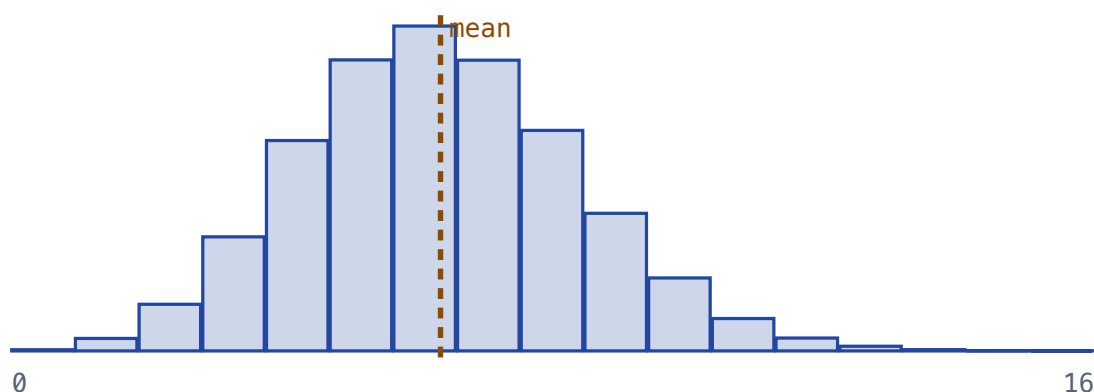
Key idea

25 independent $\text{Bernoulli}(\frac{1}{4})$ indicators.

Picture

Each element is in both sets with chance $\frac{1}{4}$, independently. So $|A \cap B|$ is $\text{Binomial}(25, \frac{1}{4})$: variance $25 \times \frac{1}{4} \times \frac{3}{4}$.

$|A \cap B|$ for 50 000 random pairs of subsets.



Sample 50,000

Mean **6.232**, variance **4.713** (exact mean 6.25, variance 4.6875).

Solution

Setup. A and B are independent uniformly random subsets of $\{1, \dots, 25\}$: each element is in each set independently with probability $\frac{1}{2}$.

1. $|A \cap B| = \sum_{k=1}^{25} J_k$, where $J_k = 1$ if $k \in A$ and $k \in B$.

Why: count the common elements.

2. The J_k are independent $\text{Bernoulli}(\frac{1}{4})$.

Why: $P(k \in A, k \in B) = \frac{1}{2} \cdot \frac{1}{2}$, and different elements are independent.

3. $\text{Var}|A \cap B| = 25 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{75}{16} = 4.6875$.

Why: variances of independent variables add, and a $\text{Bernoulli}(p)$ has variance $p(1-p)$.

Conclusion. $\frac{75}{16}$. ■

Heads from a die-many coin tosses

Roll a fair die; say it shows X . Then toss X fair coins, and let Y be the number of heads.

Question: what is the variance of Y ?

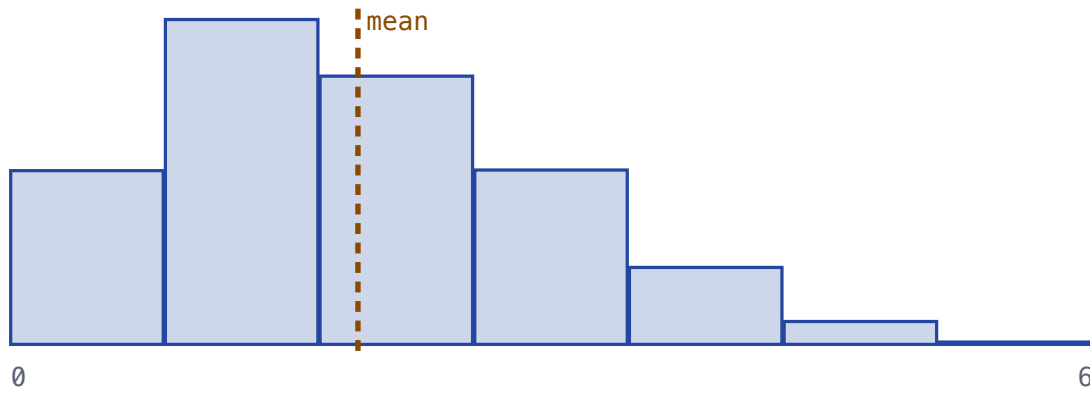
Key idea

Total variance: $\text{Var}(Y | X) = X/4$ and $\mathbb{E}[Y | X] = X/2$.

Picture

Variance = average spread within each die value + spread of the averages. Given X coins: mean $X/2$, variance $X/4$. So $3.5/4 + \text{Var}(X)/4$.

Roll a die, toss that many coins, count heads. 50 000 times.



Sample 50,000

Mean **1.745**, variance **1.607** (exact variance $77/48 \approx 1.604$).

Solution

Setup. X is a die roll; then $Y \sim \text{Bin}(X, \frac{1}{2})$ (toss X coins and count heads).

$$1. \text{Var } Y = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X]).$$

Why: law of total variance.

$$2. \text{ Given } X: \mathbb{E}[Y | X] = \frac{X}{2} \text{ and } \text{Var}(Y | X) = X \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{X}{4}.$$

Why: binomial mean np and variance $np(1-p)$.

$$3. \mathbb{E}\left[\frac{X}{4}\right] = \frac{3.5}{4} = \frac{7}{8} \text{ and } \text{Var}\left(\frac{X}{2}\right) = \frac{1}{4} \text{Var } X = \frac{1}{4} \cdot \frac{35}{12} = \frac{35}{48}.$$

Why: $\text{Var}(aX) = a^2 \text{Var } X$; a die has variance $\frac{35}{12}$.

$$4. \text{Var } Y = \frac{42}{48} + \frac{35}{48} = \frac{77}{48} \approx 1.60417.$$

Why: add, using $\frac{7}{8} = \frac{42}{48}$.

Conclusion. $\text{Var } Y = \frac{77}{48}$. ■

Mixture of two normals

With probability h , Y is drawn from a normal distribution with mean μ_h and variance σ_h^2 ; otherwise (probability $1-h$) it is drawn from a normal distribution with mean μ_t and variance σ_t^2 .

Question: find the variance of Y .

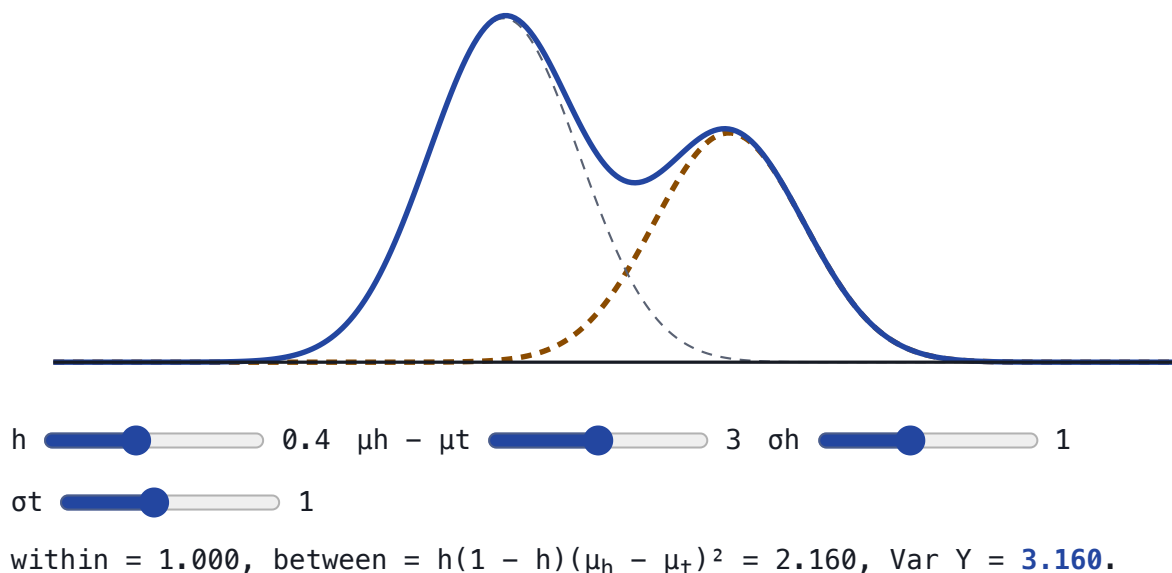
Key idea

Within-group variance + between-group variance.

Picture

Spread inside each group + spread between the groups. Within: $h \sigma_h^2 + (1 - h) \sigma_t^2$. Between: the two centres, weighted h and $1 - h$.

Two bell curves mixed. Total spread = the spread inside each curve (averaged) + the spread between the two centres.



Solution

Setup. Let $I = 1$ with probability h and $I = 0$ otherwise. Given $I = 1$, $Y \sim N(\mu_h, \sigma_h^2)$; given $I = 0$, $Y \sim N(\mu_t, \sigma_t^2)$.

$$1. \text{Var } Y = \mathbb{E}[\text{Var}(Y \mid I)] + \text{Var}(\mathbb{E}[Y \mid I]).$$

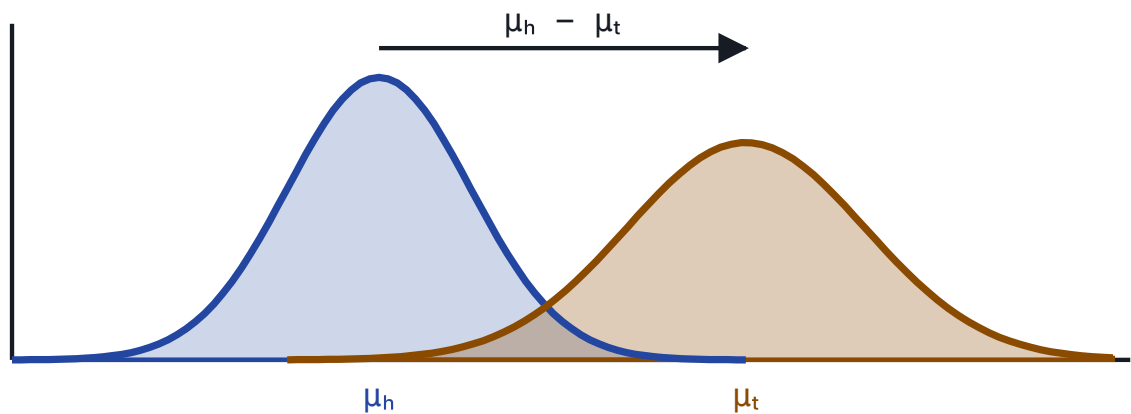
Why: law of total variance.

$$2. \mathbb{E}[\text{Var}(Y \mid I)] = h\sigma_h^2 + (1 - h)\sigma_t^2.$$

Why: the conditional variance is σ_h^2 or σ_t^2 , weighted by the probabilities.

3. $\mathbb{E}[Y | I]$ equals μ_h with probability h and μ_t otherwise; its variance is $h(1-h)(\mu_h - \mu_t)^2$.

Why: for a two-point variable with values a (prob h) and b : the mean is $m = ha + (1-h)b$, $a - m = (1-h)(a - b)$, $b - m = -h(a - b)$, so the variance is $h(1-h)^2(a - b)^2 + (1-h)h^2(a - b)^2 = h(1-h)(a - b)^2$.



$$\text{Var} = \text{within } (h\sigma_h^2 + (1-h)\sigma_t^2) + \text{between } h(1-h)(\mu_h - \mu_t)^2$$

4. $\text{Var } Y = h\sigma_h^2 + (1-h)\sigma_t^2 + h(1-h)(\mu_h - \mu_t)^2$.

Why: add steps 2 and 3.

Conclusion. Variance = within-group + between-group.

■